T3 intermediate-EFT one-loop RGE term comparison: EFT_1 after F

This stage has active field content SM+$S_1 + S_2$. The report combines the renormalisable RGBeta running with the dimension-five $\psi^2\phi^2$ Wilson-coefficient running generated after integrating out F. Throughout, $16\pi^2\,\mu\,dX/d\mu = \beta_X^{(1)}$. For the Wilson sector only the tree-generated stage-1 coefficient $C^{(0)}$ is inserted into the one-loop anomalous dimension; the one-loop threshold piece is retained as a boundary term and is not run again at this order.

Coupling and interaction notation

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\mathrm{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\mathrm{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\mathrm{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$

$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C^{ij;AB}_{LLS_1S_2}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Additional general-interaction couplings

Symbol	Field content	Stored coupling / contraction
$\lambda^{(1)}_{X_5}$	$(S_1^\dagger)^2\, S_2\, S_2^\dagger$	`lambdaT3GeneralQ000211I1`
λ_{X_6}	$(S_1^\dagger)^4$	`lambdaT3GeneralQ000400I1`
λ_{X_9}	$S_1\, (S_1^\dagger)^3$	`lambdaT3GeneralQ001300I1`
$\lambda_{X_{14}}$	$(H^\dagger)^2\, S_1\, S_2$	`lambdaT3GeneralQ021010I1`
$\lambda_{X_{18}}$	$H\, H^\dagger\, (S_1^\dagger)^2$	`lambdaT3GeneralQ110200I1`
λ_{X_1}	$(S_2^\dagger)^4$	`lambdaT3GeneralQ000004I1`
λ_{X_2}	$S_2\, (S_2^\dagger)^3$	`lambdaT3GeneralQ000013I1`
$\lambda^{(1)}_{X_7}$	$S_1\, S_1^\dagger\, (S_2^\dagger)^2$	`lambdaT3GeneralQ001102I1`
$\lambda_{X_{12}}$	$(H^\dagger)^2\, S_1^\dagger\, S_2^\dagger$	`lambdaT3GeneralQ020101I1`
$\lambda_{X_{16}}$	$H\, H^\dagger\, (S_2^\dagger)^2$	`lambdaT3GeneralQ110002I1`
$\lambda_{X_{15}}$	$(H^\dagger)^2\, (S_1)^2$	`lambdaT3GeneralQ022000I1`
$\lambda^{(1)}_{X_{17}}$	$H\, H^\dagger\, S_1^\dagger\, S_2^\dagger$	`lambdaT3GeneralQ110101I1`
$\lambda_{X_{10}}$	$(H^\dagger)^2\, (S_2^\dagger)^2$	`lambdaT3GeneralQ020002I1`
$\lambda^{(2)}_{X_5}$	$(S_1^\dagger)^2\, S_2\, S_2^\dagger$	`lambdaT3GeneralQ000211I2`
$\lambda^{(2)}_{X_7}$	$S_1\, S_1^\dagger\, (S_2^\dagger)^2$	`lambdaT3GeneralQ001102I2`
$y_1^{\rm conj}$	$L\, F\, S_i^\dagger$	`y1Conj`
$y_2^{\rm conj}$	$L\, F\, S_i^\dagger$	`y2Conj`

Renormalisable EFT1 couplings

$\beta^{(1)}_{g_1}$

Quantity: g_1

$$16\pi^2\, \mu \frac{dg_1}{d\mu} = \beta^{(1)}_{g_1}$$

Column	Representative term structure
T1	$\frac{47{g_1}^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	$\frac{47{g_1}^3}{6}$
A	-2	1	N/A	3	N/A	2	N/A	$\frac{43{g_1}^3}{6}$
A	-4	1	N/A	3	N/A	2	N/A	$\frac{55{g_1}^3}{6}$
B	1	2	N/A	2	N/A	1	N/A	$\frac{17{g_1}^3}{2}$
B	-1	2	N/A	2	N/A	1	N/A	$\frac{43{g_1}^3}{6}$
B	-3	2	N/A	2	N/A	1	N/A	$\frac{17{g_1}^3}{2}$
C	1	2	N/A	2	N/A	3	N/A	$\frac{17{g_1}^3}{2}$
C	-1	2	N/A	2	N/A	3	N/A	$\frac{43{g_1}^3}{6}$
C	-3	2	N/A	2	N/A	3	N/A	$\frac{17{g_1}^3}{2}$
D	0	3	N/A	1	N/A	2	N/A	$\frac{43{g_1}^3}{6}$
D	2	3	N/A	1	N/A	2	N/A	$\frac{55{g_1}^3}{6}$
D	-2	3	N/A	1	N/A	2	N/A	$\frac{47{g_1}^3}{6}$
E	0	3	N/A	3	N/A	2	N/A	$\frac{47{g_1}^3}{6}$
E	2	3	N/A	3	N/A	2	N/A	$\frac{71{g_1}^3}{6}$
E	-2	3	N/A	3	N/A	2	N/A	$\frac{47{g_1}^3}{6}$
E	-4	3	N/A	3	N/A	2	N/A	$\frac{71{g_1}^3}{6}$

$\beta_{g_2}^{(1)}$

Quantity: g_2

$$16\pi^2\,\mu\frac{dg_2}{d\mu} = \beta_{g_2}^{(1)}$$

Column	Representative term structure
T1	$-\frac{5{g_2}^3}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	$-\frac{5{g_2}^3}{2}$
A	-2	1	N/A	3	N/A	2	N/A	$-\frac{5{g_2}^3}{2}$
A	-4	1	N/A	3	N/A	2	N/A	$-\frac{5{g_2}^3}{2}$
B	1	2	N/A	2	N/A	1	N/A	$-\frac{17{g_2}^3}{6}$
B	-1	2	N/A	2	N/A	1	N/A	$-\frac{17{g_2}^3}{6}$
B	-3	2	N/A	2	N/A	1	N/A	$-\frac{17{g_2}^3}{6}$
C	1	2	N/A	2	N/A	3	N/A	$-\frac{17{g_2}^3}{6}$
C	-1	2	N/A	2	N/A	3	N/A	$-\frac{17{g_2}^3}{6}$
C	-3	2	N/A	2	N/A	3	N/A	$-\frac{17{g_2}^3}{6}$
D	0	3	N/A	1	N/A	2	N/A	$-\frac{5{g_2}^3}{2}$
D	2	3	N/A	1	N/A	2	N/A	$-\frac{5{g_2}^3}{2}$
D	-2	3	N/A	1	N/A	2	N/A	$-\frac{5{g_2}^3}{2}$
E	0	3	N/A	3	N/A	2	N/A	$-\frac{11{g_2}^3}{6}$
E	2	3	N/A	3	N/A	2	N/A	$-\frac{11{g_2}^3}{6}$
E	-2	3	N/A	3	N/A	2	N/A	$-\frac{11{g_2}^3}{6}$
E	-4	3	N/A	3	N/A	2	N/A	$-\frac{11{g_2}^3}{6}$

$\beta_{g_3}^{(1)}$

Quantity: g_3

$$16\pi^2\,\mu\frac{dg_3}{d\mu} = \beta_{g_3}^{(1)}$$

Column	Representative term structure
T1	$-7{g_3}^3$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	$-7{g_3}^3$
A	-2	1	N/A	3	N/A	2	N/A	$-7{g_3}^3$
A	-4	1	N/A	3	N/A	2	N/A	$-7{g_3}^3$
B	1	2	N/A	2	N/A	1	N/A	$-7{g_3}^3$
B	-1	2	N/A	2	N/A	1	N/A	$-7{g_3}^3$
B	-3	2	N/A	2	N/A	1	N/A	$-7{g_3}^3$
C	1	2	N/A	2	N/A	3	N/A	$-7{g_3}^3$
C	-1	2	N/A	2	N/A	3	N/A	$-7{g_3}^3$
C	-3	2	N/A	2	N/A	3	N/A	$-7{g_3}^3$
D	0	3	N/A	1	N/A	2	N/A	$-7{g_3}^3$
D	2	3	N/A	1	N/A	2	N/A	$-7{g_3}^3$
D	-2	3	N/A	1	N/A	2	N/A	$-7{g_3}^3$
E	0	3	N/A	3	N/A	2	N/A	$-7{g_3}^3$
E	2	3	N/A	3	N/A	2	N/A	$-7{g_3}^3$
E	-2	3	N/A	3	N/A	2	N/A	$-7{g_3}^3$
E	-4	3	N/A	3	N/A	2	N/A	$-7{g_3}^3$

$\beta^{(1)}_{Y_u}$

Quantity: Y_u

$$16\pi^2\,\mu\frac{dY_u}{d\mu} = \beta^{(1)}_{Y_u}$$

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Column	Representative term structure
T1	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$
T2	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$
T3	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
B	−1	2	N/A	2	N/A	1	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
B	−3	2	N/A	2	N/A	1	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$

Quantity: Y_u

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Column	Representative term structure
T1	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$
T2	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$
T3	$-8{g_3}^2\ (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
B	−1	2	N/A	2	N/A	1	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
B	−3	2	N/A	2	N/A	1	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$

Quantity: Y_u

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
B	−1	2	N/A	2	N/A	1	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
B	−3	2	N/A	2	N/A	1	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$

$\beta^{(1)}_{Y_d}$

Quantity: Y_d

$$16\pi^2\,\mu\frac{dY_d}{d\mu} = \beta^{(1)}_{Y_d}$$

Terms 1 of 3

Column	Representative term structure
T1	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$
T2	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$
T3	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
B	−1	2	N/A	2	N/A	1	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
B	−3	2	N/A	2	N/A	1	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$

Quantity: Y_d

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$
T2	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$
T3	$-8{g_3}^2\ (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
B	−1	2	N/A	2	N/A	1	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
B	−3	2	N/A	2	N/A	1	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$

Quantity: Y_d

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
B	−1	2	N/A	2	N/A	1	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
B	−3	2	N/A	2	N/A	1	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$

$\beta_{Y_e}^{(1)}$

Quantity: Y_e

$$16\pi^2\,\mu\frac{dY_e}{d\mu} = \beta_{Y_e}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$
T2	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$
T3	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
B	−1	2	N/A	2	N/A	1	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
B	−3	2	N/A	2	N/A	1	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$

Quantity: Y_e

Terms 2 of 2

Column	Representative term structure
T1	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$
T2	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$
T3	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
A	−2	1	N/A	3	N/A	2	N/A	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
A	−4	1	N/A	3	N/A	2	N/A	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
B	1	2	N/A	2	N/A	1	N/A	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
B	−1	2	N/A	2	N/A	1	N/A	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
B	−3	2	N/A	2	N/A	1	N/A	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
C	1	2	N/A	2	N/A	3	N/A	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
C	−1	2	N/A	2	N/A	3	N/A	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
C	−3	2	N/A	2	N/A	3	N/A	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
D	0	3	N/A	1	N/A	2	N/A	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
D	2	3	N/A	1	N/A	2	N/A	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
D	−2	3	N/A	1	N/A	2	N/A	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
E	0	3	N/A	3	N/A	2	N/A	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
E	2	3	N/A	3	N/A	2	N/A	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
E	−2	3	N/A	3	N/A	2	N/A	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$
E	−4	3	N/A	3	N/A	2	N/A	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$

$\beta^{(1)}_{m_1^2}$

Quantity: m_1^2

$$16\pi^2\,\mu\frac{dm_1^2}{d\mu} = \beta^{(1)}_{m_1^2}$$

Terms 1 of 5

Column	Representative term structure
T1	$12\mu_{\mathrm{T3GeneralS1Sq}}\left(\lambda_{X_9}\right)^*$
T2	$+12\lambda_{X_9}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$
T3	$+6\lambda_{12}^{(1)}m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$12\mu_{\mathrm{T3GeneralS1Sq}}\left(\lambda_{X_9}\right)^*$	$+12\lambda_{X_9}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$	$+6\lambda_{12}^{(1)}m_2^2$
A	−2	1	N/A	3	N/A	2	N/A	—	—	$+6\lambda_{12}^{(1)}m_2^2$
A	−4	1	N/A	3	N/A	2	N/A	—	—	$+6\lambda_{12}^{(1)}m_2^2$
B	1	2	N/A	2	N/A	1	N/A	—	—	$+4\lambda_{12}^{(1)}m_2^2$
B	−1	2	N/A	2	N/A	1	N/A	—	—	$+4\lambda_{12}^{(1)}m_2^2$
B	−3	2	N/A	2	N/A	1	N/A	—	—	$+4\lambda_{12}^{(1)}m_2^2$
C	1	2	N/A	2	N/A	3	N/A	—	—	$+4\lambda_{12}^{(1)}m_2^2$
C	−1	2	N/A	2	N/A	3	N/A	—	—	$+4\lambda_{12}^{(1)}m_2^2$
C	−3	2	N/A	2	N/A	3	N/A	—	—	$+4\lambda_{12}^{(1)}m_2^2$
D	0	3	N/A	1	N/A	2	N/A	$20\mu_{\mathrm{T3GeneralS1Sq}}\left(\lambda_{X_9}\right)^*$	$+20\lambda_{X_9}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$	$+2\lambda_{12}^{(1)}m_2^2$
D	2	3	N/A	1	N/A	2	N/A	—	—	$+2\lambda_{12}^{(1)}m_2^2$
D	−2	3	N/A	1	N/A	2	N/A	—	—	$+2\lambda_{12}^{(1)}m_2^2$
E	0	3	N/A	3	N/A	2	N/A	$20\mu_{\mathrm{T3GeneralS1Sq}}\left(\lambda_{X_9}\right)^*$	$+20\lambda_{X_9}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: m_1^2

Terms 2 of 5

Column	Representative term structure
T1	$+4\lambda_{S_1}^{(1)} m_1^2$
T2	$12\mu_{\mathrm{T3GeneralS2Sq}} \left(\lambda_{X_7}^{(1)}\right)^*$
T3	$+12\lambda_{X_7}^{(1)} \left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+4\lambda_{S_1}^{(1)} m_1^2$	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+4\lambda_{S_1}^{(1)} m_1^2$	$12\mu_{\mathrm{T3GeneralS2Sq}} \left(\lambda_{X_7}^{(1)}\right)^*$	$+12\lambda_{X_7}^{(1)} \left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$
A	-4	1	N/A	3	N/A	2	N/A	$+4\lambda_{S_1}^{(1)} m_1^2$	—	—
B	1	2	N/A	2	N/A	1	N/A	$+6\lambda_{S_1}^{(1)} m_1^2$	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+6\lambda_{S_1}^{(1)} m_1^2$	—	—
B	-3	2	N/A	2	N/A	1	N/A	$+6\lambda_{S_1}^{(1)} m_1^2$	—	—
C	1	2	N/A	2	N/A	3	N/A	$+6\lambda_{S_1}^{(1)} m_1^2$	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+6\lambda_{S_1}^{(1)} m_1^2$	—	—
C	-3	2	N/A	2	N/A	3	N/A	$+6\lambda_{S_1}^{(1)} m_1^2$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	$4\mu_{\mathrm{T3GeneralS2Sq}} \left(\lambda_{X_7}^{(1)}\right)^*$	$+4\lambda_{X_7}^{(1)} \left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	$+12\lambda_{X_7}^{(1)} \left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$ $+4\lambda_{X_7}^{(2)} \left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: m_1^2

Terms 3 of 5

Column	Representative term structure
T1	$-6{g_1}^2 m_1^2$
T2	$-\frac{9{g_2}^2 m_1^2}{2}$
T3	$-\frac{3{g_1}^2 m_1^2}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$-6{g_1}^2 m_1^2$	—	—
A	−4	1	N/A	3	N/A	2	N/A	$-24{g_1}^2 m_1^2$	—	—
B	1	2	N/A	2	N/A	1	N/A	—	$-\frac{9{g_2}^2 m_1^2}{2}$	$-\frac{3{g_1}^2 m_1^2}{2}$
B	−1	2	N/A	2	N/A	1	N/A	—	$-\frac{9{g_2}^2 m_1^2}{2}$	$-\frac{3{g_1}^2 m_1^2}{2}$
B	−3	2	N/A	2	N/A	1	N/A	—	$-\frac{9{g_2}^2 m_1^2}{2}$	$-\frac{27{g_1}^2 m_1^2}{2}$
C	1	2	N/A	2	N/A	3	N/A	—	$-\frac{9{g_2}^2 m_1^2}{2}$	$-\frac{3{g_1}^2 m_1^2}{2}$
C	−1	2	N/A	2	N/A	3	N/A	—	$-\frac{9{g_2}^2 m_1^2}{2}$	$-\frac{3{g_1}^2 m_1^2}{2}$
C	−3	2	N/A	2	N/A	3	N/A	—	$-\frac{9{g_2}^2 m_1^2}{2}$	$-\frac{27{g_1}^2 m_1^2}{2}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	$-6{g_1}^2 m_1^2$	—	—
D	−2	3	N/A	1	N/A	2	N/A	$-6{g_1}^2 m_1^2$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	$-6{g_1}^2 m_1^2$	—	—
E	−2	3	N/A	3	N/A	2	N/A	$-6{g_1}^2 m_1^2$	—	—
E	−4	3	N/A	3	N/A	2	N/A	$-24{g_1}^2 m_1^2$	—	—

Quantity: m_1^2

Terms 4 of 5

Column	Representative term structure
T1	$6\mu_{\mathrm{T3GeneralS1S2}}\left(\lambda_{11}\right)^*$
T2	$+6\lambda_{11}\left(\mu_{\mathrm{T3GeneralS1S2}}\right)^*$
T3	$-12{g_2}^2 m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$6\mu_{\mathrm{T3GeneralS1S2}}\left(\lambda_{11}\right)^*$	$+6\lambda_{11}\left(\mu_{\mathrm{T3GeneralS1S2}}\right)^*$	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$6\mu_{\mathrm{T3GeneralS1S2}}\left(\lambda_{11}\right)^*$	$+6\lambda_{11}\left(\mu_{\mathrm{T3GeneralS1S2}}\right)^*$	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	$-12{g_2}^2 m_1^2$
D	2	3	N/A	1	N/A	2	N/A	—	—	$-12{g_2}^2 m_1^2$
D	−2	3	N/A	1	N/A	2	N/A	—	—	$-12{g_2}^2 m_1^2$
E	0	3	N/A	3	N/A	2	N/A	—	—	$-12{g_2}^2 m_1^2$
E	2	3	N/A	3	N/A	2	N/A	—	—	$-12{g_2}^2 m_1^2$
E	−2	3	N/A	3	N/A	2	N/A	—	—	$-12{g_2}^2 m_1^2$
E	−4	3	N/A	3	N/A	2	N/A	—	—	$-12{g_2}^2 m_1^2$

Quantity: m_1^2

Terms 5 of 5

Column	Representative term structure
T1	$+m_1^2(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$
T2	$+m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$
T3	$4\mu_{\mathrm{T3GeneralS2Sq}}(3\left(\lambda_{X_7}^{(1)}\right)^* + \left(\lambda_{X_7}^{(2)}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+m_1^2(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	—	—
D	2	3	N/A	1	N/A	2	N/A	$+m_1^2(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+m_1^2(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	—	—
E	0	3	N/A	3	N/A	2	N/A	$+m_1^2(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$+m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	—
E	2	3	N/A	3	N/A	2	N/A	$+m_1^2(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$+m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	—
E	−2	3	N/A	3	N/A	2	N/A	$+m_1^2(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$+m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	$4\mu_{\mathrm{T3GeneralS2Sq}}(3\left(\lambda_{X_7}^{(1)}\right)^* + \left(\lambda_{X_7}^{(2)}\right)^*)$
E	−4	3	N/A	3	N/A	2	N/A	$+m_1^2(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$+m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	—

$\beta^{(1)}_{m_2^2}$

Quantity: m_2^2

$$16\pi^2\,\mu\frac{dm_2^2}{d\mu} = \beta^{(1)}_{m_2^2}$$

Terms 1 of 5

Column	Representative term structure
T1	$4\mu_{\text{T3GeneralS1Sq}}\left(\lambda^{(1)}_{X_5}\right)^*$
T2	$+4\lambda^{(1)}_{X_5}\left(\mu_{\text{T3GeneralS1Sq}}\right)^*$
T3	$-12{g_2}^2 m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$4\mu_{\text{T3GeneralS1Sq}}\left(\lambda^{(1)}_{X_5}\right)^*$	$+4\lambda^{(1)}_{X_5}\left(\mu_{\text{T3GeneralS1Sq}}\right)^*$	$-12{g_2}^2 m_2^2$
A	-2	1	N/A	3	N/A	2	N/A	—	—	$-12{g_2}^2 m_2^2$
A	-4	1	N/A	3	N/A	2	N/A	—	—	$-12{g_2}^2 m_2^2$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$12\mu_{\text{T3GeneralS1Sq}}\left(\lambda^{(1)}_{X_5}\right)^*$	$+12\lambda^{(1)}_{X_5}\left(\mu_{\text{T3GeneralS1Sq}}\right)^*$	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+12\lambda^{(1)}_{X_5}\left(\mu_{\text{T3GeneralS1Sq}}\right)^*$ $+4\lambda^{(2)}_{X_5}\left(\mu_{\text{T3GeneralS1Sq}}\right)^*$	$-12{g_2}^2 m_2^2$
E	2	3	N/A	3	N/A	2	N/A	—	—	$-12{g_2}^2 m_2^2$
E	-2	3	N/A	3	N/A	2	N/A	—	—	$-12{g_2}^2 m_2^2$
E	-4	3	N/A	3	N/A	2	N/A	—	—	$-12{g_2}^2 m_2^2$

Quantity: m_2^2

Terms 2 of 5

Column	Representative term structure
T1	$-6{g_1}^2 m_2^2$
T2	$+2\lambda_{12}^{(1)} m_1^2$
T3	$+m_2^2(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-6{g_1}^2 m_2^2$	$+2\lambda_{12}^{(1)} m_1^2$	$+m_2^2(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$
A	-2	1	N/A	3	N/A	2	N/A	—	$+2\lambda_{12}^{(1)} m_1^2$	$+m_2^2(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$
A	-4	1	N/A	3	N/A	2	N/A	$-6{g_1}^2 m_2^2$	$+2\lambda_{12}^{(1)} m_1^2$	$+m_2^2(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$
B	1	2	N/A	2	N/A	1	N/A	—	$+4\lambda_{12}^{(1)} m_1^2$	—
B	-1	2	N/A	2	N/A	1	N/A	—	$+4\lambda_{12}^{(1)} m_1^2$	—
B	-3	2	N/A	2	N/A	1	N/A	—	$+4\lambda_{12}^{(1)} m_1^2$	—
C	1	2	N/A	2	N/A	3	N/A	—	$+4\lambda_{12}^{(1)} m_1^2$	—
C	-1	2	N/A	2	N/A	3	N/A	—	$+4\lambda_{12}^{(1)} m_1^2$	—
C	-3	2	N/A	2	N/A	3	N/A	—	$+4\lambda_{12}^{(1)} m_1^2$	—
D	0	3	N/A	1	N/A	2	N/A	$-6{g_1}^2 m_2^2$	$+6\lambda_{12}^{(1)} m_1^2$	—
D	2	3	N/A	1	N/A	2	N/A	$-24{g_1}^2 m_2^2$	$+6\lambda_{12}^{(1)} m_1^2$	—
D	-2	3	N/A	1	N/A	2	N/A	—	$+6\lambda_{12}^{(1)} m_1^2$	—
E	0	3	N/A	3	N/A	2	N/A	$-6{g_1}^2 m_2^2$	—	$+m_2^2(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$
E	2	3	N/A	3	N/A	2	N/A	$-24{g_1}^2 m_2^2$	—	$+m_2^2(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$
E	-2	3	N/A	3	N/A	2	N/A	—	—	$+m_2^2(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$
E	-4	3	N/A	3	N/A	2	N/A	$-6{g_1}^2 m_2^2$	—	$+m_2^2(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$

Quantity: m_2^2

Terms 3 of 5

Column	Representative term structure
T1	$20\mu_{\mathrm{T3GeneralS2Sq}}\left(\lambda_{X_2}\right)^*$
T2	$+20\lambda_{X_2}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$
T3	$-\frac{9{g_2}^2 m_2^2}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$20\mu_{\mathrm{T3GeneralS2Sq}}\left(\lambda_{X_2}\right)^*$	$+20\lambda_{X_2}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	$-\frac{9{g_2}^2 m_2^2}{2}$
B	−1	2	N/A	2	N/A	1	N/A	—	—	$-\frac{9{g_2}^2 m_2^2}{2}$
B	−3	2	N/A	2	N/A	1	N/A	—	—	$-\frac{9{g_2}^2 m_2^2}{2}$
C	1	2	N/A	2	N/A	3	N/A	—	—	$-\frac{9{g_2}^2 m_2^2}{2}$
C	−1	2	N/A	2	N/A	3	N/A	—	—	$-\frac{9{g_2}^2 m_2^2}{2}$
C	−3	2	N/A	2	N/A	3	N/A	—	—	$-\frac{9{g_2}^2 m_2^2}{2}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$12\mu_{\mathrm{T3GeneralS2Sq}}\left(\lambda_{X_2}\right)^*$	$+12\lambda_{X_2}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$20\mu_{\mathrm{T3GeneralS2Sq}}\left(\lambda_{X_2}\right)^*$	$+20\lambda_{X_2}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: m_2^2

Terms 4 of 5

Column	Representative term structure
T1	$-\frac{27{g_1}^2 m_2^2}{2}$
T2	$+6\lambda_{S_2}^{(1)} m_2^2$
T3	$6\mu_{\mathrm{T3GeneralS1S2}}\,(\lambda_{10})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$-\frac{27{g_1}^2 m_2^2}{2}$	$+6\lambda_{S_2}^{(1)} m_2^2$	—
B	−1	2	N/A	2	N/A	1	N/A	$-\frac{3{g_1}^2 m_2^2}{2}$	$+6\lambda_{S_2}^{(1)} m_2^2$	$6\mu_{\mathrm{T3GeneralS1S2}}\,(\lambda_{10})^*$
B	−3	2	N/A	2	N/A	1	N/A	$-\frac{3{g_1}^2 m_2^2}{2}$	$+6\lambda_{S_2}^{(1)} m_2^2$	—
C	1	2	N/A	2	N/A	3	N/A	$-\frac{27{g_1}^2 m_2^2}{2}$	$+6\lambda_{S_2}^{(1)} m_2^2$	—
C	−1	2	N/A	2	N/A	3	N/A	$-\frac{3{g_1}^2 m_2^2}{2}$	$+6\lambda_{S_2}^{(1)} m_2^2$	$6\mu_{\mathrm{T3GeneralS1S2}}\,(\lambda_{10})^*$
C	−3	2	N/A	2	N/A	3	N/A	$-\frac{3{g_1}^2 m_2^2}{2}$	$+6\lambda_{S_2}^{(1)} m_2^2$	—
D	0	3	N/A	1	N/A	2	N/A	—	$+4\lambda_{S_2}^{(1)} m_2^2$	—
D	2	3	N/A	1	N/A	2	N/A	—	$+4\lambda_{S_2}^{(1)} m_2^2$	—
D	−2	3	N/A	1	N/A	2	N/A	—	$+4\lambda_{S_2}^{(1)} m_2^2$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: m_2^2

Terms 5 of 5

Column	Representative term structure
T1	$+6\lambda_{10}\,(\mu_{\mathrm{T3GeneralS1S2}})^*$
T2	$4\mu_{\mathrm{T3GeneralS1Sq}}(3\left(\lambda_{X_5}^{(1)}\right)^* + \left(\lambda_{X_5}^{(2)}\right)^*)$
T3	$+m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+6\lambda_{10}\,(\mu_{\mathrm{T3GeneralS1S2}})^*$	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+6\lambda_{10}\,(\mu_{\mathrm{T3GeneralS1S2}})^*$	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$4\mu_{\mathrm{T3GeneralS1Sq}}(3\left(\lambda_{X_5}^{(1)}\right)^* + \left(\lambda_{X_5}^{(2)}\right)^*)$	$+m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$
E	2	3	N/A	3	N/A	2	N/A	—	—	$+m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$
E	-2	3	N/A	3	N/A	2	N/A	—	—	$+m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$
E	-4	3	N/A	3	N/A	2	N/A	—	—	$+m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$

$\beta^{(1)}_{\lambda_1}$

Quantity: λ_1

$$16\pi^2\,\mu\frac{d\lambda_1}{d\mu} = \beta^{(1)}_{\lambda_1}$$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$
T3	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
A	−2	1	N/A	3	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
A	−4	1	N/A	3	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
B	1	2	N/A	2	N/A	1	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
B	−1	2	N/A	2	N/A	1	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
B	−3	2	N/A	2	N/A	1	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
C	1	2	N/A	2	N/A	3	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
C	−1	2	N/A	2	N/A	3	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
C	−3	2	N/A	2	N/A	3	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
D	0	3	N/A	1	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
D	2	3	N/A	1	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
D	−2	3	N/A	1	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
E	0	3	N/A	3	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
E	2	3	N/A	3	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
E	−2	3	N/A	3	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
E	−4	3	N/A	3	N/A	2	N/A	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$

Quantity: λ_1

Terms 2 of 7

Column	Representative term structure
T1	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$
T2	$+4\lambda_5\ (\lambda_5)^*$
T3	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+4\lambda_{X_{14}}\ (\lambda_{X_{14}})^*$ $+8\lambda_{X_{18}}\ (\lambda_{X_{18}})^*$
A	−2	1	N/A	3	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+4\lambda_{X_{12}}\ (\lambda_{X_{12}})^*$ $+24\lambda_{X_{16}}\ (\lambda_{X_{16}})^*$
A	−4	1	N/A	3	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	—
B	1	2	N/A	2	N/A	1	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+8\lambda_{X_{15}}\ (\lambda_{X_{15}})^*$
B	−1	2	N/A	2	N/A	1	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	—
B	−3	2	N/A	2	N/A	1	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+8\lambda_{X_{10}}\ (\lambda_{X_{10}})^*$
C	1	2	N/A	2	N/A	3	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+8\lambda_{X_{15}}\ (\lambda_{X_{15}})^*$
C	−1	2	N/A	2	N/A	3	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	—
C	−3	2	N/A	2	N/A	3	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+8\lambda_{X_{10}}\ (\lambda_{X_{10}})^*$
D	0	3	N/A	1	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+4\lambda_{X_{14}}\ (\lambda_{X_{14}})^*$ $+24\lambda_{X_{18}}\ (\lambda_{X_{18}})^*$
D	2	3	N/A	1	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	—
D	−2	3	N/A	1	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+4\lambda_{X_{12}}\ (\lambda_{X_{12}})^*$ $+8\lambda_{X_{16}}\ (\lambda_{X_{16}})^*$
E	0	3	N/A	3	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+4\lambda_{X_{14}}\ (\lambda_{X_{14}})^*$ $+24\lambda_{X_{18}}\ (\lambda_{X_{18}})^*$
E	2	3	N/A	3	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	—
E	−2	3	N/A	3	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+4\lambda_{X_{12}}\ (\lambda_{X_{12}})^*$ $+24\lambda_{X_{16}}\ (\lambda_{X_{16}})^*$
E	−4	3	N/A	3	N/A	2	N/A	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	—

Quantity: λ_1

Terms 3 of 7

Column	Representative term structure
T1	$+\frac{9{g_2}^4}{4}$
T2	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$
T3	$-9{g_2}^2\lambda_1$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
A	−2	1	N/A	3	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
A	−4	1	N/A	3	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
B	1	2	N/A	2	N/A	1	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
B	−1	2	N/A	2	N/A	1	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
B	−3	2	N/A	2	N/A	1	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
C	1	2	N/A	2	N/A	3	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
C	−1	2	N/A	2	N/A	3	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
C	−3	2	N/A	2	N/A	3	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
D	0	3	N/A	1	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
D	2	3	N/A	1	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
D	−2	3	N/A	1	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
E	0	3	N/A	3	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
E	2	3	N/A	3	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
E	−2	3	N/A	3	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$
E	−4	3	N/A	3	N/A	2	N/A	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$

Quantity: λ_1

Terms 4 of 7

Column	Representative term structure
T1	$+\frac{3{g_1}^4}{4}$
T2	$+12{\lambda_1}^2$
T3	$+2{\lambda_{HS_1}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+2{\lambda_{HS_1}^{(1)}}^2$
A	−2	1	N/A	3	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+2{\lambda_{HS_1}^{(1)}}^2$
A	−4	1	N/A	3	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+2{\lambda_{HS_1}^{(1)}}^2$
B	1	2	N/A	2	N/A	1	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
B	−1	2	N/A	2	N/A	1	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
B	−3	2	N/A	2	N/A	1	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
C	1	2	N/A	2	N/A	3	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
C	−1	2	N/A	2	N/A	3	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
C	−3	2	N/A	2	N/A	3	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
D	0	3	N/A	1	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+6{\lambda_{HS_1}^{(1)}}^2$
D	2	3	N/A	1	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+6{\lambda_{HS_1}^{(1)}}^2$
D	−2	3	N/A	1	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+6{\lambda_{HS_1}^{(1)}}^2$
E	0	3	N/A	3	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+6{\lambda_{HS_1}^{(1)}}^2$
E	2	3	N/A	3	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+6{\lambda_{HS_1}^{(1)}}^2$
E	−2	3	N/A	3	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+6{\lambda_{HS_1}^{(1)}}^2$
E	−4	3	N/A	3	N/A	2	N/A	$+\frac{3{g_1}^4}{4}$	$+12{\lambda_1}^2$	$+6{\lambda_{HS_1}^{(1)}}^2$

Quantity: λ_1

Terms 5 of 7

Column	Representative term structure
T1	$+6{\lambda_{HS_2}^{(1)}}^2$
T2	$+3{\lambda_{HS_2}^{(A)}}^2$
T3	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+6{\lambda_{HS_2}^{(1)}}^2$	$+3{\lambda_{HS_2}^{(A)}}^2$	—
A	-2	1	N/A	3	N/A	2	N/A	$+6{\lambda_{HS_2}^{(1)}}^2$	$+3{\lambda_{HS_2}^{(A)}}^2$	—
A	-4	1	N/A	3	N/A	2	N/A	$+6{\lambda_{HS_2}^{(1)}}^2$	$+3{\lambda_{HS_2}^{(A)}}^2$	—
B	1	2	N/A	2	N/A	1	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	—	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$
B	-1	2	N/A	2	N/A	1	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	—	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$
B	-3	2	N/A	2	N/A	1	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	—	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$
C	1	2	N/A	2	N/A	3	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	—	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$
C	-1	2	N/A	2	N/A	3	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	—	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$
C	-3	2	N/A	2	N/A	3	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	—	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$
D	0	3	N/A	1	N/A	2	N/A	$+2{\lambda_{HS_2}^{(1)}}^2$	—	—
D	2	3	N/A	1	N/A	2	N/A	$+2{\lambda_{HS_2}^{(1)}}^2$	—	—
D	-2	3	N/A	1	N/A	2	N/A	$+2{\lambda_{HS_2}^{(1)}}^2$	—	—
E	0	3	N/A	3	N/A	2	N/A	$+6{\lambda_{HS_2}^{(1)}}^2$	$+3{\lambda_{HS_2}^{(A)}}^2$	—
E	2	3	N/A	3	N/A	2	N/A	$+6{\lambda_{HS_2}^{(1)}}^2$	$+3{\lambda_{HS_2}^{(A)}}^2$	—
E	-2	3	N/A	3	N/A	2	N/A	$+6{\lambda_{HS_2}^{(1)}}^2$	$+3{\lambda_{HS_2}^{(A)}}^2$	—
E	-4	3	N/A	3	N/A	2	N/A	$+6{\lambda_{HS_2}^{(1)}}^2$	$+3{\lambda_{HS_2}^{(A)}}^2$	—

Quantity: λ_1

Terms 6 of 7

Column	Representative term structure
T1	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$
T2	$+8\lambda_7\left(\lambda_7\right)^*$
T3	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	$+8\lambda_7\left(\lambda_7\right)^*$	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
B	-3	2	N/A	2	N/A	1	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	—	—
C	1	2	N/A	2	N/A	3	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	$+8\lambda_7\left(\lambda_7\right)^*$	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
C	-3	2	N/A	2	N/A	3	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_1

Terms 7 of 7

Column	Representative term structure
T1	$+8\lambda_8 \left(\lambda_8\right)^*$
T2	$+4\lambda_{X_{17}}^{(1)} \left(\lambda_{X_{17}}^{(1)}\right)^*$
T3	$+3{\lambda_{HS_1}^{(A)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+8\lambda_8 \left(\lambda_8\right)^*$	$+4\lambda_{X_{17}}^{(1)} \left(\lambda_{X_{17}}^{(1)}\right)^*$	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+8\lambda_8 \left(\lambda_8\right)^*$	$+4\lambda_{X_{17}}^{(1)} \left(\lambda_{X_{17}}^{(1)}\right)^*$	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	$+3{\lambda_{HS_1}^{(A)}}^2$
D	2	3	N/A	1	N/A	2	N/A	—	—	$+3{\lambda_{HS_1}^{(A)}}^2$
D	-2	3	N/A	1	N/A	2	N/A	—	—	$+3{\lambda_{HS_1}^{(A)}}^2$
E	0	3	N/A	3	N/A	2	N/A	—	—	$+3{\lambda_{HS_1}^{(A)}}^2$
E	2	3	N/A	3	N/A	2	N/A	—	—	$+3{\lambda_{HS_1}^{(A)}}^2$
E	-2	3	N/A	3	N/A	2	N/A	—	—	$+3{\lambda_{HS_1}^{(A)}}^2$
E	-4	3	N/A	3	N/A	2	N/A	—	—	$+3{\lambda_{HS_1}^{(A)}}^2$

$\beta^{(1)}_{\lambda^{(1)}_{S_1}}$

Quantity: $\lambda^{(1)}_{S_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{S_1}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{S_1}}$$

Terms 1 of 11

Column	Representative term structure
T1	$12\lambda^{(1)}_{X_5}\left(\lambda^{(1)}_{X_5}\right)^*$
T2	$+288\lambda_{X_6}\left(\lambda_{X_6}\right)^*$
T3	$+108\lambda_{X_9}\left(\lambda_{X_9}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$12\lambda^{(1)}_{X_5}\left(\lambda^{(1)}_{X_5}\right)^*$	$+288\lambda_{X_6}\left(\lambda_{X_6}\right)^*$	$+108\lambda_{X_9}\left(\lambda_{X_9}\right)^*$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 2 of 11

Column	Representative term structure
T1	$+8\lambda_{X_{18}}\left(\lambda_{X_{18}}\right)^*$
T2	$+6{\lambda_{12}^{(1)}}^2$
T3	$+4{\lambda_{HS_1}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+8\lambda_{X_{18}}\left(\lambda_{X_{18}}\right)^*$	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
A	-2	1	N/A	3	N/A	2	N/A	—	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
A	-4	1	N/A	3	N/A	2	N/A	—	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
B	1	2	N/A	2	N/A	1	N/A	$8\lambda_{X_{15}}\left(\lambda_{X_{15}}\right)^*$	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
B	-1	2	N/A	2	N/A	1	N/A	—	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
B	-3	2	N/A	2	N/A	1	N/A	—	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
C	1	2	N/A	2	N/A	3	N/A	$8\lambda_{X_{15}}\left(\lambda_{X_{15}}\right)^*$	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
C	-1	2	N/A	2	N/A	3	N/A	—	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
C	-3	2	N/A	2	N/A	3	N/A	—	$+4{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
D	0	3	N/A	1	N/A	2	N/A	—	$+2{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
D	2	3	N/A	1	N/A	2	N/A	—	$+2{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
D	-2	3	N/A	1	N/A	2	N/A	—	$+2{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
E	0	3	N/A	3	N/A	2	N/A	—	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
E	2	3	N/A	3	N/A	2	N/A	—	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
E	-2	3	N/A	3	N/A	2	N/A	—	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$
E	-4	3	N/A	3	N/A	2	N/A	—	$+6{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 3 of 11

Column	Representative term structure
T1	$+10{\lambda_{S_1}^{(1)}}^2$
T2	$24\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(1)}\right)^*$
T3	$+12{g_1}^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+10{\lambda_{S_1}^{(1)}}^2$	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+10{\lambda_{S_1}^{(1)}}^2$	$24\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(1)}\right)^*$	$+12{g_1}^4$
A	-4	1	N/A	3	N/A	2	N/A	$+10{\lambda_{S_1}^{(1)}}^2$	—	$192{g_1}^4$
B	1	2	N/A	2	N/A	1	N/A	$+12{\lambda_{S_1}^{(1)}}^2$	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+12{\lambda_{S_1}^{(1)}}^2$	—	—
B	-3	2	N/A	2	N/A	1	N/A	$+12{\lambda_{S_1}^{(1)}}^2$	—	—
C	1	2	N/A	2	N/A	3	N/A	$+12{\lambda_{S_1}^{(1)}}^2$	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+12{\lambda_{S_1}^{(1)}}^2$	—	—
C	-3	2	N/A	2	N/A	3	N/A	$+12{\lambda_{S_1}^{(1)}}^2$	—	—
D	0	3	N/A	1	N/A	2	N/A	$+14{\lambda_{S_1}^{(1)}}^2$	—	—
D	2	3	N/A	1	N/A	2	N/A	$+14{\lambda_{S_1}^{(1)}}^2$	—	$+12{g_1}^4$
D	-2	3	N/A	1	N/A	2	N/A	$+14{\lambda_{S_1}^{(1)}}^2$	$8\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(1)}\right)^*$	$+12{g_1}^4$
E	0	3	N/A	3	N/A	2	N/A	$+14{\lambda_{S_1}^{(1)}}^2$	—	—
E	2	3	N/A	3	N/A	2	N/A	$+14{\lambda_{S_1}^{(1)}}^2$	—	$+12{g_1}^4$
E	-2	3	N/A	3	N/A	2	N/A	$+14{\lambda_{S_1}^{(1)}}^2$	—	$+12{g_1}^4$
E	-4	3	N/A	3	N/A	2	N/A	$+14{\lambda_{S_1}^{(1)}}^2$	—	$+192{g_1}^4$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 4 of 11

Column	Representative term structure
T1	$-12{g_1}^2\lambda_{S_1}^{(1)}$
T2	$+\frac{9{g_2}^4}{4}$
T3	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$-12{g_1}^2\lambda_{S_1}^{(1)}$	—	—
A	-4	1	N/A	3	N/A	2	N/A	$-48{g_1}^2\lambda_{S_1}^{(1)}$	—	—
B	1	2	N/A	2	N/A	1	N/A	—	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$
B	-1	2	N/A	2	N/A	1	N/A	—	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$
B	-3	2	N/A	2	N/A	1	N/A	—	$\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{27{g_2}^2}{2}-27\lambda_{S_1}^{(1)}\right)$
C	1	2	N/A	2	N/A	3	N/A	—	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$
C	-1	2	N/A	2	N/A	3	N/A	—	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$
C	-3	2	N/A	2	N/A	3	N/A	—	$\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{27{g_2}^2}{2}-27\lambda_{S_1}^{(1)}\right)$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 5 of 11

Column	Representative term structure
T1	$-9{g_2}^2\lambda_{S_1}^{(1)}$
T2	$+\frac{3{g_1}^4}{4}$
T3	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$-9{g_2}^2\lambda_{S_1}^{(1)}$	$+\frac{3{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$
B	-1	2	N/A	2	N/A	1	N/A	$-9{g_2}^2\lambda_{S_1}^{(1)}$	$+\frac{3{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$
B	-3	2	N/A	2	N/A	1	N/A	$-9{g_2}^2\lambda_{S_1}^{(1)}$	$+\frac{243{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$
C	1	2	N/A	2	N/A	3	N/A	$-9{g_2}^2\lambda_{S_1}^{(1)}$	$+\frac{3{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$
C	-1	2	N/A	2	N/A	3	N/A	$-9{g_2}^2\lambda_{S_1}^{(1)}$	$+\frac{3{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$
C	-3	2	N/A	2	N/A	3	N/A	$-9{g_2}^2\lambda_{S_1}^{(1)}$	$+\frac{243{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$
D	0	3	N/A	1	N/A	2	N/A	$-24{g_2}^2\lambda_{S_1}^{(1)}$	—	—
D	2	3	N/A	1	N/A	2	N/A	$-24{g_2}^2\lambda_{S_1}^{(1)}$	—	—
D	-2	3	N/A	1	N/A	2	N/A	$-24{g_2}^2\lambda_{S_1}^{(1)}$	—	—
E	0	3	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{S_1}^{(1)}$	—	—
E	2	3	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{S_1}^{(1)}$	—	—
E	-2	3	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{S_1}^{(1)}$	—	—
E	-4	3	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{S_1}^{(1)}$	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 6 of 11

Column	Representative term structure
T1	$+\frac{\lambda_{HS_1}^{(A)\ 2}}{4}$
T2	$8\lambda_7 \left(\lambda_7\right)^*$
T3	$+8\lambda_9 \left(\lambda_9\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+\frac{\lambda_{HS_1}^{(A)\ 2}}{4}$	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+\frac{\lambda_{HS_1}^{(A)\ 2}}{4}$	$8\lambda_7 \left(\lambda_7\right)^*$	$+8\lambda_9 \left(\lambda_9\right)^*$
B	-3	2	N/A	2	N/A	1	N/A	$+\frac{\lambda_{HS_1}^{(A)\ 2}}{4}$	—	—
C	1	2	N/A	2	N/A	3	N/A	$+\frac{\lambda_{HS_1}^{(A)\ 2}}{4}$	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+\frac{\lambda_{HS_1}^{(A)\ 2}}{4}$	$8\lambda_7 \left(\lambda_7\right)^*$	$+8\lambda_9 \left(\lambda_9\right)^*$
C	-3	2	N/A	2	N/A	3	N/A	$+\frac{\lambda_{HS_1}^{(A)\ 2}}{4}$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 7 of 11

Column	Representative term structure
T1	$+24\lambda_{11}\left(\lambda_{11}\right)^{*}$
T2	$\frac{8}{3}\lambda_{X_5}^{(1)}\left(\lambda_{X_5}^{(1)}\right)^{*}$
T3	$+\frac{832}{3}\lambda_{X_6}\left(\lambda_{X_6}\right)^{*}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+24\lambda_{11}\left(\lambda_{11}\right)^{*}$	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+24\lambda_{11}\left(\lambda_{11}\right)^{*}$	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	$\frac{8}{3}\lambda_{X_5}^{(1)}\left(\lambda_{X_5}^{(1)}\right)^{*}$	$+\frac{832}{3}\lambda_{X_6}\left(\lambda_{X_6}\right)^{*}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	$+\frac{832}{3}\lambda_{X_6}\left(\lambda_{X_6}\right)^{*}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 8 of 11

Column	Representative term structure
T1	$+\frac{352}{3}\lambda_{X_9}\left(\lambda_{X_9}\right)^*$
T2	$+\frac{16}{3}\lambda_{X_{18}}\left(\lambda_{X_{18}}\right)^*$
T3	$+22{g_2}^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+\frac{352}{3}\lambda_{X_9}\left(\lambda_{X_9}\right)^*$	$+\frac{16}{3}\lambda_{X_{18}}\left(\lambda_{X_{18}}\right)^*$	$+22{g_2}^4$
D	2	3	N/A	1	N/A	2	N/A	—	—	$22{g_2}^4$
D	−2	3	N/A	1	N/A	2	N/A	—	—	$+22{g_2}^4$
E	0	3	N/A	3	N/A	2	N/A	$+\frac{352}{3}\lambda_{X_9}\left(\lambda_{X_9}\right)^*$	$+\frac{16}{3}\lambda_{X_{18}}\left(\lambda_{X_{18}}\right)^*$	$+22{g_2}^4$
E	2	3	N/A	3	N/A	2	N/A	—	—	$22{g_2}^4$
E	−2	3	N/A	3	N/A	2	N/A	—	—	$+22{g_2}^4$
E	−4	3	N/A	3	N/A	2	N/A	—	—	$22{g_2}^4$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 9 of 11

Column	Representative term structure
T1	$+8\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$
T2	$+16{\lambda_{S_1}^{(A)}}^2$
T3	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_1}^{(1)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+8\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+16{\lambda_{S_1}^{(A)}}^2$	—
D	2	3	N/A	1	N/A	2	N/A	$+8\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+16{\lambda_{S_1}^{(A)}}^2$	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_1}^{(1)}\right)$
D	-2	3	N/A	1	N/A	2	N/A	$+8\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+16{\lambda_{S_1}^{(A)}}^2$	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_1}^{(1)}\right)$
E	0	3	N/A	3	N/A	2	N/A	$+8\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+16{\lambda_{S_1}^{(A)}}^2$	—
E	2	3	N/A	3	N/A	2	N/A	$+8\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+16{\lambda_{S_1}^{(A)}}^2$	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_1}^{(1)}\right)$
E	-2	3	N/A	3	N/A	2	N/A	$+8\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+16{\lambda_{S_1}^{(A)}}^2$	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_1}^{(1)}\right)$
E	-4	3	N/A	3	N/A	2	N/A	$+8\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+16{\lambda_{S_1}^{(A)}}^2$	$+{g_1}^2\left(32{g_2}^2-48\lambda_{S_1}^{(1)}\right)$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 10 of 11

Column	Representative term structure
T1	$\frac{8}{3}\lambda_{X_5}^{(1)}(3\left(\lambda_{X_5}^{(1)}\right)^{*}+\left(\lambda_{X_5}^{(2)}\right)^{*})$
T2	$+\lambda_{X_5}^{(2)}\left(\frac{8\left(\lambda_{X_5}^{(1)}\right)^{*}}{3}+4\left(\lambda_{X_5}^{(2)}\right)^{*}\right)$
T3	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$\frac{8}{3}\lambda_{X_5}^{(1)}(3\left(\lambda_{X_5}^{(1)}\right)^{*}+\left(\lambda_{X_5}^{(2)}\right)^{*})$	$+\lambda_{X_5}^{(2)}\left(\frac{8\left(\lambda_{X_5}^{(1)}\right)^{*}}{3}+4\left(\lambda_{X_5}^{(2)}\right)^{*}\right)$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
E	2	3	N/A	3	N/A	2	N/A	—	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
E	−2	3	N/A	3	N/A	2	N/A	—	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 11 of 11

Column	Representative term structure
T1	$+\frac{20\lambda_{12}^{(\times)^2}}{3}$
T2	$8\lambda_{X_7}^{(1)}(3\left(\lambda_{X_7}^{(1)}\right)^* + \left(\lambda_{X_7}^{(2)}\right)^*)$
T3	$+\lambda_{X_7}^{(2)}\left(8\left(\lambda_{X_7}^{(1)}\right)^* + \frac{20\left(\lambda_{X_7}^{(2)}\right)^*}{3}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+\frac{20\lambda_{12}^{(\times)^2}}{3}$	—	—
E	2	3	N/A	3	N/A	2	N/A	$+\frac{20\lambda_{12}^{(\times)^2}}{3}$	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+\frac{20\lambda_{12}^{(\times)^2}}{3}$	$8\lambda_{X_7}^{(1)}(3\left(\lambda_{X_7}^{(1)}\right)^* + \left(\lambda_{X_7}^{(2)}\right)^*)$	$+\lambda_{X_7}^{(2)}\left(8\left(\lambda_{X_7}^{(1)}\right)^* + \frac{20\left(\lambda_{X_7}^{(2)}\right)^*}{3}\right)$
E	−4	3	N/A	3	N/A	2	N/A	$+\frac{20\lambda_{12}^{(\times)^2}}{3}$	—	—

$\beta^{(1)}_{\lambda^{(1)}_{S_2}}$

Quantity: $\lambda^{(1)}_{S_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{S_2}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{S_2}}$$

Terms 1 of 11

Column	Representative term structure
T1	$8\lambda^{(1)}_{X_5}\left(\lambda^{(1)}_{X_5}\right)^*$
T2	$+22{g_2}^4$
T3	$+{g_1}^2\left(8{g_2}^2-12\lambda^{(1)}_{S_2}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$8\lambda^{(1)}_{X_5}\left(\lambda^{(1)}_{X_5}\right)^*$	$+22{g_2}^4$	$+{g_1}^2\left(8{g_2}^2-12\lambda^{(1)}_{S_2}\right)$
A	−2	1	N/A	3	N/A	2	N/A	—	$+22{g_2}^4$	—
A	−4	1	N/A	3	N/A	2	N/A	—	$22{g_2}^4$	$+{g_1}^2\left(8{g_2}^2-12\lambda^{(1)}_{S_2}\right)$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$24\lambda^{(1)}_{X_5}\left(\lambda^{(1)}_{X_5}\right)^*$	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+22{g_2}^4$	$+{g_1}^2\left(8{g_2}^2-12\lambda^{(1)}_{S_2}\right)$
E	2	3	N/A	3	N/A	2	N/A	—	$22{g_2}^4$	$+{g_1}^2\left(32{g_2}^2-48\lambda^{(1)}_{S_2}\right)$
E	−2	3	N/A	3	N/A	2	N/A	—	$+22{g_2}^4$	—
E	−4	3	N/A	3	N/A	2	N/A	—	$22{g_2}^4$	$+{g_1}^2\left(8{g_2}^2-12\lambda^{(1)}_{S_2}\right)$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 2 of 11

Column	Representative term structure
T1	$-24{g_2}^2\lambda_{S_2}^{(1)}$
T2	$+12{g_1}^4$
T3	$+2\lambda_{12}^{(1)\,2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{S_2}^{(1)}$	$+12{g_1}^4$	$+2\lambda_{12}^{(1)\,2}$
A	-2	1	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{S_2}^{(1)}$	—	$+2\lambda_{12}^{(1)\,2}$
A	-4	1	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{S_2}^{(1)}$	$+12{g_1}^4$	$+2\lambda_{12}^{(1)\,2}$
B	1	2	N/A	2	N/A	1	N/A	$-9{g_2}^2\lambda_{S_2}^{(1)}$	—	$+4\lambda_{12}^{(1)\,2}$
B	-1	2	N/A	2	N/A	1	N/A	$-9{g_2}^2\lambda_{S_2}^{(1)}$	—	$+4\lambda_{12}^{(1)\,2}$
B	-3	2	N/A	2	N/A	1	N/A	$-9{g_2}^2\lambda_{S_2}^{(1)}$	—	$+4\lambda_{12}^{(1)\,2}$
C	1	2	N/A	2	N/A	3	N/A	$-9{g_2}^2\lambda_{S_2}^{(1)}$	—	$+4\lambda_{12}^{(1)\,2}$
C	-1	2	N/A	2	N/A	3	N/A	$-9{g_2}^2\lambda_{S_2}^{(1)}$	—	$+4\lambda_{12}^{(1)\,2}$
C	-3	2	N/A	2	N/A	3	N/A	$-9{g_2}^2\lambda_{S_2}^{(1)}$	—	$+4\lambda_{12}^{(1)\,2}$
D	0	3	N/A	1	N/A	2	N/A	—	$+12{g_1}^4$	$+6\lambda_{12}^{(1)\,2}$
D	2	3	N/A	1	N/A	2	N/A	—	$192{g_1}^4$	$+6\lambda_{12}^{(1)\,2}$
D	-2	3	N/A	1	N/A	2	N/A	—	—	$+6\lambda_{12}^{(1)\,2}$
E	0	3	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{S_2}^{(1)}$	$+12{g_1}^4$	$+6\lambda_{12}^{(1)\,2}$
E	2	3	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{S_2}^{(1)}$	$+192{g_1}^4$	$+6\lambda_{12}^{(1)\,2}$
E	-2	3	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{S_2}^{(1)}$	—	$+6\lambda_{12}^{(1)\,2}$
E	-4	3	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{S_2}^{(1)}$	$+12{g_1}^4$	$+6\lambda_{12}^{(1)\,2}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 3 of 11

Column	Representative term structure
T1	$+4{\lambda_{HS_2}^{(1)}}^2$
T2	$+14{\lambda_{S_2}^{(1)}}^2$
T3	$+8\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	$+14{\lambda_{S_2}^{(1)}}^2$	$+8\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
A	−2	1	N/A	3	N/A	2	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	$+14{\lambda_{S_2}^{(1)}}^2$	$+8\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
A	−4	1	N/A	3	N/A	2	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	$+14{\lambda_{S_2}^{(1)}}^2$	$+8\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
B	1	2	N/A	2	N/A	1	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	$+12{\lambda_{S_2}^{(1)}}^2$	—
B	−1	2	N/A	2	N/A	1	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	$+12{\lambda_{S_2}^{(1)}}^2$	—
B	−3	2	N/A	2	N/A	1	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	$+12{\lambda_{S_2}^{(1)}}^2$	—
C	1	2	N/A	2	N/A	3	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	$+12{\lambda_{S_2}^{(1)}}^2$	—
C	−1	2	N/A	2	N/A	3	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	$+12{\lambda_{S_2}^{(1)}}^2$	—
C	−3	2	N/A	2	N/A	3	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	$+12{\lambda_{S_2}^{(1)}}^2$	—
D	0	3	N/A	1	N/A	2	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	$+10{\lambda_{S_2}^{(1)}}^2$	—
D	2	3	N/A	1	N/A	2	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	$+10{\lambda_{S_2}^{(1)}}^2$	—
D	−2	3	N/A	1	N/A	2	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	$+10{\lambda_{S_2}^{(1)}}^2$	—
E	0	3	N/A	3	N/A	2	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	$+14{\lambda_{S_2}^{(1)}}^2$	$+8\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	$+14{\lambda_{S_2}^{(1)}}^2$	$+8\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
E	−2	3	N/A	3	N/A	2	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	$+14{\lambda_{S_2}^{(1)}}^2$	$+8\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
E	−4	3	N/A	3	N/A	2	N/A	$+4{\lambda_{HS_2}^{(1)}}^2$	$+14{\lambda_{S_2}^{(1)}}^2$	$+8\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 4 of 11

Column	Representative term structure
T1	$+16{\lambda_{S_2}^{(A)}}^2$
T2	$\frac{832}{3}\lambda_{X_1}\left(\lambda_{X_1}\right)^*$
T3	$+\frac{352}{3}\lambda_{X_2}\left(\lambda_{X_2}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+16{\lambda_{S_2}^{(A)}}^2$	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+16{\lambda_{S_2}^{(A)}}^2$	$\frac{832}{3}\lambda_{X_1}\left(\lambda_{X_1}\right)^*$	$+\frac{352}{3}\lambda_{X_2}\left(\lambda_{X_2}\right)^*$
A	-4	1	N/A	3	N/A	2	N/A	$+16{\lambda_{S_2}^{(A)}}^2$	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+16{\lambda_{S_2}^{(A)}}^2$	—	—
E	2	3	N/A	3	N/A	2	N/A	$+16{\lambda_{S_2}^{(A)}}^2$	—	—
E	-2	3	N/A	3	N/A	2	N/A	$+16{\lambda_{S_2}^{(A)}}^2$	$\frac{832}{3}\lambda_{X_1}\left(\lambda_{X_1}\right)^*$	$+\frac{352}{3}\lambda_{X_2}\left(\lambda_{X_2}\right)^*$
E	-4	3	N/A	3	N/A	2	N/A	$+16{\lambda_{S_2}^{(A)}}^2$	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 5 of 11

Column	Representative term structure
T1	$+\frac{8}{3}\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(1)}\right)^*$
T2	$+\frac{16}{3}\lambda_{X_{16}}\left(\lambda_{X_{16}}\right)^*$
T3	$\frac{9{g_2}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+\frac{8}{3}\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(1)}\right)^*$	$+\frac{16}{3}\lambda_{X_{16}}\left(\lambda_{X_{16}}\right)^*$	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	$\frac{9{g_2}^4}{4}$
B	-1	2	N/A	2	N/A	1	N/A	—	—	$+\frac{9{g_2}^4}{4}$
B	-3	2	N/A	2	N/A	1	N/A	—	—	$+\frac{9{g_2}^4}{4}$
C	1	2	N/A	2	N/A	3	N/A	—	—	$\frac{9{g_2}^4}{4}$
C	-1	2	N/A	2	N/A	3	N/A	—	—	$+\frac{9{g_2}^4}{4}$
C	-3	2	N/A	2	N/A	3	N/A	—	—	$+\frac{9{g_2}^4}{4}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	$+\frac{16}{3}\lambda_{X_{16}}\left(\lambda_{X_{16}}\right)^*$	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 6 of 11

Column	Representative term structure
T1	$+{g_1}^2\left(\frac{27{g_2}^2}{2}-27\lambda_{S_2}^{(1)}\right)$
T2	$+\frac{243{g_1}^4}{4}$
T3	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+{g_1}^2\left(\frac{27{g_2}^2}{2}-27\lambda_{S_2}^{(1)}\right)$	$+\frac{243{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$
B	−1	2	N/A	2	N/A	1	N/A	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_2}^{(1)}\right)$	$+\frac{3{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$
B	−3	2	N/A	2	N/A	1	N/A	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_2}^{(1)}\right)$	$+\frac{3{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$
C	1	2	N/A	2	N/A	3	N/A	$+{g_1}^2\left(\frac{27{g_2}^2}{2}-27\lambda_{S_2}^{(1)}\right)$	$+\frac{243{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$
C	−1	2	N/A	2	N/A	3	N/A	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_2}^{(1)}\right)$	$+\frac{3{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$
C	−3	2	N/A	2	N/A	3	N/A	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_2}^{(1)}\right)$	$+\frac{3{g_1}^4}{4}$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 7 of 11

Column	Representative term structure
T1	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$
T2	$8\lambda_8\,(\lambda_8)^*$
T3	$+8\lambda_9\,(\lambda_9)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	$8\lambda_8\,(\lambda_8)^*$	$+8\lambda_9\,(\lambda_9)^*$
B	-3	2	N/A	2	N/A	1	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	—	—
C	1	2	N/A	2	N/A	3	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	$8\lambda_8\,(\lambda_8)^*$	$+8\lambda_9\,(\lambda_9)^*$
C	-3	2	N/A	2	N/A	3	N/A	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 8 of 11

Column	Representative term structure
T1	$+24\lambda_{10}\left(\lambda_{10}\right)^{*}$
T2	$8\lambda_{X_{10}}\left(\lambda_{X_{10}}\right)^{*}$
T3	$-12{g_1}^{2}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+24\lambda_{10}\left(\lambda_{10}\right)^{*}$	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	$8\lambda_{X_{10}}\left(\lambda_{X_{10}}\right)^{*}$	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+24\lambda_{10}\left(\lambda_{10}\right)^{*}$	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	$8\lambda_{X_{10}}\left(\lambda_{X_{10}}\right)^{*}$	—
D	0	3	N/A	1	N/A	2	N/A	—	—	$-12{g_1}^{2}\lambda_{S_2}^{(1)}$
D	2	3	N/A	1	N/A	2	N/A	—	—	$-48{g_1}^{2}\lambda_{S_2}^{(1)}$
D	-2	3	N/A	1	N/A	2	N/A	—	$+8\lambda_{X_{16}}\left(\lambda_{X_{16}}\right)^{*}$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 9 of 11

Column	Representative term structure
T1	$288\lambda_{X_1}\left(\lambda_{X_1}\right)^*$
T2	$+108\lambda_{X_2}\left(\lambda_{X_2}\right)^*$
T3	$+12\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(1)}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$288\lambda_{X_1}\left(\lambda_{X_1}\right)^*$	$+108\lambda_{X_2}\left(\lambda_{X_2}\right)^*$	$+12\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(1)}\right)^*$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 10 of 11

Column	Representative term structure
T1	$8\lambda_{X_5}^{(1)}(3\left(\lambda_{X_5}^{(1)}\right)^{*}+\left(\lambda_{X_5}^{(2)}\right)^{*})$
T2	$+\lambda_{X_5}^{(2)}\left(8\left(\lambda_{X_5}^{(1)}\right)^{*}+\frac{20\left(\lambda_{X_5}^{(2)}\right)^{*}}{3}\right)$
T3	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$8\lambda_{X_5}^{(1)}(3\left(\lambda_{X_5}^{(1)}\right)^{*}+\left(\lambda_{X_5}^{(2)}\right)^{*})$	$+\lambda_{X_5}^{(2)}\left(8\left(\lambda_{X_5}^{(1)}\right)^{*}+\frac{20\left(\lambda_{X_5}^{(2)}\right)^{*}}{3}\right)$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
E	2	3	N/A	3	N/A	2	N/A	—	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
E	−2	3	N/A	3	N/A	2	N/A	—	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 11 of 11

Column	Representative term structure
T1	$+\frac{20\lambda_{12}^{(\times)^2}}{3}$
T2	$+\frac{8}{3}\lambda_{X_7}^{(1)}(3\left(\lambda_{X_7}^{(1)}\right)^* + \left(\lambda_{X_7}^{(2)}\right)^*)$
T3	$+\lambda_{X_7}^{(2)}\left(\frac{8\left(\lambda_{X_7}^{(1)}\right)^*}{3} + 4\left(\lambda_{X_7}^{(2)}\right)^*\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+\frac{20\lambda_{12}^{(\times)^2}}{3}$	—	—
E	2	3	N/A	3	N/A	2	N/A	$+\frac{20\lambda_{12}^{(\times)^2}}{3}$	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+\frac{20\lambda_{12}^{(\times)^2}}{3}$	$+\frac{8}{3}\lambda_{X_7}^{(1)}(3\left(\lambda_{X_7}^{(1)}\right)^* + \left(\lambda_{X_7}^{(2)}\right)^*)$	$+\lambda_{X_7}^{(2)}\left(\frac{8\left(\lambda_{X_7}^{(1)}\right)^*}{3} + 4\left(\lambda_{X_7}^{(2)}\right)^*\right)$
E	−4	3	N/A	3	N/A	2	N/A	$+\frac{20\lambda_{12}^{(\times)^2}}{3}$	—	—

$\beta^{(1)}_{\lambda^{(1)}_{HS_1}}$

Quantity: $\lambda^{(1)}_{HS_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{HS_1}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_1}}$$

Terms 1 of 10

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+12\lambda_5\ (\lambda_5)^*$
T3	$+12\lambda_{X_9}\left(\lambda_{X_{18}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+12\lambda_5\ (\lambda_5)^*$	$+12\lambda_{X_9}\left(\lambda_{X_{18}}\right)^*$
A	−2	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+12\lambda_5\ (\lambda_5)^*$	—
A	−4	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+12\lambda_5\ (\lambda_5)^*$	—
B	1	2	N/A	2	N/A	1	N/A	$\lambda^{(1)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+6\lambda_5\ (\lambda_5)^*$	—
B	−1	2	N/A	2	N/A	1	N/A	$\lambda^{(1)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+6\lambda_5\ (\lambda_5)^*$	—
B	−3	2	N/A	2	N/A	1	N/A	$\lambda^{(1)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+6\lambda_5\ (\lambda_5)^*$	—
C	1	2	N/A	2	N/A	3	N/A	$\lambda^{(1)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+6\lambda_5\ (\lambda_5)^*$	—
C	−1	2	N/A	2	N/A	3	N/A	$\lambda^{(1)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+6\lambda_5\ (\lambda_5)^*$	—
C	−3	2	N/A	2	N/A	3	N/A	$\lambda^{(1)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+6\lambda_5\ (\lambda_5)^*$	—
D	0	3	N/A	1	N/A	2	N/A	—	$+4\lambda_5\ (\lambda_5)^*$	$+20\lambda_{X_9}\left(\lambda_{X_{18}}\right)^*$
D	2	3	N/A	1	N/A	2	N/A	—	$+4\lambda_5\ (\lambda_5)^*$	—
D	−2	3	N/A	1	N/A	2	N/A	—	$+4\lambda_5\ (\lambda_5)^*$	—
E	0	3	N/A	3	N/A	2	N/A	—	$+4\lambda_5\ (\lambda_5)^*$	$+20\lambda_{X_9}\left(\lambda_{X_{18}}\right)^*$
E	2	3	N/A	3	N/A	2	N/A	—	$+4\lambda_5\ (\lambda_5)^*$	—
E	−2	3	N/A	3	N/A	2	N/A	—	$+4\lambda_5\ (\lambda_5)^*$	—
E	−4	3	N/A	3	N/A	2	N/A	—	$+4\lambda_5\ (\lambda_5)^*$	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 2 of 10

Column	Representative term structure
T1	$+4\lambda_{X_{18}}(3\left(\lambda_{X_9}\right)^* + 4\left(\lambda_{X_{18}}\right)^*)$
T2	$+12\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$
T3	$-\frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+4\lambda_{X_{18}}(3\left(\lambda_{X_9}\right)^* + 4\left(\lambda_{X_{18}}\right)^*)$	$+12\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$-\frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
A	−2	1	N/A	3	N/A	2	N/A	—	$+12\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	$-\frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
A	−4	1	N/A	3	N/A	2	N/A	—	—	$-\frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
B	1	2	N/A	2	N/A	1	N/A	—	$+24\lambda_{X_{15}}\left(\lambda_{X_{15}}\right)^*$	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	$+24\lambda_{X_{15}}\left(\lambda_{X_{15}}\right)^*$	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+4\lambda_{X_{18}}(5\left(\lambda_{X_9}\right)^* + 4\left(\lambda_{X_{18}}\right)^*)$	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
D	2	3	N/A	1	N/A	2	N/A	—	—	$-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
D	−2	3	N/A	1	N/A	2	N/A	—	$+4\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	$-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
E	0	3	N/A	3	N/A	2	N/A	$+4\lambda_{X_{18}}(5\left(\lambda_{X_9}\right)^* + 4\left(\lambda_{X_{18}}\right)^*)$	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
E	2	3	N/A	3	N/A	2	N/A	—	—	$-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
E	−2	3	N/A	3	N/A	2	N/A	—	$+4\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	$-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	$-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 3 of 10

Column	Representative term structure
T1	$-\frac{3{g_1}^2\lambda_{HS_1}^{(1)}}{2}$
T2	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T3	$+6\lambda_1\lambda_{HS_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-\frac{3{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
A	−2	1	N/A	3	N/A	2	N/A	$-\frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
A	−4	1	N/A	3	N/A	2	N/A	$-\frac{51{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
B	1	2	N/A	2	N/A	1	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
B	−1	2	N/A	2	N/A	1	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
B	−3	2	N/A	2	N/A	1	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
C	1	2	N/A	2	N/A	3	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
C	−1	2	N/A	2	N/A	3	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
C	−3	2	N/A	2	N/A	3	N/A	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
D	0	3	N/A	1	N/A	2	N/A	$-\frac{3{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
D	2	3	N/A	1	N/A	2	N/A	$-\frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
D	−2	3	N/A	1	N/A	2	N/A	$-\frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$
E	0	3	N/A	3	N/A	2	N/A	$-\frac{3{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	—	$+6\lambda_1\lambda_{HS_1}^{(1)}$
E	2	3	N/A	3	N/A	2	N/A	$-\frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	—	$+6\lambda_1\lambda_{HS_1}^{(1)}$
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	—	$+6\lambda_1\lambda_{HS_1}^{(1)}$
E	−4	3	N/A	3	N/A	2	N/A	$-\frac{51{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	—	$+6\lambda_1\lambda_{HS_1}^{(1)}$

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 4 of 10

Column	Representative term structure
T1	$+4{\lambda^{(1)}_{HS_1}}^2$
T2	$+4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$
T3	$+12\lambda_{X_{16}}\left(\lambda^{(1)}_{X_7}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^2$	$+4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
A	−2	1	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^2$	$+4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	$+12\lambda_{X_{16}}\left(\lambda^{(1)}_{X_7}\right)^*$
A	−4	1	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^2$	$+4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
B	1	2	N/A	2	N/A	1	N/A	$+4{\lambda^{(1)}_{HS_1}}^2$	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
B	−1	2	N/A	2	N/A	1	N/A	$+4{\lambda^{(1)}_{HS_1}}^2$	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
B	−3	2	N/A	2	N/A	1	N/A	$+4{\lambda^{(1)}_{HS_1}}^2$	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
C	1	2	N/A	2	N/A	3	N/A	$+4{\lambda^{(1)}_{HS_1}}^2$	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
C	−1	2	N/A	2	N/A	3	N/A	$+4{\lambda^{(1)}_{HS_1}}^2$	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
C	−3	2	N/A	2	N/A	3	N/A	$+4{\lambda^{(1)}_{HS_1}}^2$	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
D	0	3	N/A	1	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^2$	$+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
D	2	3	N/A	1	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^2$	$+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
D	−2	3	N/A	1	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^2$	$+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	$+4\lambda_{X_{16}}\left(\lambda^{(1)}_{X_7}\right)^*$
E	0	3	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^2$	$+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
E	2	3	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^2$	$+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
E	−2	3	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^2$	$+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—
E	−4	3	N/A	3	N/A	2	N/A	$+4{\lambda^{(1)}_{HS_1}}^2$	$+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 5 of 10

Column	Representative term structure
T1	$+12\lambda_{X_7}^{(1)}\left(\lambda_{X_{16}}\right)^*$
T2	$+3{g_1}^4$
T3	$+\frac{9{g_2}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+12\lambda_{X_7}^{(1)}\left(\lambda_{X_{16}}\right)^*$	$+3{g_1}^4$	—
A	−4	1	N/A	3	N/A	2	N/A	—	$+12{g_1}^4$	—
B	1	2	N/A	2	N/A	1	N/A	—	—	$+\frac{9{g_2}^4}{4}$
B	−1	2	N/A	2	N/A	1	N/A	—	—	$+\frac{9{g_2}^4}{4}$
B	−3	2	N/A	2	N/A	1	N/A	—	—	$+\frac{9{g_2}^4}{4}$
C	1	2	N/A	2	N/A	3	N/A	—	—	$+\frac{9{g_2}^4}{4}$
C	−1	2	N/A	2	N/A	3	N/A	—	—	$+\frac{9{g_2}^4}{4}$
C	−3	2	N/A	2	N/A	3	N/A	—	—	$+\frac{9{g_2}^4}{4}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	$+3{g_1}^4$	—
D	−2	3	N/A	1	N/A	2	N/A	$+4\lambda_{X_7}^{(1)}\left(\lambda_{X_{16}}\right)^*$	$+3{g_1}^4$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	$+3{g_1}^4$	—
E	−2	3	N/A	3	N/A	2	N/A	$+12\lambda_{X_7}^{(1)}\left(\lambda_{X_{16}}\right)^*$ $+4\lambda_{X_7}^{(2)}\left(\lambda_{X_{16}}\right)^*$	$+3{g_1}^4$	—
E	−4	3	N/A	3	N/A	2	N/A	—	$+12{g_1}^4$	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 6 of 10

Column	Representative term structure
T1	$-9{g_2}^2\lambda_{HS_1}^{(1)}$
T2	$+\frac{3{g_1}^4}{4}$
T3	$-3{g_1}^2\lambda_{HS_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$-9{g_2}^2\lambda_{HS_1}^{(1)}$	$+\frac{3{g_1}^4}{4}$	$-3{g_1}^2\lambda_{HS_1}^{(1)}$
B	-1	2	N/A	2	N/A	1	N/A	$-9{g_2}^2\lambda_{HS_1}^{(1)}$	$+\frac{3{g_1}^4}{4}$	$-3{g_1}^2\lambda_{HS_1}^{(1)}$
B	-3	2	N/A	2	N/A	1	N/A	$-9{g_2}^2\lambda_{HS_1}^{(1)}$	$+\frac{27{g_1}^4}{4}$	$-15{g_1}^2\lambda_{HS_1}^{(1)}$
C	1	2	N/A	2	N/A	3	N/A	$-9{g_2}^2\lambda_{HS_1}^{(1)}$	$+\frac{3{g_1}^4}{4}$	$-3{g_1}^2\lambda_{HS_1}^{(1)}$
C	-1	2	N/A	2	N/A	3	N/A	$-9{g_2}^2\lambda_{HS_1}^{(1)}$	$+\frac{3{g_1}^4}{4}$	$-3{g_1}^2\lambda_{HS_1}^{(1)}$
C	-3	2	N/A	2	N/A	3	N/A	$-9{g_2}^2\lambda_{HS_1}^{(1)}$	$+\frac{27{g_1}^4}{4}$	$-15{g_1}^2\lambda_{HS_1}^{(1)}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 7 of 10

Column	Representative term structure
T1	$+\frac{3\lambda^{(A)\ 2}_{HS_1}}{4}$
T2	$+24\lambda_7\,(\lambda_7)^*$
T3	$+3\lambda_{11}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - \left(\lambda^{(1)}_{X_{17}}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+\frac{3\lambda^{(A)\ 2}_{HS_1}}{4}$	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+\frac{3\lambda^{(A)\ 2}_{HS_1}}{4}$	$+24\lambda_7\,(\lambda_7)^*$	$+3\lambda_{11}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - \left(\lambda^{(1)}_{X_{17}}\right)^*)$
B	-3	2	N/A	2	N/A	1	N/A	$+\frac{3\lambda^{(A)\ 2}_{HS_1}}{4}$	—	—
C	1	2	N/A	2	N/A	3	N/A	$+\frac{3\lambda^{(A)\ 2}_{HS_1}}{4}$	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+\frac{3\lambda^{(A)\ 2}_{HS_1}}{4}$	$+24\lambda_7\,(\lambda_7)^*$	$+3\lambda_{11}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - \left(\lambda^{(1)}_{X_{17}}\right)^*)$
C	-3	2	N/A	2	N/A	3	N/A	$+\frac{3\lambda^{(A)\ 2}_{HS_1}}{4}$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 8 of 10

Column	Representative term structure
T1	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\left(\lambda_{11}\right)^* + 2\left(\lambda^{(1)}_{X_{17}}\right)^*)$
T2	$+\lambda^{(1)}_{X_{17}}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 3\left(\lambda_{11}\right)^* + 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$
T3	$\lambda^{(1)}_{HS_1}(6\,\text{Tr}\left(Y_d\ \left(Y_d\right)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ \left(Y_e\right)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ \left(Y_u\right)^\dagger\right) + 4\lambda^{(A)}_{S_1})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\left(\lambda_{11}\right)^* + 2\left(\lambda^{(1)}_{X_{17}}\right)^*)$	$+\lambda^{(1)}_{X_{17}}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 3\left(\lambda_{11}\right)^* + 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\left(\lambda_{11}\right)^* + 2\left(\lambda^{(1)}_{X_{17}}\right)^*)$	$+\lambda^{(1)}_{X_{17}}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 3\left(\lambda_{11}\right)^* + 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	$\lambda^{(1)}_{HS_1}(6\,\text{Tr}\left(Y_d\ \left(Y_d\right)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ \left(Y_e\right)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ \left(Y_u\right)^\dagger\right) + 4\lambda^{(A)}_{S_1})$
D	2	3	N/A	1	N/A	2	N/A	—	—	$\lambda^{(1)}_{HS_1}(6\,\text{Tr}\left(Y_d\ \left(Y_d\right)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ \left(Y_e\right)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ \left(Y_u\right)^\dagger\right) + 4\lambda^{(A)}_{S_1})$
D	−2	3	N/A	1	N/A	2	N/A	—	—	$\lambda^{(1)}_{HS_1}(6\,\text{Tr}\left(Y_d\ \left(Y_d\right)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ \left(Y_e\right)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ \left(Y_u\right)^\dagger\right) + 4\lambda^{(A)}_{S_1})$
E	0	3	N/A	3	N/A	2	N/A	—	—	$\lambda^{(1)}_{HS_1}(6\,\text{Tr}\left(Y_d\ \left(Y_d\right)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ \left(Y_e\right)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ \left(Y_u\right)^\dagger\right) + 4\lambda^{(A)}_{S_1})$
E	2	3	N/A	3	N/A	2	N/A	—	—	$\lambda^{(1)}_{HS_1}(6\,\text{Tr}\left(Y_d\ \left(Y_d\right)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ \left(Y_e\right)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ \left(Y_u\right)^\dagger\right) + 4\lambda^{(A)}_{S_1})$
E	−2	3	N/A	3	N/A	2	N/A	—	—	$\lambda^{(1)}_{HS_1}(6\,\text{Tr}\left(Y_d\ \left(Y_d\right)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ \left(Y_e\right)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ \left(Y_u\right)^\dagger\right) + 4\lambda^{(A)}_{S_1})$
E	−4	3	N/A	3	N/A	2	N/A	—	—	$\lambda^{(1)}_{HS_1}(6\,\text{Tr}\left(Y_d\ \left(Y_d\right)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ \left(Y_e\right)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ \left(Y_u\right)^\dagger\right) + 4\lambda^{(A)}_{S_1})$

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 9 of 10

Column	Representative term structure
T1	$+6{g_2}^4$
T2	$+6{\lambda^{(A)}_{HS_1}}^2$
T3	$+\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+6{g_2}^4$	$+6{\lambda^{(A)}_{HS_1}}^2$	—
D	2	3	N/A	1	N/A	2	N/A	$+6{g_2}^4$	$+6{\lambda^{(A)}_{HS_1}}^2$	—
D	−2	3	N/A	1	N/A	2	N/A	$+6{g_2}^4$	$+6{\lambda^{(A)}_{HS_1}}^2$	—
E	0	3	N/A	3	N/A	2	N/A	$+6{g_2}^4$	$+6{\lambda^{(A)}_{HS_1}}^2$	$+\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$
E	2	3	N/A	3	N/A	2	N/A	$+6{g_2}^4$	$+6{\lambda^{(A)}_{HS_1}}^2$	$+\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$
E	−2	3	N/A	3	N/A	2	N/A	$+6{g_2}^4$	$+6{\lambda^{(A)}_{HS_1}}^2$	$+\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$
E	−4	3	N/A	3	N/A	2	N/A	$+6{g_2}^4$	$+6{\lambda^{(A)}_{HS_1}}^2$	$+\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 10 of 10

Column	Representative term structure
T1	$+4\lambda_{X_{16}}(3\left(\lambda^{(1)}_{X_7}\right)^* + \left(\lambda^{(2)}_{X_7}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	$+4\lambda_{X_{16}}(3\left(\lambda^{(1)}_{X_7}\right)^* + \left(\lambda^{(2)}_{X_7}\right)^*)$
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda^{(1)}_{HS_2}}$

Quantity: $\lambda^{(1)}_{HS_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{HS_2}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_2}}$$

Terms 1 of 10

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$
T2	$+4\lambda_5\ (\lambda_5)^*$
T3	$+4\lambda_{X_{18}}\left(\lambda^{(1)}_{X_5}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$	$+4\lambda_5\ (\lambda_5)^*$	$+4\lambda_{X_{18}}\left(\lambda^{(1)}_{X_5}\right)^*$
A	−2	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$	$+4\lambda_5\ (\lambda_5)^*$	—
A	−4	1	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$	$+4\lambda_5\ (\lambda_5)^*$	—
B	1	2	N/A	2	N/A	1	N/A	—	$+6\lambda_5\ (\lambda_5)^*$	—
B	−1	2	N/A	2	N/A	1	N/A	—	$+6\lambda_5\ (\lambda_5)^*$	—
B	−3	2	N/A	2	N/A	1	N/A	—	$+6\lambda_5\ (\lambda_5)^*$	—
C	1	2	N/A	2	N/A	3	N/A	—	$+6\lambda_5\ (\lambda_5)^*$	—
C	−1	2	N/A	2	N/A	3	N/A	—	$+6\lambda_5\ (\lambda_5)^*$	—
C	−3	2	N/A	2	N/A	3	N/A	—	$+6\lambda_5\ (\lambda_5)^*$	—
D	0	3	N/A	1	N/A	2	N/A	—	$+12\lambda_5\ (\lambda_5)^*$	$+12\lambda_{X_{18}}\left(\lambda^{(1)}_{X_5}\right)^*$
D	2	3	N/A	1	N/A	2	N/A	—	$+12\lambda_5\ (\lambda_5)^*$	—
D	−2	3	N/A	1	N/A	2	N/A	—	$+12\lambda_5\ (\lambda_5)^*$	—
E	0	3	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$	$+4\lambda_5\ (\lambda_5)^*$	—
E	2	3	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$	$+4\lambda_5\ (\lambda_5)^*$	—
E	−2	3	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$	$+4\lambda_5\ (\lambda_5)^*$	—
E	−4	3	N/A	3	N/A	2	N/A	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$	$+4\lambda_5\ (\lambda_5)^*$	—

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 2 of 10

Column	Representative term structure
T1	$+4\lambda^{(1)}_{X_5}\left(\lambda_{X_{18}}\right)^*$
T2	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$
T3	$+6{g_2}^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+4\lambda^{(1)}_{X_5}\left(\lambda_{X_{18}}\right)^*$	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$+6{g_2}^4$
A	-2	1	N/A	3	N/A	2	N/A	—	$+4\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	$+6{g_2}^4$
A	-4	1	N/A	3	N/A	2	N/A	—	—	$+6{g_2}^4$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	$+24\lambda_{X_{10}}\left(\lambda_{X_{10}}\right)^*$	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	$+24\lambda_{X_{10}}\left(\lambda_{X_{10}}\right)^*$	—
D	0	3	N/A	1	N/A	2	N/A	$+12\lambda^{(1)}_{X_5}\left(\lambda_{X_{18}}\right)^*$	$+12\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	$+12\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	—
E	0	3	N/A	3	N/A	2	N/A	$+12\lambda^{(1)}_{X_5}\left(\lambda_{X_{18}}\right)^*$ $+4\lambda^{(2)}_{X_5}\left(\lambda_{X_{18}}\right)^*$	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$+6{g_2}^4$
E	2	3	N/A	3	N/A	2	N/A	—	—	$+6{g_2}^4$
E	-2	3	N/A	3	N/A	2	N/A	—	$+4\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	$+6{g_2}^4$
E	-4	3	N/A	3	N/A	2	N/A	—	—	$+6{g_2}^4$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 3 of 10

Column	Representative term structure
T1	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$
T2	$+3{g_1}^4$
T3	$-\frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	$+3{g_1}^4$	$-\frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2}$
A	−2	1	N/A	3	N/A	2	N/A	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	—	$-\frac{3{g_1}^2\lambda_{HS_2}^{(1)}}{2}$
A	−4	1	N/A	3	N/A	2	N/A	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	$+3{g_1}^4$	$-\frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2}$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$-\frac{9{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	$+3{g_1}^4$	$-\frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2}$
D	2	3	N/A	1	N/A	2	N/A	$-\frac{9{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	$+12{g_1}^4$	$-\frac{51{g_1}^2\lambda_{HS_2}^{(1)}}{2}$
D	−2	3	N/A	1	N/A	2	N/A	$-\frac{9{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	—	$-\frac{3{g_1}^2\lambda_{HS_2}^{(1)}}{2}$
E	0	3	N/A	3	N/A	2	N/A	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	$+3{g_1}^4$	$-\frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2}$
E	2	3	N/A	3	N/A	2	N/A	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	$+12{g_1}^4$	$-\frac{51{g_1}^2\lambda_{HS_2}^{(1)}}{2}$
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	—	$-\frac{3{g_1}^2\lambda_{HS_2}^{(1)}}{2}$
E	−4	3	N/A	3	N/A	2	N/A	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	$+3{g_1}^4$	$-\frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 4 of 10

Column	Representative term structure
T1	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$
T2	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3	$+4{\lambda_{HS_2}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4{\lambda_{HS_2}^{(1)}}^2$
A	-2	1	N/A	3	N/A	2	N/A	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4{\lambda_{HS_2}^{(1)}}^2$
A	-4	1	N/A	3	N/A	2	N/A	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4{\lambda_{HS_2}^{(1)}}^2$
B	1	2	N/A	2	N/A	1	N/A	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4{\lambda_{HS_2}^{(1)}}^2$
B	-1	2	N/A	2	N/A	1	N/A	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4{\lambda_{HS_2}^{(1)}}^2$
B	-3	2	N/A	2	N/A	1	N/A	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4{\lambda_{HS_2}^{(1)}}^2$
C	1	2	N/A	2	N/A	3	N/A	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4{\lambda_{HS_2}^{(1)}}^2$
C	-1	2	N/A	2	N/A	3	N/A	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4{\lambda_{HS_2}^{(1)}}^2$
C	-3	2	N/A	2	N/A	3	N/A	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4{\lambda_{HS_2}^{(1)}}^2$
D	0	3	N/A	1	N/A	2	N/A	$+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4{\lambda_{HS_2}^{(1)}}^2$
D	2	3	N/A	1	N/A	2	N/A	$+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4{\lambda_{HS_2}^{(1)}}^2$
D	-2	3	N/A	1	N/A	2	N/A	$+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4{\lambda_{HS_2}^{(1)}}^2$
E	0	3	N/A	3	N/A	2	N/A	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4{\lambda_{HS_2}^{(1)}}^2$
E	2	3	N/A	3	N/A	2	N/A	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4{\lambda_{HS_2}^{(1)}}^2$
E	-2	3	N/A	3	N/A	2	N/A	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4{\lambda_{HS_2}^{(1)}}^2$
E	-4	3	N/A	3	N/A	2	N/A	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4{\lambda_{HS_2}^{(1)}}^2$

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 5 of 10

Column	Representative term structure
T1	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$
T2	$+6{\lambda^{(A)}_{HS_2}}^{2}$
T3	$+20\lambda_{X_2}\left(\lambda_{X_{16}}\right)^{*}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	$+6{\lambda^{(A)}_{HS_2}}^{2}$	—
A	−2	1	N/A	3	N/A	2	N/A	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	$+6{\lambda^{(A)}_{HS_2}}^{2}$	$+20\lambda_{X_2}\left(\lambda_{X_{16}}\right)^{*}$
A	−4	1	N/A	3	N/A	2	N/A	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	$+6{\lambda^{(A)}_{HS_2}}^{2}$	—
B	1	2	N/A	2	N/A	1	N/A	$+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—	—
B	−3	2	N/A	2	N/A	1	N/A	$+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—	—
C	1	2	N/A	2	N/A	3	N/A	$+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—	—
C	−3	2	N/A	2	N/A	3	N/A	$+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—	—
D	0	3	N/A	1	N/A	2	N/A	$+4\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—	—
D	2	3	N/A	1	N/A	2	N/A	$+4\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+4\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	—	$+12\lambda_{X_2}\left(\lambda_{X_{16}}\right)^{*}$
E	0	3	N/A	3	N/A	2	N/A	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	$+6{\lambda^{(A)}_{HS_2}}^{2}$	—
E	2	3	N/A	3	N/A	2	N/A	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	$+6{\lambda^{(A)}_{HS_2}}^{2}$	—
E	−2	3	N/A	3	N/A	2	N/A	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	$+6{\lambda^{(A)}_{HS_2}}^{2}$	$+20\lambda_{X_2}\left(\lambda_{X_{16}}\right)^{*}$
E	−4	3	N/A	3	N/A	2	N/A	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	$+6{\lambda^{(A)}_{HS_2}}^{2}$	—

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 6 of 10

Column	Representative term structure
T1	$+4\lambda_{X_{16}}(5\left(\lambda_{X_2}\right)^* + 4\left(\lambda_{X_{16}}\right)^*)$
T2	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T3	$+\frac{9{g_2}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+4\lambda_{X_{16}}(5\left(\lambda_{X_2}\right)^* + 4\left(\lambda_{X_{16}}\right)^*)$	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{9{g_2}^4}{4}$
B	−1	2	N/A	2	N/A	1	N/A	—	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{9{g_2}^4}{4}$
B	−3	2	N/A	2	N/A	1	N/A	—	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{9{g_2}^4}{4}$
C	1	2	N/A	2	N/A	3	N/A	—	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{9{g_2}^4}{4}$
C	−1	2	N/A	2	N/A	3	N/A	—	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{9{g_2}^4}{4}$
C	−3	2	N/A	2	N/A	3	N/A	—	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{9{g_2}^4}{4}$
D	0	3	N/A	1	N/A	2	N/A	—	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—
D	2	3	N/A	1	N/A	2	N/A	—	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—
D	−2	3	N/A	1	N/A	2	N/A	$+4\lambda_{X_{16}}(3\left(\lambda_{X_2}\right)^* + 4\left(\lambda_{X_{16}}\right)^*)$	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+4\lambda_{X_{16}}(5\left(\lambda_{X_2}\right)^* + 4\left(\lambda_{X_{16}}\right)^*)$	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 7 of 10

Column	Representative term structure
T1	$-9{g_2}^2\lambda_{HS_2}^{(1)}$
T2	$+\frac{27{g_1}^4}{4}$
T3	$-15{g_1}^2\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$-9{g_2}^2\lambda_{HS_2}^{(1)}$	$+\frac{27{g_1}^4}{4}$	$-15{g_1}^2\lambda_{HS_2}^{(1)}$
B	-1	2	N/A	2	N/A	1	N/A	$-9{g_2}^2\lambda_{HS_2}^{(1)}$	$+\frac{3{g_1}^4}{4}$	$-3{g_1}^2\lambda_{HS_2}^{(1)}$
B	-3	2	N/A	2	N/A	1	N/A	$-9{g_2}^2\lambda_{HS_2}^{(1)}$	$+\frac{3{g_1}^4}{4}$	$-3{g_1}^2\lambda_{HS_2}^{(1)}$
C	1	2	N/A	2	N/A	3	N/A	$-9{g_2}^2\lambda_{HS_2}^{(1)}$	$+\frac{27{g_1}^4}{4}$	$-15{g_1}^2\lambda_{HS_2}^{(1)}$
C	-1	2	N/A	2	N/A	3	N/A	$-9{g_2}^2\lambda_{HS_2}^{(1)}$	$+\frac{3{g_1}^4}{4}$	$-3{g_1}^2\lambda_{HS_2}^{(1)}$
C	-3	2	N/A	2	N/A	3	N/A	$-9{g_2}^2\lambda_{HS_2}^{(1)}$	$+\frac{3{g_1}^4}{4}$	$-3{g_1}^2\lambda_{HS_2}^{(1)}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 8 of 10

Column	Representative term structure
T1	$+\frac{3{\lambda^{(A)}_{HS_2}}^2}{4}$
T2	$+3\lambda_{10}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - \left(\lambda^{(1)}_{X_{17}}\right)^*)$
T3	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\left(\lambda_{10}\right)^* + 2\left(\lambda^{(1)}_{X_{17}}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+\frac{3{\lambda^{(A)}_{HS_2}}^2}{4}$	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+\frac{3{\lambda^{(A)}_{HS_2}}^2}{4}$	$+3\lambda_{10}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - \left(\lambda^{(1)}_{X_{17}}\right)^*)$	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\left(\lambda_{10}\right)^* + 2\left(\lambda^{(1)}_{X_{17}}\right)^*)$
B	-3	2	N/A	2	N/A	1	N/A	$+\frac{3{\lambda^{(A)}_{HS_2}}^2}{4}$	—	—
C	1	2	N/A	2	N/A	3	N/A	$+\frac{3{\lambda^{(A)}_{HS_2}}^2}{4}$	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+\frac{3{\lambda^{(A)}_{HS_2}}^2}{4}$	$+3\lambda_{10}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - \left(\lambda^{(1)}_{X_{17}}\right)^*)$	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\left(\lambda_{10}\right)^* + 2\left(\lambda^{(1)}_{X_{17}}\right)^*)$
C	-3	2	N/A	2	N/A	3	N/A	$+\frac{3{\lambda^{(A)}_{HS_2}}^2}{4}$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 9 of 10

Column	Representative term structure
T1	$+\lambda^{(1)}_{X_{17}}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 3\left(\lambda_{10}\right)^* + 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$
T2	$+24\lambda_8\left(\lambda_8\right)^*$
T3	$+4\lambda_{X_{18}}(3\left(\lambda^{(1)}_{X_5}\right)^* + \left(\lambda^{(2)}_{X_5}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+\lambda^{(1)}_{X_{17}}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 3\left(\lambda_{10}\right)^* + 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$	$+24\lambda_8\left(\lambda_8\right)^*$	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+\lambda^{(1)}_{X_{17}}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 3\left(\lambda_{10}\right)^* + 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$	$+24\lambda_8\left(\lambda_8\right)^*$	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	$+4\lambda_{X_{18}}(3\left(\lambda^{(1)}_{X_5}\right)^* + \left(\lambda^{(2)}_{X_5}\right)^*)$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 10 of 10

Column	Representative term structure
T1	$+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	-2	1	N/A	3	N/A	2	N/A	—
A	-4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	-1	2	N/A	2	N/A	1	N/A	—
B	-3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	-1	2	N/A	2	N/A	3	N/A	—
C	-3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	-2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	$+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$
E	2	3	N/A	3	N/A	2	N/A	$+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$
E	-2	3	N/A	3	N/A	2	N/A	$+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$
E	-4	3	N/A	3	N/A	2	N/A	$+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$

$\beta^{(1)}_{\lambda^{(1)}_{12}}$

Quantity: $\lambda^{(1)}_{12}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{12}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{12}}$$

Terms 1 of 11

Column	Representative term structure
T1	$4\lambda_5\,(\lambda_5)^*$
T2	$+12\lambda_{X_9}\left(\lambda^{(1)}_{X_5}\right)^*$
T3	$+4\lambda^{(1)}_{X_5}(4\left(\lambda^{(1)}_{X_5}\right)^* + 3\left(\lambda_{X_9}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$4\lambda_5\,(\lambda_5)^*$	$+12\lambda_{X_9}\left(\lambda^{(1)}_{X_5}\right)^*$	$+4\lambda^{(1)}_{X_5}(4\left(\lambda^{(1)}_{X_5}\right)^* + 3\left(\lambda_{X_9}\right)^*)$
A	−2	1	N/A	3	N/A	2	N/A	$4\lambda_5\,(\lambda_5)^*$	—	—
A	−4	1	N/A	3	N/A	2	N/A	$4\lambda_5\,(\lambda_5)^*$	—	—
B	1	2	N/A	2	N/A	1	N/A	$3\lambda_5\,(\lambda_5)^*$	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+3\lambda_5\,(\lambda_5)^*$	—	—
B	−3	2	N/A	2	N/A	1	N/A	$3\lambda_5\,(\lambda_5)^*$	—	—
C	1	2	N/A	2	N/A	3	N/A	$3\lambda_5\,(\lambda_5)^*$	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+3\lambda_5\,(\lambda_5)^*$	—	—
C	−3	2	N/A	2	N/A	3	N/A	$3\lambda_5\,(\lambda_5)^*$	—	—
D	0	3	N/A	1	N/A	2	N/A	$4\lambda_5\,(\lambda_5)^*$	$+20\lambda_{X_9}\left(\lambda^{(1)}_{X_5}\right)^*$	$+4\lambda^{(1)}_{X_5}(4\left(\lambda^{(1)}_{X_5}\right)^* + 5\left(\lambda_{X_9}\right)^*)$
D	2	3	N/A	1	N/A	2	N/A	$4\lambda_5\,(\lambda_5)^*$	—	—
D	−2	3	N/A	1	N/A	2	N/A	$4\lambda_5\,(\lambda_5)^*$	—	—
E	0	3	N/A	3	N/A	2	N/A	$2\lambda_5\,(\lambda_5)^*$	—	$+4\lambda^{(1)}_{X_5}(4\left(\lambda^{(1)}_{X_5}\right)^* + 5\left(\lambda_{X_9}\right)^*)$
E	2	3	N/A	3	N/A	2	N/A	$2\lambda_5\,(\lambda_5)^*$	—	—
E	−2	3	N/A	3	N/A	2	N/A	$2\lambda_5\,(\lambda_5)^*$	—	—
E	−4	3	N/A	3	N/A	2	N/A	$2\lambda_5\,(\lambda_5)^*$	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 2 of 11

Column	Representative term structure
T1	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^{*}$
T2	$-12{g_2}^2\lambda_{12}^{(1)}$
T3	$-6{g_1}^2\lambda_{12}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^{*}$	$-12{g_2}^2\lambda_{12}^{(1)}$	$-6{g_1}^2\lambda_{12}^{(1)}$
A	−2	1	N/A	3	N/A	2	N/A	$+4\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^{*}$	$-12{g_2}^2\lambda_{12}^{(1)}$	$-6{g_1}^2\lambda_{12}^{(1)}$
A	−4	1	N/A	3	N/A	2	N/A	—	$-12{g_2}^2\lambda_{12}^{(1)}$	$-30{g_1}^2\lambda_{12}^{(1)}$
B	1	2	N/A	2	N/A	1	N/A	—	$-9{g_2}^2\lambda_{12}^{(1)}$	$-15{g_1}^2\lambda_{12}^{(1)}$
B	−1	2	N/A	2	N/A	1	N/A	—	$-9{g_2}^2\lambda_{12}^{(1)}$	$-3{g_1}^2\lambda_{12}^{(1)}$
B	−3	2	N/A	2	N/A	1	N/A	—	$-9{g_2}^2\lambda_{12}^{(1)}$	$-15{g_1}^2\lambda_{12}^{(1)}$
C	1	2	N/A	2	N/A	3	N/A	—	$-9{g_2}^2\lambda_{12}^{(1)}$	$-15{g_1}^2\lambda_{12}^{(1)}$
C	−1	2	N/A	2	N/A	3	N/A	—	$-9{g_2}^2\lambda_{12}^{(1)}$	$-3{g_1}^2\lambda_{12}^{(1)}$
C	−3	2	N/A	2	N/A	3	N/A	—	$-9{g_2}^2\lambda_{12}^{(1)}$	$-15{g_1}^2\lambda_{12}^{(1)}$
D	0	3	N/A	1	N/A	2	N/A	$+4\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^{*}$	$-12{g_2}^2\lambda_{12}^{(1)}$	$-6{g_1}^2\lambda_{12}^{(1)}$
D	2	3	N/A	1	N/A	2	N/A	—	$-12{g_2}^2\lambda_{12}^{(1)}$	$-30{g_1}^2\lambda_{12}^{(1)}$
D	−2	3	N/A	1	N/A	2	N/A	$+4\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^{*}$	$-12{g_2}^2\lambda_{12}^{(1)}$	$-6{g_1}^2\lambda_{12}^{(1)}$
E	0	3	N/A	3	N/A	2	N/A	$+2\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^{*}$	$-24{g_2}^2\lambda_{12}^{(1)}$	$-6{g_1}^2\lambda_{12}^{(1)}$
E	2	3	N/A	3	N/A	2	N/A	—	$-24{g_2}^2\lambda_{12}^{(1)}$	$-30{g_1}^2\lambda_{12}^{(1)}$
E	−2	3	N/A	3	N/A	2	N/A	$+2\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^{*}$	$-24{g_2}^2\lambda_{12}^{(1)}$	$-6{g_1}^2\lambda_{12}^{(1)}$
E	−4	3	N/A	3	N/A	2	N/A	—	$-24{g_2}^2\lambda_{12}^{(1)}$	$-30{g_1}^2\lambda_{12}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

Terms 3 of 11

Column	Representative term structure
T1	$+4\lambda_{12}^{(1)\,2}$
T2	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T3	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(1)\,2}$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
A	-2	1	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(1)\,2}$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
A	-4	1	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(1)\,2}$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
B	1	2	N/A	2	N/A	1	N/A	$+4\lambda_{12}^{(1)\,2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
B	-1	2	N/A	2	N/A	1	N/A	$+4\lambda_{12}^{(1)\,2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
B	-3	2	N/A	2	N/A	1	N/A	$+4\lambda_{12}^{(1)\,2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
C	1	2	N/A	2	N/A	3	N/A	$+4\lambda_{12}^{(1)\,2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
C	-1	2	N/A	2	N/A	3	N/A	$+4\lambda_{12}^{(1)\,2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
C	-3	2	N/A	2	N/A	3	N/A	$+4\lambda_{12}^{(1)\,2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
D	0	3	N/A	1	N/A	2	N/A	$+4\lambda_{12}^{(1)\,2}$	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
D	2	3	N/A	1	N/A	2	N/A	$+4\lambda_{12}^{(1)\,2}$	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
D	-2	3	N/A	1	N/A	2	N/A	$+4\lambda_{12}^{(1)\,2}$	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
E	0	3	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(1)\,2}$	—	—
E	2	3	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(1)\,2}$	—	—
E	-2	3	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(1)\,2}$	—	—
E	-4	3	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(1)\,2}$	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 4 of 11

Column	Representative term structure
T1	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(A)}$
T2	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3	$+20\lambda_{X_2}\left(\lambda_{X_7}^{(1)}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(A)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
A	−2	1	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(A)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	$+20\lambda_{X_2}\left(\lambda_{X_7}^{(1)}\right)^*$
A	−4	1	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(A)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
B	1	2	N/A	2	N/A	1	N/A	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
B	−1	2	N/A	2	N/A	1	N/A	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
B	−3	2	N/A	2	N/A	1	N/A	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
C	1	2	N/A	2	N/A	3	N/A	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
C	−1	2	N/A	2	N/A	3	N/A	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
C	−3	2	N/A	2	N/A	3	N/A	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
D	0	3	N/A	1	N/A	2	N/A	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
D	2	3	N/A	1	N/A	2	N/A	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
D	−2	3	N/A	1	N/A	2	N/A	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	$+12\lambda_{X_2}\left(\lambda_{X_7}^{(1)}\right)^*$
E	0	3	N/A	3	N/A	2	N/A	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
E	2	3	N/A	3	N/A	2	N/A	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
E	−2	3	N/A	3	N/A	2	N/A	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
E	−4	3	N/A	3	N/A	2	N/A	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—

Quantity: $\lambda_{12}^{(1)}$

Terms 5 of 11

Column	Representative term structure
T1	$+4\lambda_{X_7}^{(1)}(5\left(\lambda_{X_2}\right)^* + 4\left(\lambda_{X_7}^{(1)}\right)^*)$
T2	$+48{g_1}^4$
T3	$+\frac{9{g_2}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+4\lambda_{X_7}^{(1)}(5\left(\lambda_{X_2}\right)^* + 4\left(\lambda_{X_7}^{(1)}\right)^*)$	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	$+48{g_1}^4$	—
B	1	2	N/A	2	N/A	1	N/A	—	—	$+\frac{9{g_2}^4}{4}$
B	−1	2	N/A	2	N/A	1	N/A	—	—	$+\frac{9{g_2}^4}{4}$
B	−3	2	N/A	2	N/A	1	N/A	—	—	$+\frac{9{g_2}^4}{4}$
C	1	2	N/A	2	N/A	3	N/A	—	—	$+\frac{9{g_2}^4}{4}$
C	−1	2	N/A	2	N/A	3	N/A	—	—	$+\frac{9{g_2}^4}{4}$
C	−3	2	N/A	2	N/A	3	N/A	—	—	$+\frac{9{g_2}^4}{4}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	$+48{g_1}^4$	—
D	−2	3	N/A	1	N/A	2	N/A	$+4\lambda_{X_7}^{(1)}(3\left(\lambda_{X_2}\right)^* + 4\left(\lambda_{X_7}^{(1)}\right)^*)$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	$+\frac{63{g_2}^4}{5}$
E	2	3	N/A	3	N/A	2	N/A	—	$+48{g_1}^4$	$+\frac{63{g_2}^4}{5}$
E	−2	3	N/A	3	N/A	2	N/A	$+4\lambda_{X_7}^{(1)}(5\left(\lambda_{X_2}\right)^* + 4\left(\lambda_{X_7}^{(1)}\right)^*)$	—	$+\frac{63{g_2}^4}{5}$
E	−4	3	N/A	3	N/A	2	N/A	—	$+48{g_1}^4$	$+\frac{63{g_2}^4}{5}$

Quantity: $\lambda_{12}^{(1)}$

Terms 6 of 11

Column	Representative term structure
T1	$+\frac{27{g_1}^4}{4}$
T2	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$
T3	$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+\frac{27{g_1}^4}{4}$	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$	—
B	−1	2	N/A	2	N/A	1	N/A	$+\frac{3{g_1}^4}{4}$	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$	$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$
B	−3	2	N/A	2	N/A	1	N/A	$+\frac{27{g_1}^4}{4}$	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$	—
C	1	2	N/A	2	N/A	3	N/A	$+\frac{27{g_1}^4}{4}$	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$	—
C	−1	2	N/A	2	N/A	3	N/A	$+\frac{3{g_1}^4}{4}$	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$	$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$
C	−3	2	N/A	2	N/A	3	N/A	$+\frac{27{g_1}^4}{4}$	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 7 of 11

Column	Representative term structure
T1	$+\lambda_{X_{17}}^{(1)}(\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + 2\left(\lambda_{X_{17}}^{(1)}\right)^*)$
T2	$+24\lambda_9\,(\lambda_9)^*$
T3	$+9\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+\lambda_{X_{17}}^{(1)}(\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + 2\left(\lambda_{X_{17}}^{(1)}\right)^*)$	$+24\lambda_9\,(\lambda_9)^*$	$+9\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$ $+9\lambda_{11}((\lambda_{10})^* + (\lambda_{11})^*)$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+\lambda_{X_{17}}^{(1)}(\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + 2\left(\lambda_{X_{17}}^{(1)}\right)^*)$	$+24\lambda_9\,(\lambda_9)^*$	$+9\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$ $+9\lambda_{11}((\lambda_{10})^* + (\lambda_{11})^*)$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 8 of 11

Column	Representative term structure
T1	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(A)}$
T2	$+4\lambda_{X_9}(5\left(\lambda_{X_5}^{(1)}\right)^{*}+\left(\lambda_{X_5}^{(2)}\right)^{*})$
T3	$+4\lambda_{X_5}^{(2)}(\left(\lambda_{X_5}^{(2)}\right)^{*}+\left(\lambda_{X_9}\right)^{*})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(A)}$	—	—
D	2	3	N/A	1	N/A	2	N/A	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(A)}$	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(A)}$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+4\lambda_{X_9}(5\left(\lambda_{X_5}^{(1)}\right)^{*}+\left(\lambda_{X_5}^{(2)}\right)^{*})$	$+4\lambda_{X_5}^{(2)}(\left(\lambda_{X_5}^{(2)}\right)^{*}+(\lambda_{X_9})^{*})$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 9 of 11

Column	Representative term structure
T1	$+\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T2	$+\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3	$+\lambda_{12}^{(1)}(4\lambda_{S_1}^{(A)}+4\lambda_{S_2}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	$+\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	$+\lambda_{12}^{(1)}(4\lambda_{S_1}^{(A)}+4\lambda_{S_2}^{(A)})$
E	2	3	N/A	3	N/A	2	N/A	$+\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	$+\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	$+\lambda_{12}^{(1)}(4\lambda_{S_1}^{(A)}+4\lambda_{S_2}^{(A)})$
E	−2	3	N/A	3	N/A	2	N/A	$+\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	$+\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	$+\lambda_{12}^{(1)}(4\lambda_{S_1}^{(A)}+4\lambda_{S_2}^{(A)})$
E	−4	3	N/A	3	N/A	2	N/A	$+\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	$+\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	$+\lambda_{12}^{(1)}(4\lambda_{S_1}^{(A)}+4\lambda_{S_2}^{(A)})$

Quantity: $\lambda_{12}^{(1)}$

Terms 10 of 11

Column	Representative term structure
T1	$+16{\lambda_{12}^{(A)}}^2$
T2	$+4{\lambda_{12}^{(\times)}}^2$
T3	$+4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+16{\lambda_{12}^{(A)}}^2$	$+4{\lambda_{12}^{(\times)}}^2$	$+4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	$+16{\lambda_{12}^{(A)}}^2$	$+4{\lambda_{12}^{(\times)}}^2$	$+4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$
E	-2	3	N/A	3	N/A	2	N/A	$+16{\lambda_{12}^{(A)}}^2$	$+4{\lambda_{12}^{(\times)}}^2$	$+4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$
E	-4	3	N/A	3	N/A	2	N/A	$+16{\lambda_{12}^{(A)}}^2$	$+4{\lambda_{12}^{(\times)}}^2$	$+4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$

Quantity: $\lambda_{12}^{(1)}$

Terms 11 of 11

Column	Representative term structure
T1	$+4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$
T2	$+4\lambda_{X_2}(5\left(\lambda_{X_7}^{(1)}\right)^{*}+\left(\lambda_{X_7}^{(2)}\right)^{*})$
T3	$+4\lambda_{X_7}^{(2)}((\lambda_{X_2})^{*}+\left(\lambda_{X_7}^{(2)}\right)^{*})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	—	—
E	2	3	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	$+4\lambda_{X_2}(5\left(\lambda_{X_7}^{(1)}\right)^{*}+\left(\lambda_{X_7}^{(2)}\right)^{*})$	$+4\lambda_{X_7}^{(2)}((\lambda_{X_2})^{*}+\left(\lambda_{X_7}^{(2)}\right)^{*})$
E	−4	3	N/A	3	N/A	2	N/A	$+4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	—	—

$\beta^{(1)}_{\lambda_5}$

Quantity: λ_5

$$16\pi^2\,\mu\frac{d\lambda_5}{d\mu} = \beta^{(1)}_{\lambda_5}$$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_5(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{HS_2})$
T2	$+4\left(\lambda_{X_{14}}\right)^*(\left(\lambda^{(1)}_{X_5}\right)^* + 2\left(\lambda_{X_{18}}\right)^*)$
T3	$-\frac{21{g_2}^2\lambda_5}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda_5(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{HS_2})$	$+4\left(\lambda_{X_{14}}\right)^*(\left(\lambda^{(1)}_{X_5}\right)^* + 2\left(\lambda_{X_{18}}\right)^*)$	$-\frac{21{g_2}^2\lambda_5}{2}$
A	−2	1	N/A	3	N/A	2	N/A	$\lambda_5(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{HS_2})$	—	$-\frac{21{g_2}^2\lambda_5}{2}$
A	−4	1	N/A	3	N/A	2	N/A	$\lambda_5(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{HS_2})$	—	$-\frac{21{g_2}^2\lambda_5}{2}$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	$+4\left(\lambda_{X_{14}}\right)^*(\left(\lambda^{(1)}_{X_5}\right)^* + 2\left(\lambda_{X_{18}}\right)^*)$	$-\frac{21{g_2}^2\lambda_5}{2}$
D	2	3	N/A	1	N/A	2	N/A	—	—	$-\frac{21{g_2}^2\lambda_5}{2}$
D	−2	3	N/A	1	N/A	2	N/A	—	—	$-\frac{21{g_2}^2\lambda_5}{2}$
E	0	3	N/A	3	N/A	2	N/A	—	—	$-\frac{33{g_2}^2\lambda_5}{2}$
E	2	3	N/A	3	N/A	2	N/A	—	—	$-\frac{33{g_2}^2\lambda_5}{2}$
E	−2	3	N/A	3	N/A	2	N/A	—	—	$-\frac{33{g_2}^2\lambda_5}{2}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	$-\frac{33{g_2}^2\lambda_5}{2}$

Quantity: λ_5

Terms 2 of 6

Column	Representative term structure
T1	$-\frac{9{g_1}^2\lambda_5}{2}$
T2	$+2\lambda_{12}^{(1)}\lambda_5$
T3	$+2\lambda_1\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-\frac{9{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$
A	-2	1	N/A	3	N/A	2	N/A	$-\frac{9{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$
A	-4	1	N/A	3	N/A	2	N/A	$-\frac{33{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$
B	1	2	N/A	2	N/A	1	N/A	—	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$
B	-1	2	N/A	2	N/A	1	N/A	—	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$
B	-3	2	N/A	2	N/A	1	N/A	—	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$
C	1	2	N/A	2	N/A	3	N/A	—	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$
C	-1	2	N/A	2	N/A	3	N/A	—	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$
C	-3	2	N/A	2	N/A	3	N/A	—	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$
D	0	3	N/A	1	N/A	2	N/A	$-\frac{9{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$
D	2	3	N/A	1	N/A	2	N/A	$-\frac{33{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$
D	-2	3	N/A	1	N/A	2	N/A	$-\frac{9{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$
E	0	3	N/A	3	N/A	2	N/A	$-\frac{9{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$
E	2	3	N/A	3	N/A	2	N/A	$-\frac{33{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$
E	-2	3	N/A	3	N/A	2	N/A	$-\frac{9{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$
E	-4	3	N/A	3	N/A	2	N/A	$-\frac{33{g_1}^2\lambda_5}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$

Quantity: λ_5

Terms 3 of 6

Column	Representative term structure
T1	$+4\lambda^{(1)}_{HS_1}\lambda_5$
T2	$+4\lambda^{(1)}_{HS_2}\lambda_5$
T3	$+4\lambda^{(1)}_{X_7}\left(\lambda_{X_{12}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$	—
A	-2	1	N/A	3	N/A	2	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$	$+4\lambda^{(1)}_{X_7}\left(\lambda_{X_{12}}\right)^*$
A	-4	1	N/A	3	N/A	2	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$	—
B	1	2	N/A	2	N/A	1	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$	—
B	-1	2	N/A	2	N/A	1	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$	—
B	-3	2	N/A	2	N/A	1	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$	—
C	1	2	N/A	2	N/A	3	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$	—
C	-1	2	N/A	2	N/A	3	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$	—
C	-3	2	N/A	2	N/A	3	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$	—
D	0	3	N/A	1	N/A	2	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$	—
D	2	3	N/A	1	N/A	2	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$	—
D	-2	3	N/A	1	N/A	2	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$	$+4\lambda^{(1)}_{X_7}\left(\lambda_{X_{12}}\right)^*$
E	0	3	N/A	3	N/A	2	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$	—
E	2	3	N/A	3	N/A	2	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$	—
E	-2	3	N/A	3	N/A	2	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$	$+4\lambda^{(1)}_{X_7}\left(\lambda_{X_{12}}\right)^*$ $-2\lambda^{(2)}_{X_7}\left(\lambda_{X_{12}}\right)^*$
E	-4	3	N/A	3	N/A	2	N/A	$+4\lambda^{(1)}_{HS_1}\lambda_5$	$+4\lambda^{(1)}_{HS_2}\lambda_5$	—

Quantity: λ_5

Terms 4 of 6

Column	Representative term structure
T1	$+8\lambda_{X_{16}}\left(\lambda_{X_{12}}\right)^*$
T2	$\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$
T3	$-9{g_2}^2\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+8\lambda_{X_{16}}\left(\lambda_{X_{12}}\right)^*$	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	$\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	$-9{g_2}^2\lambda_5$
B	-1	2	N/A	2	N/A	1	N/A	—	$\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	$-9{g_2}^2\lambda_5$
B	-3	2	N/A	2	N/A	1	N/A	—	$\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	$-9{g_2}^2\lambda_5$
C	1	2	N/A	2	N/A	3	N/A	—	$\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	$-9{g_2}^2\lambda_5$
C	-1	2	N/A	2	N/A	3	N/A	—	$\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	$-9{g_2}^2\lambda_5$
C	-3	2	N/A	2	N/A	3	N/A	—	$\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	$-9{g_2}^2\lambda_5$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$+8\lambda_{X_{16}}\left(\lambda_{X_{12}}\right)^*$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$+8\lambda_{X_{16}}\left(\lambda_{X_{12}}\right)^*$	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_5

Terms 5 of 6

Column	Representative term structure
T1	$-9{g_1}^2\lambda_5$
T2	$\lambda_5(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - 4\lambda_{HS_1}^{(A)})$
T3	$\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$-9{g_1}^2\lambda_5$	—	—
B	-1	2	N/A	2	N/A	1	N/A	$-3{g_1}^2\lambda_5$	—	—
B	-3	2	N/A	2	N/A	1	N/A	$-9{g_1}^2\lambda_5$	—	—
C	1	2	N/A	2	N/A	3	N/A	$-9{g_1}^2\lambda_5$	—	—
C	-1	2	N/A	2	N/A	3	N/A	$-3{g_1}^2\lambda_5$	—	—
C	-3	2	N/A	2	N/A	3	N/A	$-9{g_1}^2\lambda_5$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	$\lambda_5(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - 4\lambda_{HS_1}^{(A)})$	—
D	2	3	N/A	1	N/A	2	N/A	—	$\lambda_5(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - 4\lambda_{HS_1}^{(A)})$	—
D	-2	3	N/A	1	N/A	2	N/A	—	$\lambda_5(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - 4\lambda_{HS_1}^{(A)})$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	$\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3}\right)$
E	2	3	N/A	3	N/A	2	N/A	—	—	$\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3}\right)$
E	-2	3	N/A	3	N/A	2	N/A	—	—	$\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3}\right)$
E	-4	3	N/A	3	N/A	2	N/A	—	—	$\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3}\right)$

Quantity: λ_5

Terms 6 of 6

Column	Representative term structure
T1	$+2\left(\lambda_{X_{14}}\right)^{*}\left(2\left(\lambda_{X_5}^{(1)}\right)^{*}-\left(\lambda_{X_5}^{(2)}\right)^{*}+4\left(\lambda_{X_{18}}\right)^{*}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	-2	1	N/A	3	N/A	2	N/A	—
A	-4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	-1	2	N/A	2	N/A	1	N/A	—
B	-3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	-1	2	N/A	2	N/A	3	N/A	—
C	-3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	-2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	$+2\left(\lambda_{X_{14}}\right)^{*}\left(2\left(\lambda_{X_5}^{(1)}\right)^{*}-\left(\lambda_{X_5}^{(2)}\right)^{*}+4\left(\lambda_{X_{18}}\right)^{*}\right)$
E	2	3	N/A	3	N/A	2	N/A	—
E	-2	3	N/A	3	N/A	2	N/A	—
E	-4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda_{12}^{(A)}}$

Quantity: $\lambda_{12}^{(A)}$

$$16\pi^2\,\mu\frac{d\lambda_{12}^{(A)}}{d\mu} = \beta^{(1)}_{\lambda_{12}^{(A)}}$$

Terms 1 of 7

Column	Representative term structure
T1	$\frac{4}{3}\lambda_5\,(\lambda_5)^*$
T2	$+{g_1}^2\left(18{g_2}^2 - 15\lambda_{12}^{(A)}\right)$
T3	$-9{g_2}^2\lambda_{12}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$\frac{4}{3}\lambda_5\,(\lambda_5)^*$	$+{g_1}^2\left(18{g_2}^2 - 15\lambda_{12}^{(A)}\right)$	$-9{g_2}^2\lambda_{12}^{(A)}$
B	−1	2	N/A	2	N/A	1	N/A	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$	—	$-9{g_2}^2\lambda_{12}^{(A)}$
B	−3	2	N/A	2	N/A	1	N/A	$\frac{4}{3}\lambda_5\,(\lambda_5)^*$	$+{g_1}^2\left(18{g_2}^2 - 15\lambda_{12}^{(A)}\right)$	$-9{g_2}^2\lambda_{12}^{(A)}$
C	1	2	N/A	2	N/A	3	N/A	$\frac{4}{3}\lambda_5\,(\lambda_5)^*$	$+{g_1}^2\left(18{g_2}^2 - 15\lambda_{12}^{(A)}\right)$	$-9{g_2}^2\lambda_{12}^{(A)}$
C	−1	2	N/A	2	N/A	3	N/A	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$	—	$-9{g_2}^2\lambda_{12}^{(A)}$
C	−3	2	N/A	2	N/A	3	N/A	$\frac{4}{3}\lambda_5\,(\lambda_5)^*$	$+{g_1}^2\left(18{g_2}^2 - 15\lambda_{12}^{(A)}\right)$	$-9{g_2}^2\lambda_{12}^{(A)}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-\frac{2}{9}\lambda_5\,(\lambda_5)^*$	—	$-24{g_2}^2\lambda_{12}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	$-\frac{2}{9}\lambda_5\,(\lambda_5)^*$	$+{g_1}^2\left(16{g_2}^2 - 30\lambda_{12}^{(A)}\right)$	$-24{g_2}^2\lambda_{12}^{(A)}$
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{2}{9}\lambda_5\,(\lambda_5)^*$	—	$-24{g_2}^2\lambda_{12}^{(A)}$
E	−4	3	N/A	3	N/A	2	N/A	$-\frac{2}{9}\lambda_5\,(\lambda_5)^*$	$+{g_1}^2\left(16{g_2}^2 - 30\lambda_{12}^{(A)}\right)$	$-24{g_2}^2\lambda_{12}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 2 of 7

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T2	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$
T3	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
B	-1	2	N/A	2	N/A	1	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
B	-3	2	N/A	2	N/A	1	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
C	1	2	N/A	2	N/A	3	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
C	-1	2	N/A	2	N/A	3	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
C	-3	2	N/A	2	N/A	3	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
E	2	3	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
E	-2	3	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
E	-4	3	N/A	3	N/A	2	N/A	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 3 of 7

Column	Representative term structure
T1	$+\lambda_{HS_1}^{(A)}\,\lambda_{HS_2}^{(A)}$
T2	$4\lambda_{X_{17}}^{(1)}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{X_{17}}^{(1)}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+\lambda_{HS_1}^{(A)}\,\lambda_{HS_2}^{(A)}$	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+\lambda_{HS_1}^{(A)}\,\lambda_{HS_2}^{(A)}$	$4\lambda_{X_{17}}^{(1)}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{X_{17}}^{(1)}\right)^*$
B	-3	2	N/A	2	N/A	1	N/A	$+\lambda_{HS_1}^{(A)}\,\lambda_{HS_2}^{(A)}$	—	—
C	1	2	N/A	2	N/A	3	N/A	$+\lambda_{HS_1}^{(A)}\,\lambda_{HS_2}^{(A)}$	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+\lambda_{HS_1}^{(A)}\,\lambda_{HS_2}^{(A)}$	$4\lambda_{X_{17}}^{(1)}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{X_{17}}^{(1)}\right)^*$
C	-3	2	N/A	2	N/A	3	N/A	$+\lambda_{HS_1}^{(A)}\,\lambda_{HS_2}^{(A)}$	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(A)}$

Terms 4 of 7

Column	Representative term structure
T1	$-64\lambda_9\,(\lambda_9)^*$
T2	$-4\lambda_{10}(5\,(\lambda_{10})^* + (\lambda_{11})^*)$
T3	$-4\lambda_{11}((\lambda_{10})^* + 5\,(\lambda_{11})^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$-64\lambda_9\,(\lambda_9)^*$	$-4\lambda_{10}(5\,(\lambda_{10})^* + (\lambda_{11})^*)$	$-4\lambda_{11}((\lambda_{10})^* + 5\,(\lambda_{11})^*)$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$-64\lambda_9\,(\lambda_9)^*$	$-4\lambda_{10}(5\,(\lambda_{10})^* + (\lambda_{11})^*)$	$-4\lambda_{11}((\lambda_{10})^* + 5\,(\lambda_{11})^*)$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(A)}$

Terms 5 of 7

Column	Representative term structure
T1	$+{g_1}^2\left(-6{g_2}^2-3\lambda_{12}^{(A)}\right)$
T2	$-\frac{1}{3}\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$
T3	$-6{g_1}^2\lambda_{12}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+{g_1}^2\left(-6{g_2}^2-3\lambda_{12}^{(A)}\right)$	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+{g_1}^2\left(-6{g_2}^2-3\lambda_{12}^{(A)}\right)$	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$-\frac{1}{3}\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$-6{g_1}^2\lambda_{12}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	$-\frac{1}{3}\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	$-6{g_1}^2\lambda_{12}^{(A)}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{12}^{(A)}$

Terms 6 of 7

Column	Representative term structure
T1	$+\dfrac{16\lambda_{12}^{(A)}\lambda_{12}^{(\times)}}{3}$
T2	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$
T3	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+\dfrac{16\lambda_{12}^{(A)}\lambda_{12}^{(\times)}}{3}$	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	$+\dfrac{16\lambda_{12}^{(A)}\lambda_{12}^{(\times)}}{3}$	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$
E	-2	3	N/A	3	N/A	2	N/A	$+\dfrac{16\lambda_{12}^{(A)}\lambda_{12}^{(\times)}}{3}$	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$
E	-4	3	N/A	3	N/A	2	N/A	$+\dfrac{16\lambda_{12}^{(A)}\lambda_{12}^{(\times)}}{3}$	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 7 of 7

Column	Representative term structure
T1	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
E	−2	3	N/A	3	N/A	2	N/A	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
E	−4	3	N/A	3	N/A	2	N/A	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

$\beta^{(1)}_{\lambda_{12}^{(\times)}}$

Quantity: $\lambda_{12}^{(\times)}$

$$16\pi^2 \, \mu \frac{d\lambda_{12}^{(\times)}}{d\mu} = \beta^{(1)}_{\lambda_{12}^{(\times)}}$$

Terms 1 of 6

Column	Representative term structure
T1	$-\lambda_5 \, (\lambda_5)^*$
T2	$+\lambda^{(2)}_{X_5} (8 \left(\lambda^{(1)}_{X_5}\right)^* + 10 \left(\lambda^{(2)}_{X_5}\right)^* + 4 \left(\lambda_{X_9}\right)^*)$
T3	$+8\lambda^{(1)}_{X_5} \left(\lambda^{(2)}_{X_5}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-\lambda_5 \, (\lambda_5)^*$	$+\lambda^{(2)}_{X_5} (8 \left(\lambda^{(1)}_{X_5}\right)^* + 10 \left(\lambda^{(2)}_{X_5}\right)^* + 4 \left(\lambda_{X_9}\right)^*)$	$+8\lambda^{(1)}_{X_5} \left(\lambda^{(2)}_{X_5}\right)^*$
E	2	3	N/A	3	N/A	2	N/A	$-\lambda_5 \, (\lambda_5)^*$	—	—
E	−2	3	N/A	3	N/A	2	N/A	$-\lambda_5 \, (\lambda_5)^*$	—	—
E	−4	3	N/A	3	N/A	2	N/A	$-\lambda_5 \, (\lambda_5)^*$	—	—

Quantity: $\lambda_{12}^{(\times)}$

Terms 2 of 6

Column	Representative term structure
T1	$+4\lambda_{X_9}\left(\lambda_{X_5}^{(2)}\right)^*$
T2	$-\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$
T3	$+\frac{51{g_2}^4}{10}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+4\lambda_{X_9}\left(\lambda_{X_5}^{(2)}\right)^*$	$-\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$+\frac{51{g_2}^4}{10}$
E	2	3	N/A	3	N/A	2	N/A	—	—	$+\frac{51{g_2}^4}{10}$
E	-2	3	N/A	3	N/A	2	N/A	—	$-\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	$+\frac{51{g_2}^4}{10}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	$+\frac{51{g_2}^4}{10}$

Quantity: $\lambda_{12}^{(\times)}$

Terms 3 of 6

Column	Representative term structure
T1	$-24{g_2}^2\lambda_{12}^{(\times)}$
T2	$-6{g_1}^2\lambda_{12}^{(\times)}$
T3	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{12}^{(\times)}$	$-6{g_1}^2\lambda_{12}^{(\times)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
E	2	3	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{12}^{(\times)}$	$-30{g_1}^2\lambda_{12}^{(\times)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
E	-2	3	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{12}^{(\times)}$	$-6{g_1}^2\lambda_{12}^{(\times)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
E	-4	3	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{12}^{(\times)}$	$-30{g_1}^2\lambda_{12}^{(\times)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Quantity: $\lambda_{12}^{(\times)}$

Terms 4 of 6

Column	Representative term structure
T1	$+10{\lambda_{12}^{(\times)}}^2$
T2	$+2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$
T3	$-2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+10{\lambda_{12}^{(\times)}}^2$	$+2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$	$-2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	$+10{\lambda_{12}^{(\times)}}^2$	$+2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$	$-2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$
E	−2	3	N/A	3	N/A	2	N/A	$+10{\lambda_{12}^{(\times)}}^2$	$+2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$	$-2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$
E	−4	3	N/A	3	N/A	2	N/A	$+10{\lambda_{12}^{(\times)}}^2$	$+2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$	$-2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$

Quantity: $\lambda_{12}^{(\times)}$

Terms 5 of 6

Column	Representative term structure
T1	$+2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$
T2	$-2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$
T3	$+\lambda_{X_7}^{(2)}(4\left(\lambda_{X_2}\right)^{*}+8\left(\lambda_{X_7}^{(1)}\right)^{*}+10\left(\lambda_{X_7}^{(2)}\right)^{*})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	$-2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	—
E	2	3	N/A	3	N/A	2	N/A	$+2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	$-2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	—
E	−2	3	N/A	3	N/A	2	N/A	$+2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	$-2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	$+\lambda_{X_7}^{(2)}(4\left(\lambda_{X_2}\right)^{*}+8\left(\lambda_{X_7}^{(1)}\right)^{*}+10\left(\lambda_{X_7}^{(2)}\right)^{*})$
E	−4	3	N/A	3	N/A	2	N/A	$+2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	$-2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	—

Quantity: $\lambda_{12}^{(\times)}$

Terms 6 of 6

Column	Representative term structure
T1	$+4\lambda_{X_2}\left(\lambda_{X_7}^{(2)}\right)^*$
T2	$+8\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(2)}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+4\lambda_{X_2}\left(\lambda_{X_7}^{(2)}\right)^*$	$+8\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(2)}\right)^*$
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda^{(A)}_{HS_1}}$

Quantity: $\lambda^{(A)}_{HS_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{HS_1}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{HS_1}}$$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$
T3	$+64\lambda_{X_{15}}\left(\lambda_{X_{15}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$	$+64\lambda_{X_{15}}\left(\lambda_{X_{15}}\right)^*$
B	−1	2	N/A	2	N/A	1	N/A	$\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$	—
B	−3	2	N/A	2	N/A	1	N/A	$\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$	—
C	1	2	N/A	2	N/A	3	N/A	$\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$	$+64\lambda_{X_{15}}\left(\lambda_{X_{15}}\right)^*$
C	−1	2	N/A	2	N/A	3	N/A	$\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$	—
C	−3	2	N/A	2	N/A	3	N/A	$\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$	—
D	0	3	N/A	1	N/A	2	N/A	—	$-\frac{8}{3}\lambda_5\ (\lambda_5)^*$	—
D	2	3	N/A	1	N/A	2	N/A	—	$-\frac{8}{3}\lambda_5\ (\lambda_5)^*$	—
D	−2	3	N/A	1	N/A	2	N/A	—	$-\frac{8}{3}\lambda_5\ (\lambda_5)^*$	—
E	0	3	N/A	3	N/A	2	N/A	—	$-\frac{4}{3}\lambda_5\ (\lambda_5)^*$	—
E	2	3	N/A	3	N/A	2	N/A	—	$-\frac{4}{3}\lambda_5\ (\lambda_5)^*$	—
E	−2	3	N/A	3	N/A	2	N/A	—	$-\frac{4}{3}\lambda_5\ (\lambda_5)^*$	—
E	−4	3	N/A	3	N/A	2	N/A	—	$-\frac{4}{3}\lambda_5\ (\lambda_5)^*$	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 2 of 7

Column	Representative term structure
T1	$+{g_1}^2\left(6{g_2}^2-3\lambda_{HS_1}^{(A)}\right)$
T2	$-9{g_2}^2\lambda_{HS_1}^{(A)}$
T3	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+{g_1}^2\left(6{g_2}^2-3\lambda_{HS_1}^{(A)}\right)$	$-9{g_2}^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
B	−1	2	N/A	2	N/A	1	N/A	—	$-9{g_2}^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
B	−3	2	N/A	2	N/A	1	N/A	—	$-9{g_2}^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
C	1	2	N/A	2	N/A	3	N/A	$+{g_1}^2\left(6{g_2}^2-3\lambda_{HS_1}^{(A)}\right)$	$-9{g_2}^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
C	−1	2	N/A	2	N/A	3	N/A	—	$-9{g_2}^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
C	−3	2	N/A	2	N/A	3	N/A	—	$-9{g_2}^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 3 of 7

Column	Representative term structure
T1	$+2\lambda_1\lambda_{HS_1}^{(A)}$
T2	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
T3	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
B	−1	2	N/A	2	N/A	1	N/A	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
B	−3	2	N/A	2	N/A	1	N/A	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
C	1	2	N/A	2	N/A	3	N/A	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
C	−1	2	N/A	2	N/A	3	N/A	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
C	−3	2	N/A	2	N/A	3	N/A	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
D	0	3	N/A	1	N/A	2	N/A	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
D	2	3	N/A	1	N/A	2	N/A	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
D	−2	3	N/A	1	N/A	2	N/A	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
E	0	3	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
E	2	3	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
E	−2	3	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
E	−4	3	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$

Quantity: $\lambda^{(A)}_{HS_1}$

Terms 4 of 7

Column	Representative term structure
T1	$-64\lambda_7\,(\lambda_7)^*$
T2	$+4\lambda_{11}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$
T3	$+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{11})^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$-64\lambda_7\,(\lambda_7)^*$	$+4\lambda_{11}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$	$+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{11})^*)$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$-64\lambda_7\,(\lambda_7)^*$	$+4\lambda_{11}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$	$+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{11})^*)$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(A)}_{HS_1}$

Terms 5 of 7

Column	Representative term structure
T1	$+4\lambda^{(1)}_{X_{17}}((\lambda_{11})^* - 2\left(\lambda^{(1)}_{X_{17}}\right)^*)$
T2	$+{g_1}^2\left(-6{g_2}^2 - 3\lambda^{(A)}_{HS_1}\right)$
T3	$\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_1})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+4\lambda^{(1)}_{X_{17}}((\lambda_{11})^* - 2\left(\lambda^{(1)}_{X_{17}}\right)^*)$	$+{g_1}^2\left(-6{g_2}^2 - 3\lambda^{(A)}_{HS_1}\right)$	—
B	−3	2	N/A	2	N/A	1	N/A	—	$+{g_1}^2\left(-18{g_2}^2 - 15\lambda^{(A)}_{HS_1}\right)$	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+4\lambda^{(1)}_{X_{17}}((\lambda_{11})^* - 2\left(\lambda^{(1)}_{X_{17}}\right)^*)$	$+{g_1}^2\left(-6{g_2}^2 - 3\lambda^{(A)}_{HS_1}\right)$	—
C	−3	2	N/A	2	N/A	3	N/A	—	$+{g_1}^2\left(-18{g_2}^2 - 15\lambda^{(A)}_{HS_1}\right)$	—
D	0	3	N/A	1	N/A	2	N/A	—	—	$\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_1})$
D	2	3	N/A	1	N/A	2	N/A	—	—	$\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_1})$
D	−2	3	N/A	1	N/A	2	N/A	—	—	$\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_1})$
E	0	3	N/A	3	N/A	2	N/A	—	—	$\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_1})$
E	2	3	N/A	3	N/A	2	N/A	—	—	$\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_1})$
E	−2	3	N/A	3	N/A	2	N/A	—	—	$\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_1})$
E	−4	3	N/A	3	N/A	2	N/A	—	—	$\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_1})$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 6 of 7

Column	Representative term structure
T1	$+\frac{8}{3}\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$
T2	$-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$
T3	$-\frac{3{g_1}^2\lambda_{HS_1}^{(A)}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+\frac{8}{3}\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$	$-\frac{3{g_1}^2\lambda_{HS_1}^{(A)}}{2}$
D	2	3	N/A	1	N/A	2	N/A	—	$-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$	—
D	−2	3	N/A	1	N/A	2	N/A	$-\frac{8}{3}\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	$-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$	—
E	0	3	N/A	3	N/A	2	N/A	$-\frac{8}{9}\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$	$-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$	$-\frac{3{g_1}^2\lambda_{HS_1}^{(A)}}{2}$
E	2	3	N/A	3	N/A	2	N/A	—	$-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$	—
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{4}{3}\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$	$-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$	—
E	−4	3	N/A	3	N/A	2	N/A	—	$-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 7 of 7

Column	Representative term structure
T1	$+{g_1}^2\left({4g_2}^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$
T2	$+{g_1}^2\left(-{4g_2}^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$
T3	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	$+{g_1}^2\left({4g_2}^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	$+{g_1}^2\left(-{4g_2}^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	$+{g_1}^2\left({4g_2}^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	—	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
E	-2	3	N/A	3	N/A	2	N/A	—	$+{g_1}^2\left(-{4g_2}^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
E	-4	3	N/A	3	N/A	2	N/A	—	$+{g_1}^2\left(-{8g_2}^2-\frac{51\lambda_{HS_1}^{(A)}}{2}\right)$	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$

$\beta^{(1)}_{\lambda^{(A)}_{HS_2}}$

Quantity: $\lambda^{(A)}_{HS_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{HS_2}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{HS_2}}$$

Terms 1 of 8

Column	Representative term structure
T1	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$
T2	$+\frac{8}{3}\lambda_5\,(\lambda_5)^*$
T3	$+\frac{8}{3}\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$	$+\frac{8}{3}\lambda_5\,(\lambda_5)^*$	$+\frac{8}{3}\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$
A	−2	1	N/A	3	N/A	2	N/A	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$	$+\frac{8}{3}\lambda_5\,(\lambda_5)^*$	$-\frac{8}{3}\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$
A	−4	1	N/A	3	N/A	2	N/A	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$	$+\frac{8}{3}\lambda_5\,(\lambda_5)^*$	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$	$-\frac{8}{9}\lambda_5\,(\lambda_5)^*$	$-\frac{8}{9}\lambda_{X_{14}}\left(\lambda_{X_{14}}\right)^*$
E	2	3	N/A	3	N/A	2	N/A	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$	$-\frac{8}{9}\lambda_5\,(\lambda_5)^*$	—
E	−2	3	N/A	3	N/A	2	N/A	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$	$-\frac{8}{9}\lambda_5\,(\lambda_5)^*$	$-\frac{4}{3}\lambda_{X_{12}}\left(\lambda_{X_{12}}\right)^*$
E	−4	3	N/A	3	N/A	2	N/A	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$	$-\frac{8}{9}\lambda_5\,(\lambda_5)^*$	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 2 of 8

Column	Representative term structure
T1	$+{g_1}^2\left({4{g_2}^2}-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$
T2	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$
T3	$+2\lambda_1\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+{g_1}^2\left({4{g_2}^2}-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$
A	-2	1	N/A	3	N/A	2	N/A	—	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$
A	-4	1	N/A	3	N/A	2	N/A	—	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$
B	1	2	N/A	2	N/A	1	N/A	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
B	-1	2	N/A	2	N/A	1	N/A	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
B	-3	2	N/A	2	N/A	1	N/A	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
C	1	2	N/A	2	N/A	3	N/A	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
C	-1	2	N/A	2	N/A	3	N/A	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
C	-3	2	N/A	2	N/A	3	N/A	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+{g_1}^2\left({4{g_2}^2}-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	$+{g_1}^2\left({8{g_2}^2}-\frac{51\lambda_{HS_2}^{(A)}}{2}\right)$	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$
E	-2	3	N/A	3	N/A	2	N/A	—	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$
E	-4	3	N/A	3	N/A	2	N/A	—	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 3 of 8

Column	Representative term structure
T1	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T2	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$
T3	$-\frac{3{g_1}^2\lambda_{HS_2}^{(A)}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
A	−2	1	N/A	3	N/A	2	N/A	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	$-\frac{3{g_1}^2\lambda_{HS_2}^{(A)}}{2}$
A	−4	1	N/A	3	N/A	2	N/A	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
B	1	2	N/A	2	N/A	1	N/A	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
B	−1	2	N/A	2	N/A	1	N/A	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
B	−3	2	N/A	2	N/A	1	N/A	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
C	1	2	N/A	2	N/A	3	N/A	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
C	−1	2	N/A	2	N/A	3	N/A	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
C	−3	2	N/A	2	N/A	3	N/A	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
E	2	3	N/A	3	N/A	2	N/A	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
E	−2	3	N/A	3	N/A	2	N/A	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	$-\frac{3{g_1}^2\lambda_{HS_2}^{(A)}}{2}$
E	−4	3	N/A	3	N/A	2	N/A	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—

Quantity: $\lambda^{(A)}_{HS_2}$

Terms 4 of 8

Column	Representative term structure
T1	$+{g_1}^2\left(-4{g_2}^2-\frac{15\lambda^{(A)}_{HS_2}}{2}\right)$
T2	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T3	$+16\lambda_5\ (\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	$+{g_1}^2\left(-4{g_2}^2-\frac{15\lambda^{(A)}_{HS_2}}{2}\right)$	—	—
B	1	2	N/A	2	N/A	1	N/A	—	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+16\lambda_5\ (\lambda_5)^*$
B	−1	2	N/A	2	N/A	1	N/A	—	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+16\lambda_5\ (\lambda_5)^*$
B	−3	2	N/A	2	N/A	1	N/A	—	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+16\lambda_5\ (\lambda_5)^*$
C	1	2	N/A	2	N/A	3	N/A	—	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+16\lambda_5\ (\lambda_5)^*$
C	−1	2	N/A	2	N/A	3	N/A	—	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+16\lambda_5\ (\lambda_5)^*$
C	−3	2	N/A	2	N/A	3	N/A	—	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+16\lambda_5\ (\lambda_5)^*$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	$+{g_1}^2\left(-4{g_2}^2-\frac{15\lambda^{(A)}_{HS_2}}{2}\right)$	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 5 of 8

Column	Representative term structure
T1	$+{g_1}^2\left(18{g_2}^2-15\lambda_{HS_2}^{(A)}\right)$
T2	$-9{g_2}^2\lambda_{HS_2}^{(A)}$
T3	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+{g_1}^2\left(18{g_2}^2-15\lambda_{HS_2}^{(A)}\right)$	$-9{g_2}^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
B	−1	2	N/A	2	N/A	1	N/A	$+{g_1}^2\left(6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$	$-9{g_2}^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
B	−3	2	N/A	2	N/A	1	N/A	—	$-9{g_2}^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
C	1	2	N/A	2	N/A	3	N/A	$+{g_1}^2\left(18{g_2}^2-15\lambda_{HS_2}^{(A)}\right)$	$-9{g_2}^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
C	−1	2	N/A	2	N/A	3	N/A	$+{g_1}^2\left(6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$	$-9{g_2}^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
C	−3	2	N/A	2	N/A	3	N/A	—	$-9{g_2}^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(A)}_{HS_2}$

Terms 6 of 8

Column	Representative term structure
T1	$-4\lambda_{10}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$
T2	$-4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{10})^*)$
T3	$+64\lambda_8\,(\lambda_8)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$-4\lambda_{10}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$	$-4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{10})^*)$	$+64\lambda_8\,(\lambda_8)^*$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$-4\lambda_{10}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\lambda^{(1)}_{X_{17}}\right)^*)$	$-4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{10})^*)$	$+64\lambda_8\,(\lambda_8)^*$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 7 of 8

Column	Representative term structure
T1	$+\lambda_{X_{17}}^{(1)}(8\left(\lambda_{X_{17}}^{(1)}\right)^{*}-4\left(\lambda_{10}\right)^{*})$
T2	$-64\lambda_{X_{10}}\left(\lambda_{X_{10}}\right)^{*}$
T3	$+{g_1}^2\left(-6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+\lambda_{X_{17}}^{(1)}(8\left(\lambda_{X_{17}}^{(1)}\right)^{*}-4\left(\lambda_{10}\right)^{*})$	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	$-64\lambda_{X_{10}}\left(\lambda_{X_{10}}\right)^{*}$	$+{g_1}^2\left(-6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+\lambda_{X_{17}}^{(1)}(8\left(\lambda_{X_{17}}^{(1)}\right)^{*}-4\left(\lambda_{10}\right)^{*})$	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	$-64\lambda_{X_{10}}\left(\lambda_{X_{10}}\right)^{*}$	$+{g_1}^2\left(-6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 8 of 8

Column	Representative term structure
T1	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	-2	1	N/A	3	N/A	2	N/A	—
A	-4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	-1	2	N/A	2	N/A	1	N/A	—
B	-3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	-1	2	N/A	2	N/A	3	N/A	—
C	-3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	-2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
E	-2	3	N/A	3	N/A	2	N/A	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
E	-4	3	N/A	3	N/A	2	N/A	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$

$\beta^{(1)}_{\lambda_7}$

Quantity: λ_7

$$16\pi^2 \, \mu \frac{d\lambda_7}{d\mu} = \beta^{(1)}_{\lambda_7}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_7(6 \operatorname{Tr}\left(Y_d \ (Y_d)^\dagger\right) + 2 \operatorname{Tr}\left(Y_e \ (Y_e)^\dagger\right) + 6 \operatorname{Tr}\left(Y_u \ (Y_u)^\dagger\right))$
T2	$-9{g_2}^2\lambda_7$
T3	$-3{g_1}^2\lambda_7$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$\lambda_7(6 \operatorname{Tr}\left(Y_d \ (Y_d)^\dagger\right) + 2 \operatorname{Tr}\left(Y_e \ (Y_e)^\dagger\right) + 6 \operatorname{Tr}\left(Y_u \ (Y_u)^\dagger\right))$	$-9{g_2}^2\lambda_7$	$-3{g_1}^2\lambda_7$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$\lambda_7(6 \operatorname{Tr}\left(Y_d \ (Y_d)^\dagger\right) + 2 \operatorname{Tr}\left(Y_e \ (Y_e)^\dagger\right) + 6 \operatorname{Tr}\left(Y_u \ (Y_u)^\dagger\right))$	$-9{g_2}^2\lambda_7$	$-3{g_1}^2\lambda_7$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_7

Terms 2 of 3

Column	Representative term structure
T1	$+2\lambda_1\lambda_7$
T2	$+8\lambda^{(1)}_{HS_1}\lambda_7$
T3	$-4\lambda^{(A)}_{HS_1}\lambda_7$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+2\lambda_1\lambda_7$	$+8\lambda^{(1)}_{HS_1}\lambda_7$	$-4\lambda^{(A)}_{HS_1}\lambda_7$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+2\lambda_1\lambda_7$	$+8\lambda^{(1)}_{HS_1}\lambda_7$	$-4\lambda^{(A)}_{HS_1}\lambda_7$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_7

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda_7\lambda_{S_1}^{(1)}$
T2	$+4\lambda_8\lambda_9$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+2\lambda_7\lambda_{S_1}^{(1)}$	$+4\lambda_8\lambda_9$
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+2\lambda_7\lambda_{S_1}^{(1)}$	$+4\lambda_8\lambda_9$
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda_{12}}$

Quantity: λ_{12}

$$16\pi^2\,\mu\frac{d\lambda_{12}}{d\mu} = \beta^{(1)}_{\lambda_{12}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+\lambda_9(8\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$
T3	$-9{g_2}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\lambda_9(8\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$	$-9{g_2}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\lambda_9(8\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 4\left(\lambda^{(1)}_{X_{17}}\right)^*)$	$-9{g_2}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{12}

Terms 2 of 4

Column	Representative term structure
T1	$-3{g_1}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$
T2	$+2\lambda_{12}^{(1)}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$
T3	$+\lambda_{12}^{(A)}\left(\lambda^{(1)}_{X_{17}} - \frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$-3{g_1}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	$+2\lambda_{12}^{(1)}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	$+\lambda_{12}^{(A)}\left(\lambda^{(1)}_{X_{17}} - \frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}\right)$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$-3{g_1}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	$+2\lambda_{12}^{(1)}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	$+\lambda_{12}^{(A)}\left(\lambda^{(1)}_{X_{17}} - \frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}\right)$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{12}

Terms 3 of 4

Column	Representative term structure
T1	$+\lambda_1(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - 2\lambda^{(1)}_{X_{17}})$
T2	$+\lambda^{(1)}_{HS_1}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 6\lambda_{11})$
T3	$+\lambda^{(A)}_{HS_1}\left(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + \frac{\lambda_{11}}{2} + \lambda^{(1)}_{X_{17}}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+\lambda_1(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - 2\lambda^{(1)}_{X_{17}})$	$+\lambda^{(1)}_{HS_1}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 6\lambda_{11})$	$+\lambda^{(A)}_{HS_1}\left(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + \frac{\lambda_{11}}{2} + \lambda^{(1)}_{X_{17}}\right)$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+\lambda_1(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - 2\lambda^{(1)}_{X_{17}})$	$+\lambda^{(1)}_{HS_1}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 6\lambda_{11})$	$+\lambda^{(A)}_{HS_1}\left(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + \frac{\lambda_{11}}{2} + \lambda^{(1)}_{X_{17}}\right)$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{12}

Terms 4 of 4

Column	Representative term structure
T1	$+\lambda^{(1)}_{HS_2}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{10})$
T2	$+\lambda^{(A)}_{HS_2}\left(-2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}-\lambda^{(1)}_{X_{17}}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+\lambda^{(1)}_{HS_2}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{10})$	$+\lambda^{(A)}_{HS_2}\left(-2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}-\lambda^{(1)}_{X_{17}}\right)$
B	-3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+\lambda^{(1)}_{HS_2}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{10})$	$+\lambda^{(A)}_{HS_2}\left(-2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}-\lambda^{(1)}_{X_{17}}\right)$
C	-3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda_8}$

Quantity: λ_8

$$16\pi^2\,\mu\frac{d\lambda_8}{d\mu} = \beta^{(1)}_{\lambda_8}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_8(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+4\lambda_7\ (\lambda_9)^*$
T3	$-9{g_2}^2\lambda_8$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$\lambda_8(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+4\lambda_7\ (\lambda_9)^*$	$-9{g_2}^2\lambda_8$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$\lambda_8(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+4\lambda_7\ (\lambda_9)^*$	$-9{g_2}^2\lambda_8$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_8

Terms 2 of 3

Column	Representative term structure
T1	$-3{g_1}^2\lambda_8$
T2	$+2\lambda_1\lambda_8$
T3	$+8\lambda^{(1)}_{HS_2}\lambda_8$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$-3{g_1}^2\lambda_8$	$+2\lambda_1\lambda_8$	$+8\lambda^{(1)}_{HS_2}\lambda_8$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$-3{g_1}^2\lambda_8$	$+2\lambda_1\lambda_8$	$+8\lambda^{(1)}_{HS_2}\lambda_8$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_8

Terms 3 of 3

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(A)}\lambda_8$
T2	$+2\lambda_8\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+4\lambda_{HS_2}^{(A)}\lambda_8$	$+2\lambda_8\lambda_{S_2}^{(1)}$
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+4\lambda_{HS_2}^{(A)}\lambda_8$	$+2\lambda_8\lambda_{S_2}^{(1)}$
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda^{(A)}_{S_1}}$

Quantity: $\lambda^{(A)}_{S_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{S_1}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{S_1}}$$

Terms 1 of 5

Column	Representative term structure
T1	$-\frac{4}{3}\lambda^{(1)}_{X_5}\left(\lambda^{(1)}_{X_5}\right)^*$
T2	$-\frac{224}{3}\lambda_{X_6}\left(\lambda_{X_6}\right)^*$
T3	$-\frac{44}{3}\lambda_{X_9}\left(\lambda_{X_9}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$-\frac{4}{3}\lambda^{(1)}_{X_5}\left(\lambda^{(1)}_{X_5}\right)^*$	$-\frac{224}{3}\lambda_{X_6}\left(\lambda_{X_6}\right)^*$	$-\frac{44}{3}\lambda_{X_9}\left(\lambda_{X_9}\right)^*$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-\frac{4}{3}\lambda^{(2)}_{X_5}\left(\lambda^{(1)}_{X_5}\right)^*$	$-\frac{224}{3}\lambda_{X_6}\left(\lambda_{X_6}\right)^*$	$-\frac{44}{3}\lambda_{X_9}\left(\lambda_{X_9}\right)^*$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_1}^{(A)}$

Terms 2 of 5

Column	Representative term structure
T1	$-\frac{8}{3}\lambda_{X_{18}}\left(\lambda_{X_{18}}\right)^{*}$
T2	$-2{g_2}^4$
T3	$-24{g_2}^2\lambda_{S_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$-\frac{8}{3}\lambda_{X_{18}}\left(\lambda_{X_{18}}\right)^{*}$	$-2{g_2}^4$	$-24{g_2}^2\lambda_{S_1}^{(A)}$
D	2	3	N/A	1	N/A	2	N/A	—	$-2{g_2}^4$	$-24{g_2}^2\lambda_{S_1}^{(A)}$
D	-2	3	N/A	1	N/A	2	N/A	—	$-2{g_2}^4$	$-24{g_2}^2\lambda_{S_1}^{(A)}$
E	0	3	N/A	3	N/A	2	N/A	$-\frac{8}{3}\lambda_{X_{18}}\left(\lambda_{X_{18}}\right)^{*}$	$-2{g_2}^4$	$-24{g_2}^2\lambda_{S_1}^{(A)}$
E	2	3	N/A	3	N/A	2	N/A	—	$-2{g_2}^4$	$-24{g_2}^2\lambda_{S_1}^{(A)}$
E	-2	3	N/A	3	N/A	2	N/A	—	$-2{g_2}^4$	$-24{g_2}^2\lambda_{S_1}^{(A)}$
E	-4	3	N/A	3	N/A	2	N/A	—	$-2{g_2}^4$	$-24{g_2}^2\lambda_{S_1}^{(A)}$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 3 of 5

Column	Representative term structure
T1	$+3{\lambda_{HS_1}^{(A)}}^2$
T2	$+12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$
T3	$+20{\lambda_{S_1}^{(A)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+3{\lambda_{HS_1}^{(A)}}^2$	$+12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+20{\lambda_{S_1}^{(A)}}^2$
D	2	3	N/A	1	N/A	2	N/A	$+3{\lambda_{HS_1}^{(A)}}^2$	$+12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+20{\lambda_{S_1}^{(A)}}^2$
D	−2	3	N/A	1	N/A	2	N/A	$+3{\lambda_{HS_1}^{(A)}}^2$	$+12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+20{\lambda_{S_1}^{(A)}}^2$
E	0	3	N/A	3	N/A	2	N/A	$+3{\lambda_{HS_1}^{(A)}}^2$	$+12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+20{\lambda_{S_1}^{(A)}}^2$
E	2	3	N/A	3	N/A	2	N/A	$+3{\lambda_{HS_1}^{(A)}}^2$	$+12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+20{\lambda_{S_1}^{(A)}}^2$
E	−2	3	N/A	3	N/A	2	N/A	$+3{\lambda_{HS_1}^{(A)}}^2$	$+12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+20{\lambda_{S_1}^{(A)}}^2$
E	−4	3	N/A	3	N/A	2	N/A	$+3{\lambda_{HS_1}^{(A)}}^2$	$+12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+20{\lambda_{S_1}^{(A)}}^2$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 4 of 5

Column	Representative term structure
T1	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_1}^{(A)}\right)$
T2	$-\frac{4}{3}\lambda_{X_5}^{(1)}(3\left(\lambda_{X_5}^{(1)}\right)^*+\left(\lambda_{X_5}^{(2)}\right)^*)$
T3	$+12{\lambda_{12}^{(A)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_1}^{(A)}\right)$	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_1}^{(A)}\right)$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$-\frac{4}{3}\lambda_{X_5}^{(1)}(3\left(\lambda_{X_5}^{(1)}\right)^*+\left(\lambda_{X_5}^{(2)}\right)^*)$	$+12{\lambda_{12}^{(A)}}^2$
E	2	3	N/A	3	N/A	2	N/A	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_1}^{(A)}\right)$	—	$+12{\lambda_{12}^{(A)}}^2$
E	−2	3	N/A	3	N/A	2	N/A	$+{g_1}^2\left(8{g_2}^2-12\lambda_{S_1}^{(A)}\right)$	—	$+12{\lambda_{12}^{(A)}}^2$
E	−4	3	N/A	3	N/A	2	N/A	$+{g_1}^2\left(32{g_2}^2-48\lambda_{S_1}^{(A)}\right)$	—	$+12{\lambda_{12}^{(A)}}^2$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 5 of 5

Column	Representative term structure
T1	$-\frac{4\lambda_{12}^{(\times)\,2}}{3}$
T2	$-\frac{4}{3}\lambda_{X_7}^{(2)}\left(\lambda_{X_7}^{(2)}\right)^{*}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	$-\frac{4\lambda_{12}^{(\times)\,2}}{3}$	—
E	2	3	N/A	3	N/A	2	N/A	$-\frac{4\lambda_{12}^{(\times)\,2}}{3}$	—
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{4\lambda_{12}^{(\times)\,2}}{3}$	$-\frac{4}{3}\lambda_{X_7}^{(2)}\left(\lambda_{X_7}^{(2)}\right)^{*}$
E	−4	3	N/A	3	N/A	2	N/A	$-\frac{4\lambda_{12}^{(\times)\,2}}{3}$	—

$\beta^{(1)}_{\lambda_{11}}$

Quantity: λ_{11}

$$16\pi^2\,\mu\frac{d\lambda_{11}}{d\mu} = \beta^{(1)}_{\lambda_{11}}$$

Terms 1 of 3

Column	Representative term structure
T1	$4\lambda_9((\lambda_{10})^* + 5\,(\lambda_{11})^*)$
T2	$-9{g_2}^2\lambda_{11}$
T3	$-3{g_1}^2\lambda_{11}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$4\lambda_9((\lambda_{10})^* + 5\,(\lambda_{11})^*)$	$-9{g_2}^2\lambda_{11}$	$-3{g_1}^2\lambda_{11}$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$4\lambda_9((\lambda_{10})^* + 5\,(\lambda_{11})^*)$	$-9{g_2}^2\lambda_{11}$	$-3{g_1}^2\lambda_{11}$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{11}

Terms 2 of 3

Column	Representative term structure
T1	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$
T2	$+\lambda_{12}^{(A)}\left(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2}\right)$
T3	$+\lambda_{HS_1}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-2\lambda_{X_{17}}^{(1)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2}\right)$	$+\lambda_{HS_1}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-2\lambda_{X_{17}}^{(1)})$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2}\right)$	$+\lambda_{HS_1}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-2\lambda_{X_{17}}^{(1)})$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{11}

Terms 3 of 3

Column	Representative term structure
T1	$+\lambda^{(A)}_{HS_1}\left(\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}+\frac{\lambda^{(1)}_{X_{17}}}{2}\right)$
T2	$+12\lambda^{(1)}_{S_1}\lambda_{11}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+\lambda^{(A)}_{HS_1}\left(\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}+\frac{\lambda^{(1)}_{X_{17}}}{2}\right)$	$+12\lambda^{(1)}_{S_1}\lambda_{11}$
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+\lambda^{(A)}_{HS_1}\left(\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}+\frac{\lambda^{(1)}_{X_{17}}}{2}\right)$	$+12\lambda^{(1)}_{S_1}\lambda_{11}$
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda_9}$

Quantity: λ_9

$$16\pi^2\,\mu\frac{d\lambda_9}{d\mu} = \beta^{(1)}_{\lambda_9}$$

Terms 1 of 4

Column	Representative term structure
T1	$4\lambda_7\,(\lambda_8)^*$
T2	$-9{g_2}^2\lambda_9$
T3	$-3{g_1}^2\lambda_9$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$4\lambda_7\,(\lambda_8)^*$	$-9{g_2}^2\lambda_9$	$-3{g_1}^2\lambda_9$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$4\lambda_7\,(\lambda_8)^*$	$-9{g_2}^2\lambda_9$	$-3{g_1}^2\lambda_9$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_9

Terms 2 of 4

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_9$
T2	$-4\lambda_{12}^{(A)}\lambda_9$
T3	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}\,2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+8\lambda_{12}^{(1)}\lambda_9$	$-4\lambda_{12}^{(A)}\lambda_9$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}\,2}$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+8\lambda_{12}^{(1)}\lambda_9$	$-4\lambda_{12}^{(A)}\lambda_9$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}\,2}$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_9

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda_{S_1}^{(1)}\lambda_9$
T2	$+2\lambda_9\lambda_{S_2}^{(1)}$
T3	$+5{\lambda_{10}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+2\lambda_{S_1}^{(1)}\lambda_9$	$+2\lambda_9\lambda_{S_2}^{(1)}$	$+5{\lambda_{10}}^2$ $+5{\lambda_{11}}^2$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+2\lambda_{S_1}^{(1)}\lambda_9$	$+2\lambda_9\lambda_{S_2}^{(1)}$	$+5{\lambda_{10}}^2$ $+5{\lambda_{11}}^2$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_9

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda_{10}\lambda_{11}$
T2	$+\lambda_{X_{17}^{(1)}}^{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+2\lambda_{10}\lambda_{11}$	$+\lambda_{X_{17}^{(1)}}^{2}$
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+2\lambda_{10}\lambda_{11}$	$+\lambda_{X_{17}^{(1)}}^{2}$
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda_{10}}$

Quantity: λ_{10}

$$16\pi^2\,\mu\frac{d\lambda_{10}}{d\mu} = \beta^{(1)}_{\lambda_{10}}$$

Terms 1 of 3

Column	Representative term structure
T1	$4\lambda_9(5\,(\lambda_{10})^* + (\lambda_{11})^*)$
T2	$-9{g_2}^2\lambda_{10}$
T3	$-3{g_1}^2\lambda_{10}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$4\lambda_9(5\,(\lambda_{10})^* + (\lambda_{11})^*)$	$-9{g_2}^2\lambda_{10}$	$-3{g_1}^2\lambda_{10}$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$4\lambda_9(5\,(\lambda_{10})^* + (\lambda_{11})^*)$	$-9{g_2}^2\lambda_{10}$	$-3{g_1}^2\lambda_{10}$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{10}

Terms 2 of 3

Column	Representative term structure
T1	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$
T2	$+\lambda_{12}^{(A)}\left(-\frac{5\lambda_{10}}{2}-\frac{\lambda_{11}}{2}\right)$
T3	$+\lambda_{HS_2}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-2\lambda_{X_{17}}^{(1)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(-\frac{5\lambda_{10}}{2}-\frac{\lambda_{11}}{2}\right)$	$+\lambda_{HS_2}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-2\lambda_{X_{17}}^{(1)})$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(-\frac{5\lambda_{10}}{2}-\frac{\lambda_{11}}{2}\right)$	$+\lambda_{HS_2}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-2\lambda_{X_{17}}^{(1)})$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{10}

Terms 3 of 3

Column	Representative term structure
T1	$+\lambda_{HS_2}^{(A)}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{X_{17}}^{(1)}}{2}\right)$
T2	$+12\lambda_{10}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+\lambda_{HS_2}^{(A)}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{X_{17}}^{(1)}}{2}\right)$	$+12\lambda_{10}\lambda_{S_2}^{(1)}$
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+\lambda_{HS_2}^{(A)}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{X_{17}}^{(1)}}{2}\right)$	$+12\lambda_{10}\lambda_{S_2}^{(1)}$
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda^{(A)}_{S_2}}$

Quantity: $\lambda^{(A)}_{S_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{S_2}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{S_2}}$$

Terms 1 of 5

Column	Representative term structure
T1	$-2{g_2}^4$
T2	$+{g_1}^2\left(8{g_2}^2 - 12\lambda^{(A)}_{S_2}\right)$
T3	$-24{g_2}^2\lambda^{(A)}_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-2{g_2}^4$	$+{g_1}^2\left(8{g_2}^2 - 12\lambda^{(A)}_{S_2}\right)$	$-24{g_2}^2\lambda^{(A)}_{S_2}$
A	−2	1	N/A	3	N/A	2	N/A	$-2{g_2}^4$	—	$-24{g_2}^2\lambda^{(A)}_{S_2}$
A	−4	1	N/A	3	N/A	2	N/A	$-2{g_2}^4$	$+{g_1}^2\left(8{g_2}^2 - 12\lambda^{(A)}_{S_2}\right)$	$-24{g_2}^2\lambda^{(A)}_{S_2}$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-2{g_2}^4$	$+{g_1}^2\left(8{g_2}^2 - 12\lambda^{(A)}_{S_2}\right)$	$-24{g_2}^2\lambda^{(A)}_{S_2}$
E	2	3	N/A	3	N/A	2	N/A	$-2{g_2}^4$	$+{g_1}^2\left(32{g_2}^2 - 48\lambda^{(A)}_{S_2}\right)$	$-24{g_2}^2\lambda^{(A)}_{S_2}$
E	−2	3	N/A	3	N/A	2	N/A	$-2{g_2}^4$	—	$-24{g_2}^2\lambda^{(A)}_{S_2}$
E	−4	3	N/A	3	N/A	2	N/A	$-2{g_2}^4$	$+{g_1}^2\left(8{g_2}^2 - 12\lambda^{(A)}_{S_2}\right)$	$-24{g_2}^2\lambda^{(A)}_{S_2}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 2 of 5

Column	Representative term structure
T1	$+3{\lambda_{HS_2}^{(A)}}^2$
T2	$+12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
T3	$+20{\lambda_{S_2}^{(A)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+3{\lambda_{HS_2}^{(A)}}^2$	$+12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	$+20{\lambda_{S_2}^{(A)}}^2$
A	-2	1	N/A	3	N/A	2	N/A	$+3{\lambda_{HS_2}^{(A)}}^2$	$+12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	$+20{\lambda_{S_2}^{(A)}}^2$
A	-4	1	N/A	3	N/A	2	N/A	$+3{\lambda_{HS_2}^{(A)}}^2$	$+12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	$+20{\lambda_{S_2}^{(A)}}^2$
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+3{\lambda_{HS_2}^{(A)}}^2$	$+12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	$+20{\lambda_{S_2}^{(A)}}^2$
E	2	3	N/A	3	N/A	2	N/A	$+3{\lambda_{HS_2}^{(A)}}^2$	$+12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	$+20{\lambda_{S_2}^{(A)}}^2$
E	-2	3	N/A	3	N/A	2	N/A	$+3{\lambda_{HS_2}^{(A)}}^2$	$+12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	$+20{\lambda_{S_2}^{(A)}}^2$
E	-4	3	N/A	3	N/A	2	N/A	$+3{\lambda_{HS_2}^{(A)}}^2$	$+12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	$+20{\lambda_{S_2}^{(A)}}^2$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 3 of 5

Column	Representative term structure
T1	$-\frac{224}{3}\lambda_{X_1}\left(\lambda_{X_1}\right)^*$
T2	$-\frac{44}{3}\lambda_{X_2}\left(\lambda_{X_2}\right)^*$
T3	$-\frac{4}{3}\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(1)}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$-\frac{224}{3}\lambda_{X_1}\left(\lambda_{X_1}\right)^*$	$-\frac{44}{3}\lambda_{X_2}\left(\lambda_{X_2}\right)^*$	$-\frac{4}{3}\lambda_{X_7}^{(1)}\left(\lambda_{X_7}^{(1)}\right)^*$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{224}{3}\lambda_{X_1}\left(\lambda_{X_1}\right)^*$	$-\frac{44}{3}\lambda_{X_2}\left(\lambda_{X_2}\right)^*$	$-\frac{4}{3}\lambda_{X_7}^{(2)}\left(\lambda_{X_7}^{(1)}\right)^*$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{S_2}^{(A)}$

Terms 4 of 5

Column	Representative term structure
T1	$-\frac{8}{3}\lambda_{X_{16}}\left(\lambda_{X_{16}}\right)^*$
T2	$-\frac{4}{3}\lambda_{X_5}^{(2)}\left(\lambda_{X_5}^{(2)}\right)^*$
T3	$+12{\lambda_{12}^{(A)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$-\frac{8}{3}\lambda_{X_{16}}\left(\lambda_{X_{16}}\right)^*$	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$-\frac{4}{3}\lambda_{X_5}^{(2)}\left(\lambda_{X_5}^{(2)}\right)^*$	$+12{\lambda_{12}^{(A)}}^2$
E	2	3	N/A	3	N/A	2	N/A	—	—	$+12{\lambda_{12}^{(A)}}^2$
E	-2	3	N/A	3	N/A	2	N/A	$-\frac{8}{3}\lambda_{X_{16}}\left(\lambda_{X_{16}}\right)^*$	—	$+12{\lambda_{12}^{(A)}}^2$
E	-4	3	N/A	3	N/A	2	N/A	—	—	$+12{\lambda_{12}^{(A)}}^2$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 5 of 5

Column	Representative term structure
T1	$-\frac{4\lambda_{12}^{(\times)\,2}}{3}$
T2	$-\frac{4}{3}\lambda_{X_7}^{(1)}(3\left(\lambda_{X_7}^{(1)}\right)^{*}+\left(\lambda_{X_7}^{(2)}\right)^{*})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	$-\frac{4\lambda_{12}^{(\times)\,2}}{3}$	—
E	2	3	N/A	3	N/A	2	N/A	$-\frac{4\lambda_{12}^{(\times)\,2}}{3}$	—
E	−2	3	N/A	3	N/A	2	N/A	$-\frac{4\lambda_{12}^{(\times)\,2}}{3}$	$-\frac{4}{3}\lambda_{X_7}^{(1)}(3\left(\lambda_{X_7}^{(1)}\right)^{*}+\left(\lambda_{X_7}^{(2)}\right)^{*})$
E	−4	3	N/A	3	N/A	2	N/A	$-\frac{4\lambda_{12}^{(\times)\,2}}{3}$	—

$\beta^{(1)}_{\lambda_{X_1}}$

Quantity: λ_{X_1}

$$16\pi^2\,\mu\frac{d\lambda_{X_1}}{d\mu} = \beta^{(1)}_{\lambda_{X_1}}$$

Terms 1 of 3

Column	Representative term structure
T1	$-24{g_2}^2\lambda_{X_1}$
T2	$+12\lambda^{(1)}_{S_2}\lambda_{X_1}$
T3	$-8\lambda^{(A)}_{S_2}\lambda_{X_1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{X_1}$	$+12\lambda^{(1)}_{S_2}\lambda_{X_1}$	$-8\lambda^{(A)}_{S_2}\lambda_{X_1}$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	$12\lambda^{(1)}_{S_2}\lambda_{X_1}$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$-24{g_2}^2\lambda_{X_1}$	$+12\lambda^{(1)}_{S_2}\lambda_{X_1}$	$-8\lambda^{(A)}_{S_2}\lambda_{X_1}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_1}

Terms 2 of 3

Column	Representative term structure
T1	$+11{\lambda_{X_2}}^2$
T2	$+{\lambda_{X_7}^{(1)}}^2$
T3	$+2{\lambda_{X_{16}}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+11{\lambda_{X_2}}^2$	$+{\lambda_{X_7}^{(1)}}^2$	$+2{\lambda_{X_{16}}}^2$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+9{\lambda_{X_2}}^2$	—	$+2{\lambda_{X_{16}}}^2$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+11{\lambda_{X_2}}^2$	$+{\lambda_{X_7}^{(2)}}^2$	$+2{\lambda_{X_{16}}}^2$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_1}

Terms 3 of 3

Column	Representative term structure
T1	$+3{\lambda_{X_7}^{(1)}}^2$
T2	$+2\lambda_{X_7}^{(1)}\lambda_{X_7}^{(2)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+3{\lambda_{X_7}^{(1)}}^2$	—
E	0	3	N/A	3	N/A	2	N/A	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+3{\lambda_{X_7}^{(1)}}^2$	$+2\lambda_{X_7}^{(1)}\lambda_{X_7}^{(2)}$
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda_{X_2}}$

Quantity: λ_{X_2}

$$16\pi^2\,\mu\frac{d\lambda_{X_2}}{d\mu} = \beta^{(1)}_{\lambda_{X_2}}$$

Terms 1 of 3

Column	Representative term structure
T1	$88\lambda_{X_1}\left(\lambda_{X_2}\right)^*$
T2	$-24{g_2}^2\lambda_{X_2}$
T3	$+2\lambda^{(1)}_{12}\lambda^{(1)}_{X_7}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$88\lambda_{X_1}\left(\lambda_{X_2}\right)^*$	$-24{g_2}^2\lambda_{X_2}$	$+2\lambda^{(1)}_{12}\lambda^{(1)}_{X_7}$
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$72\lambda_{X_1}\left(\lambda_{X_2}\right)^*$	—	$+6\lambda^{(1)}_{12}\lambda^{(1)}_{X_7}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$88\lambda_{X_1}\left(\lambda_{X_2}\right)^*$	$-24{g_2}^2\lambda_{X_2}$	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_2}

Terms 2 of 3

Column	Representative term structure
T1	$+4\lambda^{(1)}_{HS_2}\lambda_{X16}$
T2	$+22\lambda^{(1)}_{S_2}\lambda_{X_2}$
T3	$+\lambda^{(1)}_{12}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+4\lambda^{(1)}_{HS_2}\lambda_{X16}$	$+22\lambda^{(1)}_{S_2}\lambda_{X_2}$	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+4\lambda^{(1)}_{HS_2}\lambda_{X16}$	$+18\lambda^{(1)}_{S_2}\lambda_{X_2}$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+4\lambda^{(1)}_{HS_2}\lambda_{X16}$	$+22\lambda^{(1)}_{S_2}\lambda_{X_2}$	$+\lambda^{(1)}_{12}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_2}

Terms 3 of 3

Column	Representative term structure
T1	$+\lambda_{12}^{(\times)}(4\lambda_{X_7}^{(1)} + 4\lambda_{X_7}^{(2)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	$+\lambda_{12}^{(\times)}(4\lambda_{X_7}^{(1)} + 4\lambda_{X_7}^{(2)})$
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda^{(1)}_{X_5}}$

Quantity: $\lambda^{(1)}_{X_5}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{X_5}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{X_5}}$$

Terms 1 of 5

Column	Representative term structure
T1	$4\,(\lambda_5)^*\left(\lambda_{X_{14}}\right)^*$
T2	$+24\lambda_{X_6}\left(\lambda^{(1)}_{X_5}\right)^*$
T3	$-12{g_2}^2\lambda^{(1)}_{X_5}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$4\,(\lambda_5)^*\left(\lambda_{X_{14}}\right)^*$	$+24\lambda_{X_6}\left(\lambda^{(1)}_{X_5}\right)^*$	$-12{g_2}^2\lambda^{(1)}_{X_5}$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$4\,(\lambda_5)^*\left(\lambda_{X_{14}}\right)^*$	$+40\lambda_{X_6}\left(\lambda^{(1)}_{X_5}\right)^*$	$-12{g_2}^2\lambda^{(1)}_{X_5}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$2\,(\lambda_5)^*\left(\lambda_{X_{14}}\right)^*$	—	$-24{g_2}^2\lambda^{(1)}_{X_5}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{X_5}$

Terms 2 of 5

Column	Representative term structure
T1	$-6{g_1}^2\lambda^{(1)}_{X_5}$
T2	$+\lambda^{(1)}_{12}(8\lambda^{(1)}_{X_5}+6\lambda_{X_9})$
T3	$+4\lambda^{(1)}_{HS_2}\lambda_{X_{18}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-6{g_1}^2\lambda^{(1)}_{X_5}$	$+\lambda^{(1)}_{12}(8\lambda^{(1)}_{X_5}+6\lambda_{X_9})$	$+4\lambda^{(1)}_{HS_2}\lambda_{X_{18}}$
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$-6{g_1}^2\lambda^{(1)}_{X_5}$	$+\lambda^{(1)}_{12}(8\lambda^{(1)}_{X_5}+10\lambda_{X_9})$	$+4\lambda^{(1)}_{HS_2}\lambda_{X_{18}}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-6{g_1}^2\lambda^{(1)}_{X_5}$	$+\lambda^{(1)}_{12}(8\lambda^{(1)}_{X_5}+10\lambda_{X_9})$	$+4\lambda^{(1)}_{HS_2}\lambda_{X_{18}}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{X_5}$

Terms 3 of 5

Column	Representative term structure
T1	$+2\lambda^{(1)}_{S_1}\lambda^{(1)}_{X_5}$
T2	$+8\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_5}$
T3	$+4\lambda^{(A)}_{S_2}\lambda^{(1)}_{X_5}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+2\lambda^{(1)}_{S_1}\lambda^{(1)}_{X_5}$	$+8\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_5}$	$+4\lambda^{(A)}_{S_2}\lambda^{(1)}_{X_5}$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+2\lambda^{(1)}_{S_1}\lambda^{(1)}_{X_5}$	$+4\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_5}$	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+2\lambda^{(1)}_{S_1}\lambda^{(1)}_{X_5}$	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{X_5}$

Terms 4 of 5

Column	Representative term structure
T1	$-4\lambda^{(A)}_{S_1}\lambda^{(1)}_{X_5}$
T2	$+8\lambda_{X_6}(5\left(\lambda^{(1)}_{X_5}\right)^{*}+\left(\lambda^{(2)}_{X_5}\right)^{*})$
T3	$+\lambda^{(\times)}_{12}(4\lambda^{(2)}_{X_5}+4\lambda_{X_9})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$-4\lambda^{(A)}_{S_1}\lambda^{(1)}_{X_5}$	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+8\lambda_{X_6}(5\left(\lambda^{(1)}_{X_5}\right)^{*}+\left(\lambda^{(2)}_{X_5}\right)^{*})$	$+\lambda^{(\times)}_{12}(4\lambda^{(2)}_{X_5}+4\lambda_{X_9})$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{X_5}$

Terms 5 of 5

Column	Representative term structure
T1	$+\lambda^{(A)}_{S_1}(-4\lambda^{(1)}_{X_5} - 2\lambda^{(2)}_{X_5})$
T2	$+\lambda^{(1)}_{S_2}(8\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$
T3	$+\lambda^{(A)}_{S_2}(4\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+\lambda^{(A)}_{S_1}(-4\lambda^{(1)}_{X_5} - 2\lambda^{(2)}_{X_5})$	$+\lambda^{(1)}_{S_2}(8\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$	$+\lambda^{(A)}_{S_2}(4\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

$\beta^{(1)}_{\lambda^{(2)}_{X_5}}$

Quantity: $\lambda^{(2)}_{X_5}$

$$16\pi^2\,\mu\frac{d\lambda^{(2)}_{X_5}}{d\mu} = \beta^{(1)}_{\lambda^{(2)}_{X_5}}$$

Terms 1 of 4

Column	Representative term structure
T1	$-2\,(\lambda_5)^*\left(\lambda_{X_{14}}\right)^*$
T2	$+16\lambda_{X_6}\left(\lambda^{(2)}_{X_5}\right)^*$
T3	$-24{g_2}^2\lambda^{(2)}_{X_5}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-2\,(\lambda_5)^*\left(\lambda_{X_{14}}\right)^*$	$+16\lambda_{X_6}\left(\lambda^{(2)}_{X_5}\right)^*$	$-24{g_2}^2\lambda^{(2)}_{X_5}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_5}^{(2)}$

Terms 2 of 4

Column	Representative term structure
T1	$-6{g_1}^2\lambda_{X_5}^{(2)}$
T2	$+8\lambda_{12}^{(1)}\lambda_{X_5}^{(2)}$
T3	$+\lambda_{12}^{(\times)}(16\lambda_{X_5}^{(1)}+20\lambda_{X_5}^{(2)}+8\lambda_{X_9})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-6{g_1}^2\lambda_{X_5}^{(2)}$	$+8\lambda_{12}^{(1)}\lambda_{X_5}^{(2)}$	$+\lambda_{12}^{(\times)}(16\lambda_{X_5}^{(1)}+20\lambda_{X_5}^{(2)}+8\lambda_{X_9})$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(2)}_{X_5}$

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda^{(1)}_{S_1}\lambda^{(2)}_{X_5}$
T2	$+2\lambda^{(A)}_{S_1}\lambda^{(2)}_{X_5}$
T3	$+2\lambda^{(1)}_{S_2}\lambda^{(2)}_{X_5}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+2\lambda^{(1)}_{S_1}\lambda^{(2)}_{X_5}$	$+2\lambda^{(A)}_{S_1}\lambda^{(2)}_{X_5}$	$+2\lambda^{(1)}_{S_2}\lambda^{(2)}_{X_5}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(2)}_{X_5}$

Terms 4 of 4

Column	Representative term structure
T1	$-2\lambda^{(A)}_{S_2}\lambda^{(2)}_{X_5}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	-2	1	N/A	3	N/A	2	N/A	—
A	-4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	-1	2	N/A	2	N/A	1	N/A	—
B	-3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	-1	2	N/A	2	N/A	3	N/A	—
C	-3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	-2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	$-2\lambda^{(A)}_{S_2}\lambda^{(2)}_{X_5}$
E	2	3	N/A	3	N/A	2	N/A	—
E	-2	3	N/A	3	N/A	2	N/A	—
E	-4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda_{X_6}}$

Quantity: λ_{X_6}

$$16\pi^2 \, \mu \frac{d\lambda_{X_6}}{d\mu} = \beta^{(1)}_{\lambda_{X_6}}$$

Terms 1 of 3

Column	Representative term structure
T1	$12\lambda_{S_1}^{(1)}\lambda_{X_6}$
T2	$+3\lambda_{X_5}^{(1)\,2}$
T3	$+9\lambda_{X_9}{}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$12\lambda_{S_1}^{(1)}\lambda_{X_6}$	$+3\lambda_{X_5}^{(1)\,2}$	$+9\lambda_{X_9}{}^2$
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+12\lambda_{S_1}^{(1)}\lambda_{X_6}$	—	$+11\lambda_{X_9}{}^2$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+12\lambda_{S_1}^{(1)}\lambda_{X_6}$	$+3\lambda_{X_5}^{(1)\,2}$	$+11\lambda_{X_9}{}^2$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_6}

Terms 2 of 3

Column	Representative term structure
T1	$+2{\lambda_{X_{18}}}^2$
T2	$-24{g_2}^2\lambda_{X_6}$
T3	$-8\lambda_{S_1}^{(A)}\lambda_{X_6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+2{\lambda_{X_{18}}}^2$	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+2{\lambda_{X_{18}}}^2$	$-24{g_2}^2\lambda_{X_6}$	$-8\lambda_{S_1}^{(A)}\lambda_{X_6}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+2{\lambda_{X_{18}}}^2$	$-24{g_2}^2\lambda_{X_6}$	$-8\lambda_{S_1}^{(A)}\lambda_{X_6}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_6}

Terms 3 of 3

Column	Representative term structure
T1	$+{\lambda_{X_5}^{(1)}}^2$
T2	$+2\lambda_{X_5}^{(1)}\lambda_{X_5}^{(2)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	$+{\lambda_{X_5}^{(1)}}^2$	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	$+{\lambda_{X_5}^{(2)}}^2$	$+2\lambda_{X_5}^{(1)}\lambda_{X_5}^{(2)}$
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda^{(1)}_{X_7}}$

Quantity: $\lambda^{(1)}_{X_7}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{X_7}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{X_7}}$$

Terms 1 of 5

Column	Representative term structure
T1	$40\lambda_{X_1}\left(\lambda^{(1)}_{X_7}\right)^*$
T2	$-12{g_2}^2\lambda^{(1)}_{X_7}$
T3	$-6{g_1}^2\lambda^{(1)}_{X_7}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$40\lambda_{X_1}\left(\lambda^{(1)}_{X_7}\right)^*$	$-12{g_2}^2\lambda^{(1)}_{X_7}$	$-6{g_1}^2\lambda^{(1)}_{X_7}$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$24\lambda_{X_1}\left(\lambda^{(1)}_{X_7}\right)^*$	$-12{g_2}^2\lambda^{(1)}_{X_7}$	$-6{g_1}^2\lambda^{(1)}_{X_7}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	$-24{g_2}^2\lambda^{(1)}_{X_7}$	$-6{g_1}^2\lambda^{(1)}_{X_7}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{X_7}$

Terms 2 of 5

Column	Representative term structure
T1	$+\lambda^{(1)}_{12}(10\lambda_{X_2}+8\lambda^{(1)}_{X_7})$
T2	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{16}}$
T3	$+4\lambda^{(1)}_{S_1}\lambda^{(1)}_{X_7}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+\lambda^{(1)}_{12}(10\lambda_{X_2}+8\lambda^{(1)}_{X_7})$	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{16}}$	$+4\lambda^{(1)}_{S_1}\lambda^{(1)}_{X_7}$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+\lambda^{(1)}_{12}(6\lambda_{X_2}+8\lambda^{(1)}_{X_7})$	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{16}}$	$+8\lambda^{(1)}_{S_1}\lambda^{(1)}_{X_7}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+\lambda^{(1)}_{12}(10\lambda_{X_2}+8\lambda^{(1)}_{X_7})$	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{16}}$	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{X_7}$

Terms 3 of 5

Column	Representative term structure
T1	$+2\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_7}$
T2	$-4\lambda^{(A)}_{S_2}\lambda^{(1)}_{X_7}$
T3	$+4\lambda_5\lambda_{X_{12}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$+2\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_7}$	$-4\lambda^{(A)}_{S_2}\lambda^{(1)}_{X_7}$	$+4\lambda_5\lambda_{X_{12}}$
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$+2\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_7}$	—	$+4\lambda_5\lambda_{X_{12}}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$+2\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_7}$	—	$+2\lambda_5\lambda_{X_{12}}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_7}^{(1)}$

Terms 4 of 5

Column	Representative term structure
T1	$+4\lambda_{S_1}^{(A)}\lambda_{X_7}^{(1)}$
T2	$8\lambda_{X_1}(5\left(\lambda_{X_7}^{(1)}\right)^{*}+\left(\lambda_{X_7}^{(2)}\right)^{*})$
T3	$+\lambda_{12}^{(\times)}(4\lambda_{X_2}+4\lambda_{X_7}^{(2)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+4\lambda_{S_1}^{(A)}\lambda_{X_7}^{(1)}$	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	$8\lambda_{X_1}(5\left(\lambda_{X_7}^{(1)}\right)^{*}+\left(\lambda_{X_7}^{(2)}\right)^{*})$	$+\lambda_{12}^{(\times)}(4\lambda_{X_2}+4\lambda_{X_7}^{(2)})$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{X_7}$

Terms 5 of 5

Column	Representative term structure
T1	$+\lambda^{(1)}_{S_1}(8\lambda^{(1)}_{X_7} + 2\lambda^{(2)}_{X_7})$
T2	$+\lambda^{(A)}_{S_1}(4\lambda^{(1)}_{X_7} + 2\lambda^{(2)}_{X_7})$
T3	$+\lambda^{(A)}_{S_2}(-4\lambda^{(1)}_{X_7} - 2\lambda^{(2)}_{X_7})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+\lambda^{(1)}_{S_1}(8\lambda^{(1)}_{X_7} + 2\lambda^{(2)}_{X_7})$	$+\lambda^{(A)}_{S_1}(4\lambda^{(1)}_{X_7} + 2\lambda^{(2)}_{X_7})$	$+\lambda^{(A)}_{S_2}(-4\lambda^{(1)}_{X_7} - 2\lambda^{(2)}_{X_7})$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

$\beta^{(1)}_{\lambda^{(2)}_{X_7}}$

Quantity: $\lambda^{(2)}_{X_7}$

$$16\pi^2\,\mu\frac{d\lambda^{(2)}_{X_7}}{d\mu} = \beta^{(1)}_{\lambda^{(2)}_{X_7}}$$

Terms 1 of 4

Column	Representative term structure
T1	$16\lambda_{X_1}\left(\lambda^{(2)}_{X_7}\right)^*$
T2	$-24{g_2}^2\lambda^{(2)}_{X_7}$
T3	$-6{g_1}^2\lambda^{(2)}_{X_7}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$16\lambda_{X_1}\left(\lambda^{(2)}_{X_7}\right)^*$	$-24{g_2}^2\lambda^{(2)}_{X_7}$	$-6{g_1}^2\lambda^{(2)}_{X_7}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(2)}_{X_7}$

Terms 2 of 4

Column	Representative term structure
T1	$+8\lambda^{(1)}_{12}\lambda^{(2)}_{X_7}$
T2	$+\lambda^{(\times)}_{12}(8\lambda_{X_2}+16\lambda^{(1)}_{X_7}+20\lambda^{(2)}_{X_7})$
T3	$+2\lambda^{(1)}_{S_1}\lambda^{(2)}_{X_7}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+8\lambda^{(1)}_{12}\lambda^{(2)}_{X_7}$	$+\lambda^{(\times)}_{12}(8\lambda_{X_2}+16\lambda^{(1)}_{X_7}+20\lambda^{(2)}_{X_7})$	$+2\lambda^{(1)}_{S_1}\lambda^{(2)}_{X_7}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(2)}_{X_7}$

Terms 3 of 4

Column	Representative term structure
T1	$-2\lambda^{(A)}_{S_1}\lambda^{(2)}_{X_7}$
T2	$+2\lambda^{(1)}_{S_2}\lambda^{(2)}_{X_7}$
T3	$+2\lambda^{(A)}_{S_2}\lambda^{(2)}_{X_7}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$-2\lambda^{(A)}_{S_1}\lambda^{(2)}_{X_7}$	$+2\lambda^{(1)}_{S_2}\lambda^{(2)}_{X_7}$	$+2\lambda^{(A)}_{S_2}\lambda^{(2)}_{X_7}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(2)}_{X_7}$

Terms 4 of 4

Column	Representative term structure
T1	$-2\lambda_5\lambda_{X_{12}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	$-2\lambda_5\lambda_{X_{12}}$
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda_{X_9}}$

Quantity: λ_{X_9}

$$16\pi^2 \, \mu \frac{d\lambda_{X_9}}{d\mu} = \beta^{(1)}_{\lambda_{X_9}}$$

Terms 1 of 3

Column	Representative term structure
T1	$72\lambda_{X_6} \left(\lambda_{X_9}\right)^*$
T2	$+6\lambda_{12}^{(1)} \lambda_{X_5}^{(1)}$
T3	$+4\lambda_{HS_1}^{(1)} \lambda_{X_{18}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$72\lambda_{X_6} \left(\lambda_{X_9}\right)^*$	$+6\lambda_{12}^{(1)} \lambda_{X_5}^{(1)}$	$+4\lambda_{HS_1}^{(1)} \lambda_{X_{18}}$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$88\lambda_{X_6} \left(\lambda_{X_9}\right)^*$	$+2\lambda_{12}^{(1)} \lambda_{X_5}^{(1)}$	$+4\lambda_{HS_1}^{(1)} \lambda_{X_{18}}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$88\lambda_{X_6} \left(\lambda_{X_9}\right)^*$	—	$+4\lambda_{HS_1}^{(1)} \lambda_{X_{18}}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_9}

Terms 2 of 3

Column	Representative term structure
T1	$+18\lambda_{S_1}^{(1)}\lambda_{X_9}$
T2	$-24{g_2}^2\lambda_{X_9}$
T3	$+\lambda_{12}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+18\lambda_{S_1}^{(1)}\lambda_{X_9}$	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+22\lambda_{S_1}^{(1)}\lambda_{X_9}$	$-24{g_2}^2\lambda_{X_9}$	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+22\lambda_{S_1}^{(1)}\lambda_{X_9}$	$-24{g_2}^2\lambda_{X_9}$	$+\lambda_{12}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X_9}

Terms 3 of 3

Column	Representative term structure
T1	$+\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(1)} + 4\lambda_{X_5}^{(2)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	$+\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(1)} + 4\lambda_{X_5}^{(2)})$
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	—
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda_{X10}}$

Quantity: λ_{X10}

$$16\pi^2\,\mu\frac{d\lambda_{X10}}{d\mu} = \beta^{(1)}_{\lambda_{X10}}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_{X10}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$-9{g_2}^2\lambda_{X10}$
T3	$-3{g_1}^2\lambda_{X10}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	$\lambda_{X10}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-9{g_2}^2\lambda_{X10}$	$-3{g_1}^2\lambda_{X10}$
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	$\lambda_{X10}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-9{g_2}^2\lambda_{X10}$	$-3{g_1}^2\lambda_{X10}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{10}}$

Terms 2 of 3

Column	Representative term structure
T1	$+2\lambda_1\lambda_{X_{10}}$
T2	$+8\lambda^{(1)}_{HS_2}\lambda_{X_{10}}$
T3	$-4\lambda^{(A)}_{HS_2}\lambda_{X_{10}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	$+2\lambda_1\lambda_{X_{10}}$	$+8\lambda^{(1)}_{HS_2}\lambda_{X_{10}}$	$-4\lambda^{(A)}_{HS_2}\lambda_{X_{10}}$
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	$+2\lambda_1\lambda_{X_{10}}$	$+8\lambda^{(1)}_{HS_2}\lambda_{X_{10}}$	$-4\lambda^{(A)}_{HS_2}\lambda_{X_{10}}$
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{10}}$

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda_{S_2}^{(1)}\lambda_{X_{10}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	$+2\lambda_{S_2}^{(1)}\lambda_{X_{10}}$
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	$+2\lambda_{S_2}^{(1)}\lambda_{X_{10}}$
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	—
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda_{X_{12}}}$

Quantity: $\lambda_{X_{12}}$

$$16\pi^2\,\mu\frac{d\lambda_{X_{12}}}{d\mu} = \beta^{(1)}_{\lambda_{X_{12}}}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{X_{12}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+4\lambda^{(1)}_{X_7}\ (\lambda_5)^*$
T3	$+8\lambda_{X_{16}}\ (\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$\lambda_{X_{12}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+4\lambda^{(1)}_{X_7}\ (\lambda_5)^*$	$+8\lambda_{X_{16}}\ (\lambda_5)^*$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$\lambda_{X_{12}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+4\lambda^{(1)}_{X_7}\ (\lambda_5)^*$	$+8\lambda_{X_{16}}\ (\lambda_5)^*$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$\lambda_{X_{12}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+4\lambda^{(1)}_{X_7}\ (\lambda_5)^*$ $-2\lambda^{(2)}_{X_7}\ (\lambda_5)^*$	$+8\lambda_{X_{16}}\ (\lambda_5)^*$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{12}}$

Terms 2 of 5

Column	Representative term structure
T1	$-\frac{21{g_2}^2\lambda_{X_{12}}}{2}$
T2	$-\frac{9{g_1}^2\lambda_{X_{12}}}{2}$
T3	$+2\lambda_{12}^{(1)}\lambda_{X_{12}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$-\frac{21{g_2}^2\lambda_{X_{12}}}{2}$	$-\frac{9{g_1}^2\lambda_{X_{12}}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{X_{12}}$
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$-\frac{21{g_2}^2\lambda_{X_{12}}}{2}$	$-\frac{9{g_1}^2\lambda_{X_{12}}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{X_{12}}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$-\frac{33{g_2}^2\lambda_{X_{12}}}{2}$	$-\frac{9{g_1}^2\lambda_{X_{12}}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{X_{12}}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{12}}$

Terms 3 of 5

Column	Representative term structure
T1	$+2\lambda_1\lambda_{X_{12}}$
T2	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{12}}$
T3	$+4\lambda^{(1)}_{HS_2}\lambda_{X_{12}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_{X_{12}}$	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{12}}$	$+4\lambda^{(1)}_{HS_2}\lambda_{X_{12}}$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+2\lambda_1\lambda_{X_{12}}$	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{12}}$	$+4\lambda^{(1)}_{HS_2}\lambda_{X_{12}}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+2\lambda_1\lambda_{X_{12}}$	$+4\lambda^{(1)}_{HS_1}\lambda_{X_{12}}$	$+4\lambda^{(1)}_{HS_2}\lambda_{X_{12}}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{12}}$

Terms 4 of 5

Column	Representative term structure
T1	$-4\lambda_{HS_2}^{(A)}\lambda_{X_{12}}$
T2	$-4\lambda_{HS_1}^{(A)}\lambda_{X_{12}}$
T3	$-2\lambda_{12}^{(A)}\lambda_{X_{12}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$-4\lambda_{HS_2}^{(A)}\lambda_{X_{12}}$	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	$-4\lambda_{HS_1}^{(A)}\lambda_{X_{12}}$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$-2\lambda_{HS_2}^{(A)}\lambda_{X_{12}}$	$-2\lambda_{HS_1}^{(A)}\lambda_{X_{12}}$	$-2\lambda_{12}^{(A)}\lambda_{X_{12}}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: λ_{X12}

Terms 5 of 5

Column	Representative term structure
T1	$-2\lambda_{12}^{(\times)}\lambda_{X12}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	$-2\lambda_{12}^{(\times)}\lambda_{X12}$
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda_{X_{14}}}$

Quantity: $\lambda_{X_{14}}$

$$16\pi^2\,\mu\frac{d\lambda_{X_{14}}}{d\mu} = \beta^{(1)}_{\lambda_{X_{14}}}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{X_{14}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+4\,(\lambda_5)^*\,(\left(\lambda^{(1)}_{X_5}\right)^* + 2\,\left(\lambda_{X_{18}}\right)^*)$
T3	$-\frac{21{g_2}^2\lambda_{X_{14}}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda_{X_{14}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+4\,(\lambda_5)^*\,(\left(\lambda^{(1)}_{X_5}\right)^* + 2\,\left(\lambda_{X_{18}}\right)^*)$	$-\frac{21{g_2}^2\lambda_{X_{14}}}{2}$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$\lambda_{X_{14}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+4\,(\lambda_5)^*\,(\left(\lambda^{(1)}_{X_5}\right)^* + 2\,\left(\lambda_{X_{18}}\right)^*)$	$-\frac{21{g_2}^2\lambda_{X_{14}}}{2}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$\lambda_{X_{14}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	—	$-\frac{33{g_2}^2\lambda_{X_{14}}}{2}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{14}}$

Terms 2 of 5

Column	Representative term structure
T1	$-\frac{9{g_1}^2\lambda_{X_{14}}}{2}$
T2	$+2\lambda_{12}^{(1)}\lambda_{X_{14}}$
T3	$+2\lambda_1\lambda_{X_{14}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-\frac{9{g_1}^2\lambda_{X_{14}}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{X_{14}}$	$+2\lambda_1\lambda_{X_{14}}$
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$-\frac{9{g_1}^2\lambda_{X_{14}}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{X_{14}}$	$+2\lambda_1\lambda_{X_{14}}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-\frac{9{g_1}^2\lambda_{X_{14}}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{X_{14}}$	$+2\lambda_1\lambda_{X_{14}}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{14}}$

Terms 3 of 5

Column	Representative term structure
T1	$+4\lambda_{HS_1}^{(1)}\lambda_{X_{14}}$
T2	$+4\lambda_{HS_2}^{(1)}\lambda_{X_{14}}$
T3	$+4\lambda_{HS_2}^{(A)}\lambda_{X_{14}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+4\lambda_{HS_1}^{(1)}\lambda_{X_{14}}$	$+4\lambda_{HS_2}^{(1)}\lambda_{X_{14}}$	$+4\lambda_{HS_2}^{(A)}\lambda_{X_{14}}$
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+4\lambda_{HS_1}^{(1)}\lambda_{X_{14}}$	$+4\lambda_{HS_2}^{(1)}\lambda_{X_{14}}$	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+4\lambda_{HS_1}^{(1)}\lambda_{X_{14}}$	$+4\lambda_{HS_2}^{(1)}\lambda_{X_{14}}$	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{14}}$

Terms 4 of 5

Column	Representative term structure
T1	$+4\lambda_{HS_1}^{(A)}\lambda_{X_{14}}$
T2	$+2\left(\lambda_5\right)^*\left(2\left(\lambda_{X_5}^{(1)}\right)^*-\left(\lambda_{X_5}^{(2)}\right)^*+4\left(\lambda_{X_{18}}\right)^*\right)$
T3	$-2\lambda_{12}^{(A)}\lambda_{X_{14}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+4\lambda_{HS_1}^{(A)}\lambda_{X_{14}}$	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	$+2\left(\lambda_5\right)^*\left(2\left(\lambda_{X_5}^{(1)}\right)^*-\left(\lambda_{X_5}^{(2)}\right)^*+4\left(\lambda_{X_{18}}\right)^*\right)$	$-2\lambda_{12}^{(A)}\lambda_{X_{14}}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{14}}$

Terms 5 of 5

Column	Representative term structure
T1	$-2\lambda_{12}^{(\times)}\lambda_{X_{14}}$
T2	$-\frac{4\lambda_{HS_1}^{(A)}\lambda_{X_{14}}}{3}$
T3	$-\frac{4\lambda_{HS_2}^{(A)}\lambda_{X_{14}}}{3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-2\lambda_{12}^{(\times)}\lambda_{X_{14}}$	$-\frac{4\lambda_{HS_1}^{(A)}\lambda_{X_{14}}}{3}$	$-\frac{4\lambda_{HS_2}^{(A)}\lambda_{X_{14}}}{3}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

$\beta^{(1)}_{\lambda_{X_{15}}}$

Quantity: $\lambda_{X_{15}}$

$$16\pi^2\,\mu\frac{d\lambda_{X_{15}}}{d\mu} = \beta^{(1)}_{\lambda_{X_{15}}}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_{X_{15}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$-9{g_2}^2\lambda_{X_{15}}$
T3	$-3{g_1}^2\lambda_{X_{15}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$\lambda_{X_{15}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-9{g_2}^2\lambda_{X_{15}}$	$-3{g_1}^2\lambda_{X_{15}}$
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	$\lambda_{X_{15}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-9{g_2}^2\lambda_{X_{15}}$	$-3{g_1}^2\lambda_{X_{15}}$
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{15}}$

Terms 2 of 3

Column	Representative term structure
T1	$+2\lambda_1\lambda_{X_{15}}$
T2	$+8\lambda_{HS_1}^{(1)}\lambda_{X_{15}}$
T3	$+4\lambda_{HS_1}^{(A)}\lambda_{X_{15}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	$+2\lambda_1\lambda_{X_{15}}$	$+8\lambda_{HS_1}^{(1)}\lambda_{X_{15}}$	$+4\lambda_{HS_1}^{(A)}\lambda_{X_{15}}$
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	$+2\lambda_1\lambda_{X_{15}}$	$+8\lambda_{HS_1}^{(1)}\lambda_{X_{15}}$	$+4\lambda_{HS_1}^{(A)}\lambda_{X_{15}}$
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{15}}$

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda_{S_1}^{(1)}\lambda_{X_{15}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	$+2\lambda_{S_1}^{(1)}\lambda_{X_{15}}$
B	−1	2	N/A	2	N/A	1	N/A	—
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	$+2\lambda_{S_1}^{(1)}\lambda_{X_{15}}$
C	−1	2	N/A	2	N/A	3	N/A	—
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	—
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\lambda_{X_{16}}}$

Quantity: $\lambda_{X_{16}}$

$$16\pi^2\,\mu\frac{d\lambda_{X_{16}}}{d\mu} = \beta^{(1)}_{\lambda_{X_{16}}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{X_{16}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+40\lambda_{X_1}\left(\lambda_{X_{16}}\right)^*$
T3	$-\frac{33{g_2}^2\lambda_{X_{16}}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$\lambda_{X_{16}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+40\lambda_{X_1}\left(\lambda_{X_{16}}\right)^*$	$-\frac{33{g_2}^2\lambda_{X_{16}}}{2}$
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$\lambda_{X_{16}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+24\lambda_{X_1}\left(\lambda_{X_{16}}\right)^*$	$-\frac{9{g_2}^2\lambda_{X_{16}}}{2}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$\lambda_{X_{16}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+40\lambda_{X_1}\left(\lambda_{X_{16}}\right)^*$	$-\frac{33{g_2}^2\lambda_{X_{16}}}{2}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{16}}$

Terms 2 of 4

Column	Representative term structure
T1	$-\frac{3{g_1}^2\lambda_{X_{16}}}{2}$
T2	$+6\lambda_1\lambda_{X_{16}}$
T3	$+2\lambda_{HS_1}^{(1)}\lambda_{X_7}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$-\frac{3{g_1}^2\lambda_{X_{16}}}{2}$	$+6\lambda_1\lambda_{X_{16}}$	$+2\lambda_{HS_1}^{(1)}\lambda_{X_7}^{(1)}$
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$-\frac{3{g_1}^2\lambda_{X_{16}}}{2}$	$+6\lambda_1\lambda_{X_{16}}$	$+6\lambda_{HS_1}^{(1)}\lambda_{X_7}^{(1)}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$-\frac{3{g_1}^2\lambda_{X_{16}}}{2}$	$+6\lambda_1\lambda_{X_{16}}$	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{16}}$

Terms 3 of 4

Column	Representative term structure
T1	$+\lambda_{HS_2}^{(1)}(10\lambda_{X_2}+8\lambda_{X_{16}})$
T2	$+2\lambda_{S_2}^{(1)}\lambda_{X_{16}}$
T3	$-4\lambda_{S_2}^{(A)}\lambda_{X_{16}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+\lambda_{HS_2}^{(1)}(10\lambda_{X_2}+8\lambda_{X_{16}})$	$+2\lambda_{S_2}^{(1)}\lambda_{X_{16}}$	$-4\lambda_{S_2}^{(A)}\lambda_{X_{16}}$
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+\lambda_{HS_2}^{(1)}(6\lambda_{X_2}+8\lambda_{X_{16}})$	$+2\lambda_{S_2}^{(1)}\lambda_{X_{16}}$	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+\lambda_{HS_2}^{(1)}(10\lambda_{X_2}+8\lambda_{X_{16}})$	$+2\lambda_{S_2}^{(1)}\lambda_{X_{16}}$	$-4\lambda_{S_2}^{(A)}\lambda_{X_{16}}$
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{16}}$

Terms 4 of 4

Column	Representative term structure
T1	$+4\lambda_5\lambda_{X_{12}}$
T2	$+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	$+4\lambda_5\lambda_{X_{12}}$	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	$+12\lambda_5\lambda_{X_{12}}$	—
E	0	3	N/A	3	N/A	2	N/A	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	$+4\lambda_5\lambda_{X_{12}}$	$+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda^{(1)}_{X_{17}}}$

Quantity: $\lambda^{(1)}_{X_{17}}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{X_{17}}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{X_{17}}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda^{(1)}_{X_{17}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+\lambda_9(8\left(\lambda^{(1)}_{X_{17}}\right)^* - 4\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*)$
T3	$-9{g_2}^2\lambda^{(1)}_{X_{17}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$\lambda^{(1)}_{X_{17}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\lambda_9(8\left(\lambda^{(1)}_{X_{17}}\right)^* - 4\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*)$	$-9{g_2}^2\lambda^{(1)}_{X_{17}}$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$\lambda^{(1)}_{X_{17}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\lambda_9(8\left(\lambda^{(1)}_{X_{17}}\right)^* - 4\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*)$	$-9{g_2}^2\lambda^{(1)}_{X_{17}}$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{17}}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$-3{g_1}^2\lambda_{X_{17}}^{(1)}$
T2	$+2\lambda_{12}^{(1)}\lambda_{X_{17}}^{(1)}$
T3	$+\lambda_{12}^{(A)}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-\frac{\lambda_{X_{17}}^{(1)}}{2}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	$-3{g_1}^2\lambda_{X_{17}}^{(1)}$	$+2\lambda_{12}^{(1)}\lambda_{X_{17}}^{(1)}$	$+\lambda_{12}^{(A)}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-\frac{\lambda_{X_{17}}^{(1)}}{2}\right)$
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	$-3{g_1}^2\lambda_{X_{17}}^{(1)}$	$+2\lambda_{12}^{(1)}\lambda_{X_{17}}^{(1)}$	$+\lambda_{12}^{(A)}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-\frac{\lambda_{X_{17}}^{(1)}}{2}\right)$
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{17}}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+\lambda_1(4\lambda_{X_{17}}^{(1)} - 2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})$
T2	$+\lambda_{HS_1}^{(1)}(4\lambda_{X_{17}}^{(1)} - 6\lambda_{11})$
T3	$+\lambda_{HS_1}^{(A)}\left(-\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + \frac{\lambda_{11}}{2} - 2\lambda_{X_{17}}^{(1)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+\lambda_1(4\lambda_{X_{17}}^{(1)} - 2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})$	$+\lambda_{HS_1}^{(1)}(4\lambda_{X_{17}}^{(1)} - 6\lambda_{11})$	$+\lambda_{HS_1}^{(A)}\left(-\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + \frac{\lambda_{11}}{2} - 2\lambda_{X_{17}}^{(1)}\right)$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+\lambda_1(4\lambda_{X_{17}}^{(1)} - 2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})$	$+\lambda_{HS_1}^{(1)}(4\lambda_{X_{17}}^{(1)} - 6\lambda_{11})$	$+\lambda_{HS_1}^{(A)}\left(-\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + \frac{\lambda_{11}}{2} - 2\lambda_{X_{17}}^{(1)}\right)$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda^{(1)}_{X_{17}}$

Terms 4 of 4

Column	Representative term structure
T1	$+\lambda^{(1)}_{HS_2}(4\lambda^{(1)}_{X_{17}} - 6\lambda_{10})$
T2	$+\lambda^{(A)}_{HS_2}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\lambda_{10}}{2} + 2\lambda^{(1)}_{X_{17}}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+\lambda^{(1)}_{HS_2}(4\lambda^{(1)}_{X_{17}} - 6\lambda_{10})$	$+\lambda^{(A)}_{HS_2}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\lambda_{10}}{2} + 2\lambda^{(1)}_{X_{17}}\right)$
B	−3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+\lambda^{(1)}_{HS_2}(4\lambda^{(1)}_{X_{17}} - 6\lambda_{10})$	$+\lambda^{(A)}_{HS_2}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\lambda_{10}}{2} + 2\lambda^{(1)}_{X_{17}}\right)$
C	−3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\lambda_{X_{18}}}$

Quantity: $\lambda_{X_{18}}$

$$16\pi^2\,\mu\frac{d\lambda_{X_{18}}}{d\mu} = \beta^{(1)}_{\lambda_{X_{18}}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{X_{18}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+12\,(\lambda_5)^*\left(\lambda_{X_{14}}\right)^*$
T3	$+24\lambda_{X_6}\left(\lambda_{X_{18}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$\lambda_{X_{18}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+12\,(\lambda_5)^*\left(\lambda_{X_{14}}\right)^*$	$+24\lambda_{X_6}\left(\lambda_{X_{18}}\right)^*$
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$\lambda_{X_{18}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+4\,(\lambda_5)^*\left(\lambda_{X_{14}}\right)^*$	$+40\lambda_{X_6}\left(\lambda_{X_{18}}\right)^*$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$\lambda_{X_{18}}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+4\,(\lambda_5)^*\left(\lambda_{X_{14}}\right)^*$	$+40\lambda_{X_6}\left(\lambda_{X_{18}}\right)^*$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{18}}$

Terms 2 of 4

Column	Representative term structure
T1	$-\frac{9{g_2}^2\lambda_{X_{18}}}{2}$
T2	$-\frac{3{g_1}^2\lambda_{X_{18}}}{2}$
T3	$+6\lambda_1\lambda_{X_{18}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$-\frac{9{g_2}^2\lambda_{X_{18}}}{2}$	$-\frac{3{g_1}^2\lambda_{X_{18}}}{2}$	$+6\lambda_1\lambda_{X_{18}}$
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$-\frac{33{g_2}^2\lambda_{X_{18}}}{2}$	$-\frac{3{g_1}^2\lambda_{X_{18}}}{2}$	$+6\lambda_1\lambda_{X_{18}}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$-\frac{33{g_2}^2\lambda_{X_{18}}}{2}$	$-\frac{3{g_1}^2\lambda_{X_{18}}}{2}$	$+6\lambda_1\lambda_{X_{18}}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{18}}$

Terms 3 of 4

Column	Representative term structure
T1	$+\lambda^{(1)}_{HS_1}(6\lambda_{X_9} + 8\lambda_{X_{18}})$
T2	$+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{X_5}$
T3	$+2\lambda^{(1)}_{S_1}\lambda_{X_{18}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+\lambda^{(1)}_{HS_1}(6\lambda_{X_9} + 8\lambda_{X_{18}})$	$+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{X_5}$	$+2\lambda^{(1)}_{S_1}\lambda_{X_{18}}$
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+\lambda^{(1)}_{HS_1}(10\lambda_{X_9} + 8\lambda_{X_{18}})$	$+2\lambda^{(1)}_{HS_2}\lambda^{(1)}_{X_5}$	$+2\lambda^{(1)}_{S_1}\lambda_{X_{18}}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+\lambda^{(1)}_{HS_1}(10\lambda_{X_9} + 8\lambda_{X_{18}})$	—	$+2\lambda^{(1)}_{S_1}\lambda_{X_{18}}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\lambda_{X_{18}}$

Terms 4 of 4

Column	Representative term structure
T1	$-4\lambda_{S_1}^{(A)}\lambda_{X_{18}}$
T2	$+\lambda_{HS_2}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
A	0	1	N/A	3	N/A	2	N/A	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—
D	0	3	N/A	1	N/A	2	N/A	$-4\lambda_{S_1}^{(A)}\lambda_{X_{18}}$	—
D	2	3	N/A	1	N/A	2	N/A	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—
E	0	3	N/A	3	N/A	2	N/A	$-4\lambda_{S_1}^{(A)}\lambda_{X_{18}}$	$+\lambda_{HS_2}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$
E	2	3	N/A	3	N/A	2	N/A	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—

$\beta^{(1)}_{\mu_{\mathrm{T3GeneralS1S2}}}$

Quantity: $\mu_{\mathrm{T3GeneralS1S2}}$

$$16\pi^2\,\mu\frac{d\mu_{\mathrm{T3GeneralS1S2}}}{d\mu} = \beta^{(1)}_{\mu_{\mathrm{T3GeneralS1S2}}}$$

Terms 1 of 3

Column	Representative term structure
T1	$12\lambda_9\left(\mu_{\mathrm{T3GeneralS1S2}}\right)^*$
T2	$-\frac{9{g_2}^2\mu_{\mathrm{T3GeneralS1S2}}}{2}$
T3	$-\frac{3{g_1}^2\mu_{\mathrm{T3GeneralS1S2}}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$12\lambda_9\left(\mu_{\mathrm{T3GeneralS1S2}}\right)^*$	$-\frac{9{g_2}^2\mu_{\mathrm{T3GeneralS1S2}}}{2}$	$-\frac{3{g_1}^2\mu_{\mathrm{T3GeneralS1S2}}}{2}$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$12\lambda_9\left(\mu_{\mathrm{T3GeneralS1S2}}\right)^*$	$-\frac{9{g_2}^2\mu_{\mathrm{T3GeneralS1S2}}}{2}$	$-\frac{3{g_1}^2\mu_{\mathrm{T3GeneralS1S2}}}{2}$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\mu_{\mathrm{T3GeneralS1S2}}$

Terms 2 of 3

Column	Representative term structure
T1	$+2\lambda_{12}^{(1)}\,\mu_{\mathrm{T3GeneralS1S2}}$
T2	$-\frac{3\lambda_{12}^{(A)}\,\mu_{\mathrm{T3GeneralS1S2}}}{2}$
T3	$+6\lambda_{10}m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	$+2\lambda_{12}^{(1)}\,\mu_{\mathrm{T3GeneralS1S2}}$	$-\frac{3\lambda_{12}^{(A)}\,\mu_{\mathrm{T3GeneralS1S2}}}{2}$	$+6\lambda_{10}m_2^2$
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	$+2\lambda_{12}^{(1)}\,\mu_{\mathrm{T3GeneralS1S2}}$	$-\frac{3\lambda_{12}^{(A)}\,\mu_{\mathrm{T3GeneralS1S2}}}{2}$	$+6\lambda_{10}m_2^2$
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\mu_{\mathrm{T3GeneralS1S2}}$

Terms 3 of 3

Column	Representative term structure
T1	$+6\lambda_{11}m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	−2	1	N/A	3	N/A	2	N/A	—
A	−4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	−1	2	N/A	2	N/A	1	N/A	$+6\lambda_{11}m_1^2$
B	−3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	−1	2	N/A	2	N/A	3	N/A	$+6\lambda_{11}m_1^2$
C	−3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	−2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	−2	3	N/A	3	N/A	2	N/A	—
E	−4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\mu_{\mathrm{T3GeneralS1Sq}}}$

Quantity: $\mu_{\mathrm{T3GeneralS1Sq}}$

$$16\pi^2\,\mu\frac{d\mu_{\mathrm{T3GeneralS1Sq}}}{d\mu} = \beta^{(1)}_{\mu_{\mathrm{T3GeneralS1Sq}}}$$

Terms 1 of 3

Column	Representative term structure
T1	$24\lambda_{X_6}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$
T2	$+2\lambda^{(1)}_{S_1}\mu_{\mathrm{T3GeneralS1Sq}}$
T3	$+6\lambda^{(1)}_{X_5}m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$24\lambda_{X_6}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$	$+2\lambda^{(1)}_{S_1}\mu_{\mathrm{T3GeneralS1Sq}}$	$+6\lambda^{(1)}_{X_5}m_2^2$
A	−2	1	N/A	3	N/A	2	N/A	—	—	—
A	−4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	−1	2	N/A	2	N/A	1	N/A	—	—	—
B	−3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	−1	2	N/A	2	N/A	3	N/A	—	—	—
C	−3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$40\lambda_{X_6}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$	$+2\lambda^{(1)}_{S_1}\mu_{\mathrm{T3GeneralS1Sq}}$	$+2\lambda^{(1)}_{X_5}m_2^2$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	−2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$40\lambda_{X_6}\left(\mu_{\mathrm{T3GeneralS1Sq}}\right)^*$	$+2\lambda^{(1)}_{S_1}\mu_{\mathrm{T3GeneralS1Sq}}$	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	−2	3	N/A	3	N/A	2	N/A	—	—	—
E	−4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\mu_{\mathrm{T3GeneralS1Sq}}$

Terms 2 of 3

Column	Representative term structure
T1	$+6\lambda_{X_9} m_1^2$
T2	$-12{g_2}^2 \mu_{\mathrm{T3GeneralS1Sq}}$
T3	$-4\lambda_{S_1}^{(A)} \mu_{\mathrm{T3GeneralS1Sq}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	$+6\lambda_{X_9} m_1^2$	—	—
A	-2	1	N/A	3	N/A	2	N/A	—	—	—
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	$+10\lambda_{X_9} m_1^2$	$-12{g_2}^2 \mu_{\mathrm{T3GeneralS1Sq}}$	$-4\lambda_{S_1}^{(A)} \mu_{\mathrm{T3GeneralS1Sq}}$
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	—	—
E	0	3	N/A	3	N/A	2	N/A	$+10\lambda_{X_9} m_1^2$	$-12{g_2}^2 \mu_{\mathrm{T3GeneralS1Sq}}$	$-4\lambda_{S_1}^{(A)} \mu_{\mathrm{T3GeneralS1Sq}}$
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	—	—	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\mu_{\mathrm{T3GeneralS1Sq}}$

Terms 3 of 3

Column	Representative term structure
T1	$+m_2^2(6\lambda_{X_5}^{(1)} + 2\lambda_{X_5}^{(2)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	-2	1	N/A	3	N/A	2	N/A	—
A	-4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	-1	2	N/A	2	N/A	1	N/A	—
B	-3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	-1	2	N/A	2	N/A	3	N/A	—
C	-3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	-2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	$+m_2^2(6\lambda_{X_5}^{(1)} + 2\lambda_{X_5}^{(2)})$
E	2	3	N/A	3	N/A	2	N/A	—
E	-2	3	N/A	3	N/A	2	N/A	—
E	-4	3	N/A	3	N/A	2	N/A	—

$\beta^{(1)}_{\mu_{\mathrm{T3GeneralS2Sq}}}$

Quantity: $\mu_{\mathrm{T3GeneralS2Sq}}$

$$16\pi^2\,\mu\frac{d\mu_{\mathrm{T3GeneralS2Sq}}}{d\mu} = \beta^{(1)}_{\mu_{\mathrm{T3GeneralS2Sq}}}$$

Terms 1 of 3

Column	Representative term structure
T1	$40\lambda_{X_1}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$
T2	$-12{g_2}^2\mu_{\mathrm{T3GeneralS2Sq}}$
T3	$+2\lambda^{(1)}_{S_2}\mu_{\mathrm{T3GeneralS2Sq}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$40\lambda_{X_1}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$	$-12{g_2}^2\mu_{\mathrm{T3GeneralS2Sq}}$	$+2\lambda^{(1)}_{S_2}\mu_{\mathrm{T3GeneralS2Sq}}$
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	$24\lambda_{X_1}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$	—	$+2\lambda^{(1)}_{S_2}\mu_{\mathrm{T3GeneralS2Sq}}$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$40\lambda_{X_1}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^*$	$-12{g_2}^2\mu_{\mathrm{T3GeneralS2Sq}}$	$+2\lambda^{(1)}_{S_2}\mu_{\mathrm{T3GeneralS2Sq}}$
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\mu_{\mathrm{T3GeneralS2Sq}}$

Terms 2 of 3

Column	Representative term structure
T1	$-4\lambda_{S_2}^{(A)}\mu_{\mathrm{T3GeneralS2Sq}}$
T2	$+10\lambda_{X_2}m_2^2$
T3	$+2\lambda_{X_7}^{(1)}m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
A	0	1	N/A	3	N/A	2	N/A	—	—	—
A	-2	1	N/A	3	N/A	2	N/A	$-4\lambda_{S_2}^{(A)}\mu_{\mathrm{T3GeneralS2Sq}}$	$+10\lambda_{X_2}m_2^2$	$+2\lambda_{X_7}^{(1)}m_1^2$
A	-4	1	N/A	3	N/A	2	N/A	—	—	—
B	1	2	N/A	2	N/A	1	N/A	—	—	—
B	-1	2	N/A	2	N/A	1	N/A	—	—	—
B	-3	2	N/A	2	N/A	1	N/A	—	—	—
C	1	2	N/A	2	N/A	3	N/A	—	—	—
C	-1	2	N/A	2	N/A	3	N/A	—	—	—
C	-3	2	N/A	2	N/A	3	N/A	—	—	—
D	0	3	N/A	1	N/A	2	N/A	—	—	—
D	2	3	N/A	1	N/A	2	N/A	—	—	—
D	-2	3	N/A	1	N/A	2	N/A	—	$+6\lambda_{X_2}m_2^2$	$+6\lambda_{X_7}^{(1)}m_1^2$
E	0	3	N/A	3	N/A	2	N/A	—	—	—
E	2	3	N/A	3	N/A	2	N/A	—	—	—
E	-2	3	N/A	3	N/A	2	N/A	$-4\lambda_{S_2}^{(A)}\mu_{\mathrm{T3GeneralS2Sq}}$	$+10\lambda_{X_2}m_2^2$	—
E	-4	3	N/A	3	N/A	2	N/A	—	—	—

Quantity: $\mu_{\mathrm{T3GeneralS2Sq}}$

Terms 3 of 3

Column	Representative term structure
T1	$+m_1^2(6\lambda_{X_7}^{(1)} + 2\lambda_{X_7}^{(2)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
A	0	1	N/A	3	N/A	2	N/A	—
A	-2	1	N/A	3	N/A	2	N/A	—
A	-4	1	N/A	3	N/A	2	N/A	—
B	1	2	N/A	2	N/A	1	N/A	—
B	-1	2	N/A	2	N/A	1	N/A	—
B	-3	2	N/A	2	N/A	1	N/A	—
C	1	2	N/A	2	N/A	3	N/A	—
C	-1	2	N/A	2	N/A	3	N/A	—
C	-3	2	N/A	2	N/A	3	N/A	—
D	0	3	N/A	1	N/A	2	N/A	—
D	2	3	N/A	1	N/A	2	N/A	—
D	-2	3	N/A	1	N/A	2	N/A	—
E	0	3	N/A	3	N/A	2	N/A	—
E	2	3	N/A	3	N/A	2	N/A	—
E	-2	3	N/A	3	N/A	2	N/A	$+m_1^2(6\lambda_{X_7}^{(1)} + 2\lambda_{X_7}^{(2)})$
E	-4	3	N/A	3	N/A	2	N/A	—

Dimension-five $\psi^2\phi^2$ Wilson coefficients

The Wilson tensor is reported in the real-scalar component basis C_{ijab}. Each component with a nonzero one-loop beta function has its own table. Components absent at the matching boundary but generated by operator mixing are included automatically.

No successful EFT1 Wilson-RGE results are available.